Disease-Adaptive Heterogeneous Graph Learning with Anatomically Aligned Multi-Sequence MRI Representations for Lumbar Degenerative Disease

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Status of this build. This document contains Chapters 1 to 5: introduction, literature review, methodology, results, and discussion and conclusions. Chapters are numbered naturally; the counter advance used while Chapter 1 was unwritten has been removed. Research question RQ4, cross-institutional transfer, was not executed and is reported as such rather than omitted. Data collection is conducted at Rizgary Teaching Hospital; no patient-identifiable data appears in this document or in the repository that produces it.

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Abstract

Magnetic resonance grading of lumbar degenerative disease is slow, and expert readers agree with one another only moderately: reported inter-reader κ ranges from 0.49 for subarticular stenosis to 0.73 for the central canal. This thesis asks whether explicit anatomical structure, imposed on a multi-sequence convolutional model, improves automated grading on the LumbarDISC benchmark of 1,974 studies and 48,657 annotated targets.

Three mechanisms were proposed and each was tested against a control matched in capacity rather than against its own absence: anatomically aligned cross-sequence self-supervision, disease-conditioned sequence routing, and a heterogeneous disease–anatomy graph. An eight-rung ablation ladder was executed over seven training seeds, seventy runs in total, with inference on a frozen patient-level test partition and false discovery rate control across a pre-specified comparison family.

The complete system improves on a single-sequence baseline by +0.0177 quadratic weighted kappa (95% CI [+0.0097, +0.0260]) on every seed, and Severe→Normal confusions fall from 5.7% to 3.8%. Two further comparisons survive correction: typed heterogeneous graph edges (+0.0093, $p_{\mathrm{FDR}} = 0.009$) and a clinically cost-sensitive ordinal objective (+0.0082, $p_{\mathrm{FDR}} = 0.009$), the latter carrying the largest standardised effect in the study.

These differences are statistically detectable and clinically negligible, and the thesis reports them as such. Counted on the test predictions, the complete system grades 42 more of 7,310 targets correctly than the baseline, one additional correct target for every seven patients, and the aggregate improvement is modest relative to reported inter-radiologist variability. No claim is made that the architecture would alter radiological practice.

The anatomical mechanisms do not. Cross-sequence self-supervision confers no measurable benefit on accuracy, on robustness to a withheld sequence, or on where the model attends. Disease-conditioned routing reproduces the clinically expected sequence preference in every run, yet a router-free model shows the same preference under intervention, identifying the pattern as a property of the benchmark's annotation structure rather than of routing. Anatomically correct graph topology does not outperform a degree-preserving random shuffle. These are bounds rather than absences: the paired intervals constrain the three mechanisms to at most 1.32%, 1.55% and 1.05% of baseline performance.

Attribution analysis supplies a mechanism for that pattern. A plain single-sequence encoder already concentrates 2.4 times the chance share of its evidence on the annotated target, so the localisation these priors were introduced to supply is largely accomplished before any of them is added.

The contributions are therefore a verified multi-sequence grading system, a controlled ablation protocol in which three analyses that initially produced positive conclusions were overturned by their own controls, and evidence that anatomical structural priors do not pay at this data scale together with a measured explanation of why.

Keywords: lumbar spinal stenosis, magnetic resonance imaging, ordinal classification, graph neural networks, self-supervised learning, ablation study, negative results.

Contents

Chapter 1

Introduction

1.1 Introduction and Thesis Position

Lumbar degenerative disease presents an unusually demanding problem for medical image analysis. The anatomical region is compact, but the diagnostic task is not: one examination contains several intervertebral levels, several pathological compartments, left–right distinctions, multiple magnetic resonance imaging (MRI) sequences acquired in different planes, and severity labels whose boundaries are partly subjective. A clinically meaningful automated system must therefore do more than recognise that degeneration is present. It must identify *what* is abnormal, *where* it is abnormal, *how severe* the abnormality is, and *which imaging evidence* supports that judgement. It must also remain interpretable as a measurement of radiological grading rather than be mistaken for an autonomous clinical diagnosis.

The need for such systems arises from both disease burden and reporting workload. Low back pain remains one of the largest contributors to disability worldwide, and lumbar degenerative change is a frequent structural correlate examined by MRI [1, 2]. Routine interpretation is repetitive because the reader must inspect multiple levels across sagittal and axial acquisitions, while the severity of central canal, foraminal and subarticular narrowing is judged against morphological criteria that are not perfectly reproducible between human readers [3–6]. Automation is attractive in this setting not because a neural network is assumed to possess a superior biological truth, but because a reproducible system may help standardise a high-volume structured task whose human reference process itself has known variability.

The technical field has advanced rapidly. Early systems demonstrated that lumbar anatomy could be localised and individual discs graded with convolutional networks; more recent pipelines combine explicit localisation, 2.5D regions of interest, multiview fusion, attention and large public benchmarks [7–10]. The central problem has consequently shifted. The thesis no longer asks whether a convolutional neural network or transformer can classify a lumbar MRI in isolation. It asks whether the *structure of the grading problem itself* has been represented faithfully enough.

This thesis takes the position that lumbar MRI grading is a structured prediction problem with four coupled sources of information. First, disease occurs at ordered anatomical levels rather than at unrelated image locations. Second, several conditions coexist within one motion segment and lateral compartments occur as paired left–right structures. Third, the diagnostic value of an MRI sequence is target-dependent: sagittal T1, sagittal T2/STIR and axial T2 provide complementary rather than interchangeable evidence [11]. Fourth, a model that performs well on one benchmark may degrade when scanners, protocols, population and reference-standard conventions change across institutions [12–14].

The research therefore evaluates three methodological contributions and one translational validation programme. The integrated framework evaluating these components is designated **AMOG-Net** (**A**natomy-aware **M**ulti-sequence **O**rdinal **G**raph Network). The first contribution is anatomically aligned cross-sequence self-supervision, in which DICOM patient-space correspondence defines which views should agree during representation learning. The second is disease-conditioned adaptive sequence routing, in which the model learns target- and patient-dependent weights over available MRI sequences and is explicitly tested under both missing and degraded inputs. The third is a heterogeneous disease–anatomy graph whose nodes represent level–condition–laterality targets and whose typed edges encode adjacent-level, same-level cross-condition and bilateral relations. These are then evaluated under a deliberately separate transfer programme: development occurs on a multinational public benchmark; the selected model is frozen; zero-shot performance is measured on an independent Rizgary Hospital cohort; and controlled few-shot adaptation is performed only afterwards.

Several related techniques are necessary but are not claimed as doctoral novelty. Anatomical localisation, segmentation, 2.5D region construction, conventional CNN or transformer encoders, class-imbalance handling, ordinal loss functions, calibration and parameter-efficient fine-tuning are established tools. They are used to construct a fair experiment around the thesis contributions rather than presented as original merely because they appear in the same architecture. This novelty boundary is important in a field where anatomy-guided contextual grading, cross-sequence attention and graph-based disc modelling have already been reported [15–17]. The question is not whether any one of those mechanisms exists elsewhere; it is whether the particular representation tested here—typed relationships among the benchmark's disease outputs, anatomical cross-sequence supervision, target-conditioned evidence allocation and explicit institutional transfer—adds measurable knowledge beyond strong matched baselines.

The principal empirical outcome is that the complete system improves agreement with the reference standard over a single-sequence baseline by +0.0177 quadratic weighted kappa, on every one of seven training seeds and after correction for multiple comparisons, but that the anatomical mechanisms proposed to achieve this do not account for the improvement: attribution analysis indicates the convolutional encoder already localises the graded structure without them.

1.2 Background and Motivation

1.2.1 Clinical Motivation: A Repetitive but Structured Imaging Task

Lumbar degeneration is not a single lesion. It is a family of related structural changes involving the intervertebral disc, facet joints, ligamentum flavum, vertebral endplates and the neural spaces that these structures surround. Disc dehydration and height loss alter load distribution; annular bulging or herniation may encroach on the canal and foramina; posterior element hypertrophy may further reduce the space available to neural tissue. These changes are assessed at several levels, most commonly from L1–L2 through L5–S1, and their significance depends on which anatomical compartment is narrowed.

Three stenotic compartments are especially relevant to the benchmark task used in this thesis. Central canal stenosis concerns the thecal sac and cauda equina, foraminal narrowing concerns the exiting nerve roots, and subarticular or lateral recess stenosis concerns the traversing roots. Their visual criteria are related but not identical, and neither are the acquisitions from which they are best judged. Central canal and lateral recess morphology depend heavily on axial T2 information, whereas neural foraminal assessment benefits particularly from sagittal or parasagittal T1 because perineural fat provides the relevant contrast [4, 6, 11]. A model that treats all sequences as equivalent therefore ignores a clinical property of the task before learning has even begun.

The labels themselves are another source of complexity. Morphological grading systems convert continuous anatomical variation into ordinal categories. Human agreement is strongest for some compartments and weaker for others: the reliability study of Lurie et al. found substantially better inter-reader agreement for central canal stenosis than for foraminal and subarticular findings [3]. This matters methodologically. If the reference standard contains boundary uncertainty, model evaluation must distinguish gross errors from one-grade disagreements and must interpret performance relative to the reliability of the grading instrument. Apparent model error and reference-standard disagreement are not always separable.

This combination—high examination volume, repeated level-wise assessment, complementary sequence evidence and imperfect categorical boundaries—makes lumbar MRI a suitable domain for artificial intelligence, but only if the automation is framed correctly. The objective is not to replace full clinical interpretation. Symptoms, neurological findings, prior imaging and management context remain outside the information available to an image-only grading system. The narrower objective is to make a defined radiological grading task more consistent, measurable and portable. Reader-assistance studies support this framing: deep-learning assistance has been shown to reduce reporting time and improve or preserve inter-reader agreement, with the greatest gains often occurring among less experienced readers [18]. This thesis therefore treats the system as a potential component of decision support, not as an autonomous treatment decision-maker.

1.2.2 Technical Motivation: From Classification to Structured Prediction

The evolution of lumbar MRI automation has progressively narrowed the unit of prediction. Early work often asked whether a lesion or abnormality was present in an image. Subsequent systems localised discs, graded individual levels, and shared encoders across several pathologies. Public benchmark design has now made the task more explicit. The RSNA Lumbar Degenerative Imaging Spine Classification resource, published as LumbarDISC, contains 2,697 patients and 8,593 MRI series collected across eight institutions in six countries [10]. The associated challenge requires five condition/laterality targets at five lumbar levels, producing 25 ordinal outputs per study [10, 19]. A correct model therefore does not simply output "mild", "moderate" or "severe" for an image; it attaches one of those grades to a particular condition at a particular level and, for lateral compartments, a particular side.

This target structure exposes a limitation in the prevailing formulation. Most systems share image features but ultimately emit independent output heads. Shared features allow several tasks to benefit from a common representation, but a prediction at L4–L5 does not explicitly inform an uncertain prediction at L3–L4; a confident left foraminal prediction does not interact structurally with its right counterpart; and the central, foraminal and subarticular targets at one motion segment do not exchange typed messages simply because they coexist anatomically. The representation is therefore multi-task but not fully relational.

Graph learning provides a natural language for testing whether those relationships carry useful information. Graph neural networks are not novel by themselves, and a lumbar application has already reconstructed an individual disc as a point cloud and graded it with a graph network [17]. That precedent is important because it prevents an unjustified claim that "using a graph in lumbar MRI" is new. The unresolved question is narrower: whether the *prediction targets* can be represented as nodes connected by clinically motivated relation types, and whether those relations improve grading beyond independent heads, an ordered level transformer and a homogeneous graph. The distinction between a graph over points within one disc and a graph over disease–anatomy targets across the spine is the novelty boundary this thesis tests.

A second technical motivation concerns multimodal evidence. Recent methods have already shown that cross-view and cross-sequence attention can improve lumbar assessment [9, 16], while anatomy-guided inter-level context has also become an active direction [15]. These developments mean that novelty cannot rest on "using multiple sequences" or "using attention". The remaining question is whether evidence allocation should be *conditioned on the target and the patient*, and whether one trained model can remain operational when an expected sequence is unavailable or present but of poor quality. A fixed fusion rule assumes the same contribution of sagittal T1, sagittal T2/STIR and axial T2 for every target and every study. Clinical sequence–pathology correspondence makes that assumption doubtful.

A third motivation is representation learning. Generic ImageNet pretraining is convenient but does not exploit the particular correspondence available in MRI: the same anatomical level can be observed under different contrasts and planes. Earlier spinal work demonstrated that self-supervision derived from patient-level or longitudinal information can reduce dependence on manual labels [20], and disc-centric contrastive work continues that line [21]. DICOM geometry makes a stronger, directly anatomical pairing possible. If sagittal and axial observations can be matched in patient space, then the correspondence itself can serve as supervision without pretending that the images should look visually alike.

Finally, there is a translational motivation. Systematic reviews repeatedly find that internally impressive medical-imaging models underperform when re-evaluated outside their development distribution [1, 2]. Lumbar-specific external validations show the same pattern, with fine-grained grading generally less portable than localisation or detection [12, 13]. A model intended for eventual use in Iraq or the wider region therefore cannot be justified solely by performance on a multinational benchmark. The model must be transported to a genuinely independent hospital cohort, its zero-shot degradation measured before local tuning, and the amount of local annotation required to recover performance quantified explicitly. That annotation–performance curve is more actionable than a single statement that "domain adaptation improved accuracy".

1.2.3 Why the Problem Should Be Addressed Now

Three developments make the present study timely. First, LumbarDISC supplies a large, multi-institutional and anatomically structured benchmark whose target schema is rich enough to test relational modelling rather than generic image classification [10]. Second, public anatomical resources such as SPIDER reduce the need to claim localisation or segmentation as the central contribution [22]. Third, the current methodological literature is sufficiently mature to provide strong controls: multi-view cross-attention, anatomy-guided contextual models, contemporary CNN and transformer backbones, self-supervised pretraining, calibration and parameter-efficient transfer all exist as comparators or supporting tools. The research can therefore ask a more precise question than whether a new architecture attains a high score.

The resulting opportunity is methodological rather than merely computational. The field now has enough data and enough baseline competence to test whether explicit clinical structure provides value over capacity alone. If a typed target graph, anatomical self-supervision or disease-conditioned router fails to outperform its control, that negative result is still informative. Conversely, if a gain disappears when topology is shuffled, when a simpler transformer is matched for capacity, or when the model is transported to another hospital, the interpretation must change. This insistence on falsifiable components is part of the motivation for the study, not an afterthought added to evaluation.

1.3 Problem Statement

Despite rapid progress in automated lumbar MRI analysis, four connected methodological problems remain.

1.3.1 Problem 1: The Output Space Is Structured but Commonly Treated as Independent

The benchmark output space contains ordered lumbar levels, multiple conditions at each level and bilateral counterparts. These targets arise from one connected anatomical system, yet the dominant design predicts them independently after a shared encoder. This separation is convenient for implementation but does not test whether adjacent-level context, within-level co-occurrence or bilateral relations carry predictive information. Existing graph work in lumbar MRI does not resolve this question because its graph is constructed within a disc rather than across the disease targets [17]. CrossSpine introduces anatomical priors and level-specific behaviour but still does not formulate the full 25 target outputs as a typed communication graph [16]. The first problem is therefore not the absence of contextual modelling in general; it is the absence of a controlled test of explicit disease–anatomy relations among the prediction targets.

A relational model also carries a specific risk: useful context can become "information contagion". A severe lesion at one level should not cause the model to elevate an adjacent normal level simply because degeneration often co-occurs. Any graph proposed as a solution must therefore preserve local visual evidence, include non-relational and homogeneous controls, and be stress-tested on naturally isolated lesions. Without those controls, improved aggregate performance would not establish that the anatomical topology is beneficial.

1.3.2 Problem 2: Sequence Fusion Is Usually Fixed While Evidence Is Target- and Quality-Dependent

Sagittal T1, sagittal T2/STIR and axial T2 are complementary acquisitions. The foramina, central canal and lateral recess are not equally well represented in each view, and routine data may be incomplete or technically degraded. Strong recent systems address multi-view fusion through cross-attention [9, 16], but a remaining deployment question is whether one model can dynamically allocate evidence by condition, level, patient and sequence quality. A model trained only on complete, high-quality benchmark studies may fail when one sequence is missing, truncated or motion-corrupted. Missingness and corruption are also different problems: an absent sequence can be masked, whereas a degraded sequence remains present and can attract misleading attention. The second problem is therefore to learn adaptive sequence allocation without turning the router into an unexamined source of instability.

1.3.3 Problem 3: Cross-Sequence Correspondence Is Available but Underused as Supervision

Self-supervised learning can reduce dependence on costly expert labels, but common contrastive objectives construct positive pairs through augmentations of the same image. Multi-sequence MRI contains a more domain-specific relationship: different acquisitions can depict the same anatomical level even when their intensity pattern, orientation and diagnostic role differ. DICOM patient-space geometry provides the physical correspondence required to identify those pairs. Prior spinal self-supervision demonstrates the value of metadata-derived or longitudinal relationships [20], but anatomical cross-sequence correspondence has not been established as the supervision signal for the target formulation studied here.

The problem is not solved simply by forcing all sequence representations to become identical. T1 and T2 contrasts contain sequence-specific diagnostic information, so excessive invariance could improve embedding alignment while damaging grading. The study must therefore separate the self-supervised projection space from downstream features and test explicitly whether sequence-specific information survives pretraining.

1.3.4 Problem 4: Internal Accuracy Does Not Establish Transportability or Clinical Error Alignment

The strongest evidence against uncritical internal benchmarking is the repeated performance decline under external validation [1, 12, 13]. Scanner manufacturer, field strength, resolution, axial prescription, preprocessing and patient mix can all change. More subtly, the reference standard may change: a public benchmark may use structured expert annotation while a hospital's routine reports reflect local reporting habits and may omit findings that were not considered clinically salient. An external performance drop can therefore combine imaging-domain shift, localisation failure and reference-standard shift.

A defensible transfer study must measure zero-shot performance before local tuning, keep the local test set fixed, separate it from the adaptation pool, and harmonise the local reference standard as far as resources allow. The present work therefore uses the Rizgary Hospital cohort for external evaluation rather than for iterative architecture development. The principal local comparison is restricted to central canal stenosis, for which a defensible overlap with the public benchmark can be created; local disc bulge, protrusion and extrusion remain a separate morphology task rather than being relabelled as stenosis.

The error objective also requires care. Severity grades are ordinal, and distant undergrading is clinically different from adjacent disagreement. Yet the literature does not establish that ordinal losses automatically improve classification: Niemeyer et al. found several ordinal formulations unable to outperform standard cross-entropy for Pfirrmann grading [23]. A recent preprint has begun to evaluate distance-aware ordinal assessment on LumbarDISC [24]; however, the combination of asymmetric clinical cost with target-level relational modelling appears unaddressed in the literature reviewed for this thesis. At the same time, reducing Severe→Normal/Mild errors by simply shifting the model toward predicting Severe would create an unacceptable false-positive burden. The fourth problem is therefore to evaluate error direction, calibration and selective prediction without converting radiological grading into an unsupported claim of treatment utility.

1.3.5 Integrated Problem Statement

Taken together, the research problem can be stated as follows:

> Current automated lumbar MRI grading systems can localise anatomy and predict per-level disease severity with increasingly strong internal performance, but the prevailing formulation does not fully exploit the relational structure of the 25-target output space, does not explicitly learn target- and quality-dependent sequence allocation, underuses DICOM-defined cross-sequence anatomical correspondence as a self-supervised signal, and provides limited evidence on how these design choices behave under true institutional transfer. A controlled methodology is therefore required to determine whether explicitly modelling these properties improves grading, robustness, label efficiency and portability beyond strong matched baselines, while preserving local evidence and respecting the uncertainty of the clinical reference standard.

This protocol anticipated that implementation might reveal a previously unrecognised but reproducible problem, and required that such a finding be recorded here rather than absorbed silently. It did. Verification of the DICOM correspondence found a reproducible failure mode affecting a minority of studies, in which the axial acquisition does not cover the annotated level at all, so a projected point lands tens of millimetres away in unrelated anatomy. Nine studies fail on every projection tested. The check, its distribution and the affected studies are reported in Section 4.3. It is recorded because it is a property of the data that any similar pipeline will encounter, not because a proposed contribution underperformed.

1.4 Research Gap and Novelty Boundary

Chapter 2 develops the evidence in detail; the purpose here is to state the gap concisely enough to define the thesis.

1.4.1 Gap A: Explicit Relational Modelling of Disease–Anatomy Targets

The reviewed literature contains multi-task systems, inter-level attention, anatomical priors and one lumbar graph model, but no reviewed study constructs the benchmark's level–condition–laterality targets as a heterogeneous graph with typed relations for adjacent levels, within-level condition coexistence and bilateral symmetry. Baur et al. use a graph within one reconstructed disc [17]; Nguyen et al. introduce level-specific priors and cross-sequence attention without target-to-target graph communication [16]; Chai et al. demonstrate anatomy-guided contextual multi-level modelling without the disease-target graph formulation proposed here [15]. The intended contribution is therefore not "graph neural networks for lumbar MRI" but a controlled test of an explicit typed graph over the prediction targets themselves.

1.4.2 Gap B: Anatomically Aligned Self-Supervision and Disease-Conditioned Evidence Routing

The literature establishes both the complementarity of MRI sequences and the value of self-supervised pretraining, but these strands remain only partially connected in lumbar grading. Anatomical correspondence across sequences is usually treated as a preprocessing requirement for fusion rather than as supervision. At the same time, multisequence systems generally learn a common fusion mechanism rather than a condition- and quality-dependent allocation that is deliberately challenged under missing and corrupted modalities. The thesis therefore tests two linked ideas: that DICOM geometry can define cross-sequence positive relationships, and that a single model can use the resulting features through a target-conditioned router whose failure modes are directly measurable.

1.4.3 Gap C: Quantified Transfer and Adaptation Under a Harmonised External Task

Existing external validations establish that transport is difficult, but they usually report one transported model at one point in the adaptation process—often with no adaptation at all. The operational question for a receiving hospital is instead how performance changes before adaptation and how quickly it recovers as local labels are supplied. The literature reviewed for this thesis contains no Middle Eastern evaluation of the same scope and no quantified parameter-efficient annotation curve for this lumbar target formulation. The gap is therefore not simply "lack of external validation"; it is lack of a staged zero-shot-to-few-shot transfer experiment with fixed local testing, controlled annotation budgets and explicit reference-standard harmonisation.

1.4.4 Supporting Gap: Directional Error and Calibrated Uncertainty

Ordinal severity and asymmetric error are clinically meaningful, but ordinal objectives have not consistently outperformed categorical baselines [23]. Calibration is also rarely reported in lumbar MRI despite the known overconfidence of modern neural networks [25]. These are therefore treated as supporting questions rather than headline novelty. A cost-sensitive objective is retained only if it reduces clinically consequential under-grading without creating indiscriminate Severe false positives, and selective prediction is evaluated as a model behaviour rather than proof of a validated clinical workflow.

The novelty claim throughout the thesis will remain appropriately bounded. The phrasing "to our knowledge" or "in the reviewed literature" is used where necessary, and the final claim will be updated against the literature available at the time of submission. A new paper that adopts one element of the proposed system does not invalidate the PhD if the controlled experimental question remains distinct; it does, however, change how originality must be described.

[**LATER INTEGRATION: Immediately before thesis submission, update this novelty boundary against publications appearing after the current literature-search cut-off. Record which claims remain original and which should be reframed as independent validation, extension or comparative evidence.**]

1.5 Research Aim

The overall aim of this PhD is:

> *to determine whether explicitly modelling anatomical relationships among lumbar disease targets, anatomically aligned evidence across MRI sequences, and*

target-dependent sequence availability improves the accuracy, robustness, label efficiency and cross-institutional portability of automated lumbar spine MRI severity grading, relative to strong non-relational and fixed-fusion baselines.

The aim is intentionally phrased as a determination rather than a promise of superiority. The thesis succeeds scientifically if it establishes under controlled conditions where these mechanisms help, where they fail, and what their limits are.

1.6 Research Objectives

The aim is operationalised through the following objectives.

1. To construct a reproducible patient-level lumbar MRI analysis pipeline that preserves DICOM patient-space geometry, performs reliable level localisation, and supplies identical anatomically defined regions of interest to all grading baselines and proposed models.

2. To establish competitive independent and contextual grading baselines on the public LumbarDISC development data, including fixed fusion, cross-attention and ordered inter-level context where reproducibility permits, so that improvements cannot be attributed to a weak comparator.

3. To develop and evaluate anatomically aligned cross-sequence self-supervision in which positive relationships are defined by the same patient and lumbar level across MRI sequences using DICOM geometry.

4. To quantify whether this anatomical self-supervision improves supervised grading under 10%, 25%, 50% and 100% label availability and whether it preserves sequence-specific diagnostic information.

5. To develop and evaluate disease-conditioned adaptive sequence routing that learns patient- and target-dependent weights over available MRI evidence while explicitly representing sequence availability and acquisition quality.

6. To test the router under both missing-sequence and corrupted-sequence conditions, and to compare its behaviour with concatenation, averaging and ordinary cross-attention under matched inputs and training budgets.

7. To construct a heterogeneous disease–anatomy graph over the 25 level–condition–laterality targets, with typed adjacent-level, same-level cross-condition and bilateral relations, and to compare it with independent heads, an ordered level transformer, a homogeneous graph, ungated message passing and shuffled topology.

8. To test whether relational reasoning introduces adjacent-level false positives in naturally isolated lesions and to retain the graph only where any performance or robustness benefit is not purchased by clinically incoherent information propagation.

9. To compare categorical, ordinal and clinically asymmetric cost-sensitive grading objectives, together with probability calibration and uncertainty analysis, while explicitly reporting both severe under-grading and severe over-grading.

10. To quantify zero-shot transfer from the multinational public benchmark to the independent Rizgary Hospital cohort on the shared central-canal grading task, separating localisation, image-domain and reference-standard effects as far as the retrospective data allow.

11. To estimate an annotation–performance recovery curve using controlled local adaptation sizes $N \in \{10, 25, 50, 100\}$ and to compare where a limited trainable-parameter budget is most effectively placed.

12. To report all principal comparisons with patient-level uncertainty, repeated training where feasible, contribution-specific ablations, negative results and explicit methodological limitations.

These objectives define a staged experiment rather than a monolithic architecture. A component is not retained because it appears in the proposal; it is retained only if the corresponding experiment demonstrates a reproducible contribution or a meaningful robustness, calibration or interpretability advantage.

1.7 Research Questions

The research questions are fixed prospectively and correspond directly to the methodological tests in Chapter 3.

RQ1 — Relational disease modelling. Does representing lumbar disease targets as a typed heterogeneous graph improve grading over independent target heads, an ordered five-level transformer and a homogeneous graph, and are any gains concentrated where single-target evidence is weakest?

RQ2 — Anatomically aligned self-supervision. Does cross-sequence pretraining based on DICOM-defined anatomical correspondence improve grading, label efficiency and transfer compared with ImageNet initialisation, generic medical pretraining and ordinary augmentation-based contrastive learning?

RQ3 — Disease-conditioned modality routing. Can one model learn target- and patient-dependent sequence weights while remaining robust to both missing and quality-degraded MRI sequences, without requiring a separately trained network for every combination of sagittal T1, sagittal T2/STIR and axial T2?

RQ4 — Cross-institutional transfer. What is the zero-shot degradation when a benchmark-trained model is applied to a previously unseen Middle Eastern clinical cohort, how much of that degradation is attributable to localisation versus grading/reference-standard shift, and how does performance recover as the amount of local annotation available for adaptation increases?

RQ5 — Clinically asymmetric error and uncertainty. Do ordinal and cost-sensitive objectives reduce clinically consequential distant errors, particularly Severe→Normal/Mild, and can calibrated uncertainty support selective prediction without overstating clinical safety?

These questions are intentionally answerable by negative as well as positive results. If a typed graph performs no better than a homogeneous graph, RQ1 is still answered. If anatomical self-supervision improves label efficiency but not external transfer, RQ2 yields a more limited but still informative result. This design avoids defining doctoral success as the requirement that every proposed component win.

1.8 Research Hypotheses

The study uses directional hypotheses where the literature justifies a prediction, but avoids arbitrary target accuracies that would have no defensible basis before implementation.

H1 — Anatomical self-supervision. Anatomical cross-sequence self-supervision will improve label-efficient grading and cross-institutional robustness relative to generic pretraining, particularly when only a fraction of benchmark labels is used.

H2 — Adaptive routing. Disease-conditioned routing with explicit availability masking, modality dropout and quality-aware gating will degrade less under missing-sequence and corruption experiments than fixed concatenation or ordinary cross-attention trained only on clean complete studies.

H3 — Heterogeneous relational structure. A typed heterogeneous graph will outperform independent target heads and a homogeneous graph if the proposed anatomical edge types carry useful information; performance under a shuffled-edge control should deteriorate if the topology itself matters.

H4 — External degradation. Zero-shot performance on the Rizgary Hospital cohort will be lower than internal benchmark performance because of scanner, protocol, population, localisation and reference-standard shift; the magnitude and pattern of that degradation are treated as findings rather than as failure of the research programme.

H5 — Annotation-dependent recovery. Local adaptation will improve performance as annotation increases, but no predetermined percentage of recovery is required. The shape of the recovery curve and the location in the network at which limited adaptation is most effective are empirical outcomes.

H6 — Asymmetric grading error. Cost-sensitive and ordinal objectives may reduce distant Severe→Normal/Mild errors even if they do not improve aggregate macro-F1; any reduction must be evaluated against false-positive Severe predictions and against the cross-entropy baseline [23].

The hypotheses will not be rewritten after inspection of the held-out public or Rizgary test results. If an implementation decision changes the operational test of a hypothesis, that change will be entered into the protocol changelog with its date and rationale.

1.9 Conceptual Framework

The thesis can be understood as a progression from *local evidence* to *structured evidence* and finally to *transported evidence*.

At the first level, a model receives anatomically localised image evidence for a single level and target. This is the baseline condition: if local evidence alone is sufficient, more complex mechanisms should not be retained. At the second level, the model is permitted to combine complementary MRI sequences, but the combination must respect anatomical correspondence and sequence availability. Anatomical self-supervision asks whether the same structure under different acquisitions can produce a more label-efficient representation, while disease-conditioned routing asks how strongly each sequence should contribute to a given target.

At the third level, a target is no longer treated as an isolated endpoint. Its representation becomes a node in a typed graph whose relations correspond to three known structures of the lumbar grading problem: the ordered chain of levels, coexisting compartments at a level, and bilateral pairing. Message passing is soft rather than deterministic; no rule states that disease at one level forces disease at its neighbour. The graph is therefore a hypothesis about useful context, not a hard-coded disease progression model.

At the fourth level, predictions are evaluated as ordinal and probabilistic outputs. The model is judged not only by aggregate discrimination but by error distance, Severe-class performance, calibration and the false-positive cost of any cost-sensitive objective. This stage asks whether a statistically improved model is also better aligned with the shape of clinically consequential grading error.

The final level is institutional transport. The representation, router, graph and output head developed on public data are frozen before external evaluation. The Rizgary Hospital cohort then asks a different question from internal validation: what remains true when the acquisition and reporting environment changes? Only after zero-shot performance has been measured is local adaptation permitted. This ordering prevents the local test cohort from becoming an informal development set.

Conceptually, the thesis therefore tests the proposition

$$\text{robust grading} = f(\text{local anatomy, cross-sequence evidence, target relations, uncertainty, domain}), \tag{1.1}$$

where the equation is explanatory rather than a literal computational model. Each term corresponds to a controlled experiment, and no term is assumed necessary until its ablation supports that conclusion.

1.10 Significance of the Study

1.10.1 Scientific Significance

The principal scientific significance lies in changing the unit of reasoning from an isolated image label to a structured set of disease targets. Many improvements in medical imaging can be produced by additional parameters, larger backbones or more aggressive tuning. Such gains are difficult to interpret mechanistically. By contrast, this thesis attaches each proposed contribution to a specific hypothesis and a destructive control: anatomical edges are compared with homogeneous and shuffled topology; adaptive routing is compared with fixed fusion and challenged by missing and corrupted modalities; anatomical self-supervision is compared with generic pretraining and tested for over-invariance. The resulting evidence can show not merely whether the full model scored higher, but *why* it did or did not.

This matters because recent lumbar literature already demonstrates sophisticated contextual models [9, 15, 16]. The research contribution cannot therefore be the generic use of attention, graphs or multiple sequences. Its value lies in isolating the anatomical assumptions and measuring them against alternatives under one controlled pipeline.

1.10.2 Methodological Significance

The study places unusually strong emphasis on validity controls. Patient identity, not slice or target, defines the data split; self-supervised learning is prohibited from seeing held-out patients; localisation is evaluated separately from grading; zero-shot transfer is measured before adaptation; the external reference standard is kept distinct from routine report extraction; and every claimed model contribution requires an ablation. These choices respond directly to recurring weaknesses in the medical-imaging literature, where leakage, non-comparable splits and weak external validation can produce optimistic performance estimates [1, 2].

The transfer design is methodologically significant for a second reason. The local hospital cohort is not used simply as another test set. The experiment attempts to disentangle localisation failure, image-domain shift and reference-standard shift, then measures recovery as local labels are added. This converts "generalisation" from a binary claim into a measurable process.

1.10.3 Translational and Regional Significance

The regional significance follows from the use of an independent Iraqi hospital cohort that was not curated to mirror the public benchmark. Models developed on large international datasets are often assumed to transfer into settings whose scanner mix, acquisition practice, disease prevalence and reporting conventions were not represented in development. Testing that assumption is particularly important for institutions that do not have the resources to reproduce a large annotation campaign locally. If useful performance can be recovered from tens rather than hundreds of local annotations, that is a practical finding; if not, the negative result is equally important because it quantifies the annotation burden required for responsible transfer.

The study does not claim that one external hospital represents all of Iraq or the Middle East. Rizgary Hospital provides evidence of transport to one genuinely independent setting. Regional generalisation would require additional hospitals and is therefore a future study rather than an inference drawn from one centre.

1.10.4 Clinical Significance and its Boundary

The intended clinical significance is improved technical support for structured MRI grading: consistency, reproducibility, calibrated confidence and explicit handling of incomplete imaging. The thesis does not claim that this will improve treatment outcomes, referral decisions or radiologist productivity. Those claims require a prospective reader-assistance or impact study, as demonstrated by work that measures reader performance directly [18]. The present study therefore uses language such as "decision-support potential" and “technical portability”, not “clinical benefit”.

Of the benefits anticipated above, the executed experiments support *discrimination* and, within it, the reduction of clinically distant errors. Quadratic weighted kappa improves by +0.0177 over a single-sequence baseline, Severe→Normal confusions fall from 5.7% to 3.8% and recall on Severe targets rises from 61.1% to 63.1%, the last two attributable to the ordinal and cost-sensitive objective rather than to any proposed structural mechanism.

Robustness to missing sequences improves, but the improvement is attributable to modality dropout rather than to disease-conditioned routing, which changes sequence reliance by an amount indistinguishable from zero. Label efficiency and cross-institutional transfer were not evaluated and no claim is made for either. Calibration is reported but was not the object of a confirmatory comparison.

No inference is drawn from these figures about patient benefit. The study measures agreement with a reference standard on a retrospective benchmark; it does not measure any clinical outcome.

1.11 Proposed Contributions

At the protocol stage, the contributions below collectively form the proposed **AMOG-Net** framework and are *proposed contributions*. They become demonstrated thesis contributions only if the experiments support them.

1.11.1 Contribution 1: Anatomically Aligned Cross-Sequence Self-Supervision

The first proposed contribution uses DICOM-defined patient-space correspondence to construct positive relationships between different MRI sequences depicting the same patient and lumbar level. The objective is not to invent a new generic contrastive loss; established contrastive or non-contrastive objectives may be used. The contribution under test is the anatomical definition of correspondence and its value for label efficiency, grading and transport. A dedicated preservation experiment checks whether alignment erases sequence-specific diagnostic information.

1.11.2 Contribution 2: Disease-Conditioned and Quality-Aware Sequence Routing

The second proposed contribution allows one study to be fused differently for different disease targets. Routing depends on target identity, patient features, sequence availability and quality evidence, and is evaluated under genuine or simulated missingness and corruption. The contribution is not the existence of a softmax gate or mixture-of-experts mechanism; it is the target-conditioned allocation of clinically complementary evidence and the explicit robustness evaluation that accompanies it.

1.11.3 Contribution 3: Heterogeneous Disease–Anatomy Graph Reasoning

The third proposed contribution represents the 25 benchmark outputs as nodes in a heterogeneous graph. Typed edges encode adjacent-level, within-level cross-condition and bilateral relations. The graph includes a mechanism for preserving local evidence when neighbour messages are unhelpful and is evaluated against independent, ordered-transformer, homogeneous, ungated and shuffled-topology controls. A focal isolated-lesion test determines whether relational reasoning improves uncertain predictions without artificially spreading disease to adjacent normal levels.

1.11.4 Contribution 4: A Structured Cross-Institutional Transfer Evaluation

The translational contribution is an experimental design rather than a new network layer. The selected public-data model is evaluated zero-shot on a fixed Rizgary Hospital test set before any local adaptation. A separate local adaptation pool is then sampled at controlled annotation sizes and used to compare constrained update strategies. The analysis is strengthened by a verified-localisation control and, where resources permit, harmonised independent reader grading of central canal stenosis. This design estimates both the cost of transport and the cost of recovery.

1.11.5 Supporting Contributions

Ordinal and cost-sensitive objectives, calibration, uncertainty estimation and parameter-efficient adaptation support the principal contributions but are not claimed as original methods. Their value lies in testing whether the proposed structured model behaves sensibly under clinically relevant error definitions and limited local annotation. They are retained only when their contribution is measurable.

1.11.6 Demonstrated Contributions

The experiments reported in Chapter 4 support three contributions. They are not the three proposed above, and the difference is itself part of what the thesis contributes.

D1 — A verified anatomy-aware multi-sequence grading system. AMOG-Net improves quadratic weighted kappa over a single-sequence baseline by +0.0177 (95% CI

$[+0.0097, +0.0260]$), on every one of seven training seeds and surviving false discovery rate correction across the pre-specified comparison family. Two further comparisons also survive: typed heterogeneous edges and the ordinal, cost-sensitive objective. The system, its frozen data partition and its behavioural test suite are released.

D2 — A controlled ablation protocol for anatomical priors. Each claimed mechanism is tested against a control matched in capacity rather than against its own absence: a degree-preserving edge shuffle for graph topology, a router-free fixed-fusion arm for sequence routing, an architecture-matched untrained network as an attribution floor, and a label-shuffled model as a pipeline-level negative control. Section 4.12 documents three analyses that produced incorrect positive conclusions before the corresponding control was introduced, in each case in the direction of the hypothesis under test.

D3 — Evidence that anatomical structural priors do not improve grading at this data scale, with a mechanism. Neither anatomically aligned cross-sequence self-supervision nor disease-conditioned routing produces a detectable improvement, and anatomically correct graph topology does not outperform a degree-preserving random shuffle. These are bounds rather than absences: the paired intervals constrain the three mechanisms to at most 1.32%, 1.55% and 1.05% of baseline performance respectively. Gradient-weighted class activation mapping shows why: the convolutional encoder already concentrates 2.4 times the chance share of its attention on the annotated target before any prior is added. The localisation these priors were designed to supply is largely accomplished upstream of them.

1.11.7 Proposed Contributions Not Supported

Contribution 1 (anatomical cross-sequence self-supervision) is rejected on accuracy, on cross-sequence robustness under controlled input ablation, and on attention concentration. The pretext task is learnable and was learned — validation InfoNCE 1.0771 against a chance value of 3.4657 — but the representation confers no measurable downstream benefit.

Contribution 2 (disease-conditioned routing) divides. The router does learn clinically sensible target-dependent sequence weights, reproducing the expected allocation in all fifteen runs in which a gate is present. It does not improve grading (+0.0053 QWK, five seeds of seven, bounded at 1.55% of baseline) and it does not improve robustness to missing sequences. Its allocation tracks the annotation structure of the benchmark rather than creating a dependence of its own.

Contribution 3 (heterogeneous disease–anatomy graph) is partly supported. Typed relational structure improves grading over a homogeneous graph (+0.0093 QWK on every one of seven seeds, surviving correction at $p_{\mathrm{FDR}} = 0.009$); the anatomical identity of the relations does not. The resulting claim is narrower than proposed and more specific than the original: relational typing carries information, anatomical adjacency does not.

Contribution 4 (cross-institutional transfer evaluation) was not executed. Cohort preparation is complete and is reported in Section 4.9; adjudication of conflicting reference labels and a correct de-identification procedure remain outstanding. It is retained here as unexecuted rather than withdrawn.

1.12 Scope and Delimitations

Clear boundaries are necessary because lumbar MRI supports a much broader clinical and computational agenda than one PhD can test.

1.12.1 Clinical Scope

The primary public task is degenerative lumbar disease severity grading at five levels from L1–L2 through L5–S1. On LumbarDISC, the full target space contains central canal stenosis, left and right neural foraminal narrowing, and left and right subarticular stenosis, each graded as Normal/Mild, Moderate or Severe [10]. The study does not attempt to diagnose every cause of low back pain, infer symptoms, determine surgical candidacy or predict long-term patient outcomes.

The primary local transfer task is narrower. Rizgary Hospital reports support a defensible central-canal comparison more readily than the complete benchmark schema. The confirmatory external analysis is therefore restricted to central canal stenosis at the five lumbar levels unless new expert grading expands the local reference standard. Disc bulge, protrusion and extrusion are retained as a separate local morphology task and are not heuristically converted into stenosis grades.

1.12.2 Imaging Scope

The principal imaging inputs are sagittal T1, sagittal T2/STIR and axial T2 MRI. Other sequences may appear in clinical studies but do not define the primary model unless a justified later protocol amendment is made. DICOM geometry is preserved because cross-plane correspondence is part of the scientific design. PNG or other rendered formats may be used for visual inspection but do not replace the geometric metadata required for alignment.

1.12.3 Methodological Scope

Anatomical localisation and segmentation are enabling technologies. They are measured and controlled because a poor crop can invalidate grading, but a new localisation architecture is not the headline doctoral contribution. Likewise, backbone selection is controlled rather than treated as a novelty competition. The research may use ResNet-, EfficientNet-, Swin- or other contemporary encoders, but the primary backbone is frozen before the main contribution-specific ablations.

The graph topology is defined anatomically rather than discovered freely from the test data. Adjacent levels, same-level conditions and bilateral counterparts form the pre-specified relation families. Learning the globally optimal topology is a different research problem and is excluded from the primary scope.

1.12.4 Data and Institutional Scope

Model development uses the public LumbarDISC benchmark and, where permitted, SPIDER for anatomical localisation or segmentation support. The locally held labelled competition subset contained approximately 1,975 studies at the protocol stage. After quality control — retaining studies for which all three required sequences could be located and the DICOM geometry needed for anatomically corresponding crops was recoverable — the executed cohort is 1,974 studies carrying 48,657 annotated targets, partitioned at patient level into 1,381 training, 296 validation and 297 held-out test studies. The test partition contributes the 7,310 scored targets on which every result in Chapter 4 is computed. The partition is written once and reloaded by every run. The local project inventory currently identifies 299 narrative reports and 294 candidate matched multi-sequence cases, while a wider raw DICOM inventory contains additional studies. These numbers are not presented as the final cohort until one-to-one reconciliation is complete: [**TO RECORD AFTER IMPLEMENTATION: final Rizgary case-flow counts and reasons for exclusion**].

Before analytical use of the local cohort, the Rizgary data-preparation phase will follow the approved institutional governance protocol. A controlled research copy will be de-identified before access by the model-development team, and the permitted storage and compute environment will be verified before local processing begins. These safeguards precede, rather than form part of, model development.

Rizgary Hospital is a single external institution. The study can therefore support a claim of transport to that independent setting, not universal Middle Eastern performance. Additional regional hospitals would be required for a multicentre regional generalisation claim.

1.12.5 Ethical and Governance Scope

The local component is retrospective and does not alter patient care. Formal use of the local data remains conditional on the relevant institutional permissions: [**TO CONFIRM: hospital/ethics approval reference, date and responsible institution**]; [**TO CON-**

FIRM: retrospective consent or waiver basis]; and [**TO CONFIRM: permitted compute environment, including whether external cloud/GPU processing is authorised**]. Identifiable raw DICOM is not supplied to students or external compute; a controlled de-identified research copy is required before local analysis.

1.12.6 Evaluation Scope

The thesis evaluates technical grading performance, robustness, calibration, computational cost and cross-institutional portability. The unit of statistical independence is the patient. It does not claim prospective clinical impact, radiologist replacement or treatment benefit. Reader-assistance evaluation is a logical future study if the technical system proves sufficiently reliable.

1.13 Assumptions and Operational Definitions

The following terms are used consistently throughout the thesis.

Term	**Operational meaning in this thesis**
Lumbar level	One of L1–L2, L2–L3, L3–L4, L4–L5 or L5–S1.
Target	A level–condition–laterality output on LumbarDISC; 25 targets per study at full benchmark scope.
Severity grade	Normal/Mild, Moderate or Severe, encoded as an ordered three-class outcome.
Local evidence	Image-derived representation for one target before graph message passing.
Anatomical correspondence	Physical correspondence between acquisitions defined through DICOM patient-space geometry and lumbar localisation, not through image appearance alone.
Sequence routing	Learned allocation of weight over available MRI sequence features for a specific patient and target.
Missing modality	An expected MRI sequence is absent and explicitly unavailable to the model.
Degraded modality	A sequence is present but technically compromised, for example by motion, truncation, noise, slice loss or poor coverage.
Heterogeneous graph	A graph whose nodes are disease–anatomy targets and whose edges belong to clinically defined relation types.

Term	Operational meaning in this thesis
Zero-shot transfer	Evaluation of the frozen public-data model on the fixed local test set before local fine-tuning, threshold selection or local calibration.
Few-shot adaptation	Model adaptation using only the separate local adaptation pool under pre-specified annotation budgets.
Reference-standard shift	Difference between the process used to generate labels in the benchmark and the process used to establish the local external reference.
Selective prediction	Model abstention or referral based on uncertainty; this is a technical behaviour, not evidence of a validated clinical workflow.

These definitions are intentionally narrower than ordinary-language usage. In particular, "external validation" does not mean that a model has been proven safe for deployment, and "uncertainty" does not mean that the network has measured a patient's clinical risk.

1.14 Overview of the Research Design

The research is a retrospective, multi-dataset medical-imaging methods study. Its logic is cumulative, but the conclusions remain component-specific.

1. **Data and anatomy foundation.** Public data are inventoried, partitioned by patient, reconstructed in DICOM patient space, anatomically localised and converted into disease-specific 2.5D regions of interest.

2. **Strong baseline establishment.** An independent ROI classifier and competitive multiview/context baselines establish what can be achieved without the proposed relational formulation.

3. **Representation experiment.** Anatomically aligned cross-sequence self-supervision is evaluated against generic and medical pretraining under several label fractions and sequence-specific preservation probes.

4. **Fusion experiment.** Disease-conditioned sequence routing is tested against fixed and attention-based fusion under complete, missing and corrupted sequence conditions.

5. **Relational experiment.** Homogeneous and heterogeneous target graphs are compared with independent and ordered-transformer controls, including shuffled topology and isolated-lesion stress testing.

6. **Output and uncertainty experiment.** Categorical, ordinal and cost-sensitive objectives are evaluated together with calibration and uncertainty, with both under-grading and over-grading reported.

7. **External transport experiment.** The selected public-data model is frozen and evaluated zero-shot on the fixed Rizgary Hospital test set. A verified-ROI control and harmonised reader reference are used where feasible to interpret the source of degradation.

8. **Adaptation experiment.** Controlled local annotation budgets are sampled from a separate adaptation pool to estimate how quickly performance recovers and where a limited trainable-parameter budget is most effective.

This order deliberately prevents the final architecture from becoming an indivisible engineering object. Each stage introduces one assumption that can be rejected without invalidating the remainder of the thesis. The final model is therefore the simplest configuration supported by the evidence, not necessarily the configuration containing every proposed module.

1.15 Expected Outcomes and Criteria for Scientific Success

The protocol does not define success as reaching a predetermined accuracy. Such a threshold would be difficult to justify because performance depends on target, severity distribution, reference-standard reliability and the final patient split. Scientific success is instead defined by whether the research questions can be answered under controlled conditions.

For RQ1, success means establishing whether typed anatomical relations add value beyond capacity-matched controls and whether they avoid adjacent-level contamination. A null result is informative if the comparison is sufficiently powered and the topology controls are valid. For RQ2, success means establishing whether anatomical cross-sequence pairing improves label efficiency, transfer or neither, while documenting whether sequence-specific information is preserved. For RQ3, success means quantifying robustness under missing and degraded modalities and determining whether adaptive routing adds value over simpler fusion. For RQ4, success means producing an interpretable zero-shot transfer estimate and a controlled adaptation curve, even if the external drop is large. For RQ5, success means characterising the trade-off between severe under-grading, severe over-grading and probability reliability rather than selecting a loss solely because it raises one aggregate metric.

At final submission, the thesis will distinguish three categories explicitly: *supported contributions*, *negative findings*, and *unresolved questions*. This is preferable to rewriting the original proposal so that only successful components appear to have been planned.

RQ1 — Relational disease modelling. Partly supported. A typed heterogeneous graph improves grading over independent heads and over a homogeneous graph, on every seed. The anatomical identity of the relations does not contribute: an anatomically correct topology does not outperform a degree-preserving random shuffle.

RQ2 — Anatomically aligned self-supervision. Not supported. The pretext task is learnable and was learned, but the resulting representation yields no detectable improvement in grading, in robustness to withheld sequences, or in where the model attends.

RQ3 — Disease-conditioned modality routing. Mechanism supported, benefit not. The router reliably learns the clinically expected sequence weights, but a router-free model shows the same sequence dependence under intervention, and routing improves neither accuracy nor missing-sequence robustness.

RQ4 — Cross-institutional transfer. Not executed. Cohort preparation is complete; reference-label adjudication and de-identification remain outstanding.

RQ5 — Clinically asymmetric error and uncertainty. Supported for error asymmetry. Ordinal and cost-sensitive objectives improve both aggregate agreement and recall on Severe targets, on every seed, and form the most reliable single contributor to final performance. Selective prediction under calibrated uncertainty was not evaluated confirmatorily.

1.16 Thesis Organisation

At the current protocol stage, the thesis is organised around the progression from problem definition to evidence.

Chapter 1 — Introduction defines the clinical and methodological motivation, research problem, gaps, aim, objectives, research questions, hypotheses, proposed contributions, significance and scope of the thesis.

Chapter 2 — Literature Review examines the clinical grading systems that define the reference labels, MRI sequence–pathology correspondence, localisation and segmentation, backbone and contextual architectures, class imbalance and ordinal grading, public datasets, external validation, interpretability and the specific gaps that motivate the present study.

Chapter 3 — Methodology translates those gaps into a protocol-grade experimental design. It defines the public and local cohorts, leakage controls, DICOM reconstruction, localisation and ROI construction, self-supervised pretraining, adaptive routing, heterogeneous graph reasoning, grading objectives, calibration, ablations, external validation, domain adaptation, statistical analysis and reproducibility safeguards.

Chapter 4 — Results and Analysis will report the staged E0–E8 experiments, including component-specific ablations, robustness analyses, calibration, zero-shot external transfer, few-shot adaptation and failure analysis. The retained experiment groups are the eight-rung ablation ladder E0–E7 with two matched controls on the graph configuration, a controlled input ablation testing sequence dependence by intervention rather than by learned allocation, and a Grad-CAM attribution analysis against an architecture-matched untrained floor. The cross-institutional transfer group was prepared but not executed and is reported as such. The research questions are unchanged from those fixed in Section 1.7.

Chapter 5 — Discussion and Conclusions will interpret the findings against the literature, distinguish demonstrated contributions from negative results, state limitations, discuss implications for transport to other institutions and identify the next studies required before any clinical-impact claim.

[**TO CONFIRM: final institutional thesis structure if the university requires results and discussion as separate chapters, publication-based chapters, or an additional conclusion chapter**]. The scientific ordering above should remain recognisable even if the administrative chapter numbering changes.

1.17 Rules for Completing Prospective Placeholders

Because this chapter is written before completion of implementation, several fields are intentionally prospective. The following rules govern their later completion.

1. [**TO CONFIRM**] is used only for institutional or governance facts that are currently unknown, such as an approval reference or permitted compute environment. These fields are replaced only with documented facts.

2. [**TO RECORD AFTER IMPLEMENTATION**] is used for executed technical or cohort quantities that cannot responsibly be specified before the pipeline runs, such as final quality-controlled sample counts.

3. [**LATER INTEGRATION**] is used for a small number of places where the final thesis may state the principal findings or update the novelty boundary. These additions may report evidence; they must not alter the prospective research questions or make an unsuccessful hypothesis disappear.

4. Any substantive protocol change made after results become visible is documented with date, reason and affected analysis. Confirmatory and exploratory analyses remain distinguishable.

5. The introduction will not be retrofitted to imply that the final retained architecture was inevitable. The final thesis should preserve the fact that ACSSL, adaptive routing, heterogeneous graph reasoning and supporting losses were hypotheses subjected to testing.

These rules turn placeholders into a transparency mechanism rather than an invitation to hindsight bias.

1.18 Chapter Summary

This chapter has defined the thesis as a study of structured and transferable lumbar MRI grading rather than as a competition between image backbones. The clinical task is inherently multi-level, multi-condition, bilateral and multi-sequence, while its reference labels are ordinal and imperfectly reproducible. The technical literature has established that localisation should precede grading, that sophisticated multiview systems can perform strongly on public data, and that external performance frequently degrades. Those achievements narrow rather than remove the research gap.

The thesis therefore asks whether three explicit structural mechanisms add value: anatomically aligned cross-sequence self-supervision, disease-conditioned and quality-aware sequence routing, and a typed heterogeneous graph over level–condition–laterality targets. These mechanisms are evaluated component by component against strong controls rather than bundled into a single architecture and assumed beneficial. Supporting ordinal, cost-sensitive and uncertainty methods are used to test the clinical shape of error without being advertised as standalone novelty.

The translational component separates model development from external evaluation. LumbarDISC supplies the public development benchmark; SPIDER may support anatomical localisation; and the independent Rizgary Hospital cohort is reserved for zero-shot central-canal transfer and controlled local adaptation. The local reference process is treated explicitly as a potential source of shift rather than assumed equivalent to the public benchmark. Clinical impact remains outside the scope until a separate reader-assistance or prospective study is conducted.

The aim, objectives, RQ1–RQ5 and H1–H6 stated here form the prospective frame for Chapters 2 and 3. The remainder of the thesis is therefore not required to prove that every proposed mechanism succeeds. It is required to determine, with reproducible evidence, which assumptions survive controlled testing and institutional transfer.

In summary, the complete system improves on a single-sequence baseline reproducibly and by a margin that survives correction for multiple comparisons while remaining far too small to alter clinical practice, and the objective function designed around clinically asymmetric error is the most reliable single contributor to that improvement. Of the three proposed structural mechanisms, relational typing is supported, anatomical topology and disease-conditioned routing are not, and anatomically aligned self-supervision is rejected on every axis on which it was tested; attribution analysis indicates that the convolutional encoder already performs the localisation these mechanisms were intended to supply. The principal limitations are that all results are internal to one benchmark, that cross-institutional transfer was not executed, and that the study is underpowered for single-component effects of the magnitude observed.

Chapter 2

Literature Review

2.1 Introduction

2.1.1 Scope and Objectives of the Review

Lumbar degenerative disease is among the most common indications for magnetic resonance imaging (MRI) worldwide, and the interpretation of those images is a repetitive, subjective and time-consuming task that has attracted sustained attention from the medical image analysis community [26, 27]. This chapter reviews that body of work. Its purpose is not to catalogue every published model, but to establish what is genuinely settled, what remains contested, and where a controlled experiment can still add knowledge.

The review is organised around a single guiding question: *what determines whether an automated system can grade lumbar degenerative severity reliably enough to be clinically useful?* Four candidate answers recur in the literature—the quality and definition of the reference labels, the degree to which the anatomy is localised before it is classified, the representation and context supplied to the model, and the extent to which the resulting system transfers beyond the institution that produced it. Each is examined in turn.

The chapter closes by identifying what the field has *not* attempted. The recurring assumption across this literature is that the lumbar spine can be graded as a collection of independent predictions—each level, each condition and each side treated separately. That assumption is a modelling convenience rather than a clinical claim, and the gaps this thesis addresses follow from relaxing it.

Three specific objectives follow from this. First, to characterise the clinical grading systems that supply the ground truth for every supervised model in this field, and to quantify the reliability ceiling those systems impose. Second, to trace the methodological evolution from handcrafted features to contemporary convolutional and attention-based architectures, with particular attention to comparative claims between the two. Third, to assess the evidence on how much diagnostic information is carried by each MRI sequence and plane, and thus whether the computational expense of multi-sequence processing is warranted.

2.1.2 Search Strategy, Sources, and Inclusion Criteria

The literature underpinning this chapter was assembled into a structured inventory of 92 records spanning 2001 to 2026. Sources were identified through PubMed, Scopus, IEEE Xplore and arXiv, using combinations of the terms *lumbar spine*, *MRI*, *spinal stenosis*, *disc degeneration*, *deep learning*, *convolutional neural network*, *transformer*, *segmentation* and *grading*. Reference lists of the retrieved systematic reviews [1, 2, 28, 29] were searched by hand for records the database queries had missed, and the citation graphs of the most influential frameworks were traced forward.

Records were retained if they satisfied at least one of four criteria: they define or validate a clinical grading standard for lumbar degenerative disease; they present an automated method for detecting, segmenting or grading lumbar pathology on MRI; they externally validate such a method; or they describe a dataset or benchmark on which such methods are trained. Studies confined to computed tomography or plain radiography were excluded unless they establish a methodological principle transferable to MRI, as were studies of the cervical or thoracic spine except where explicitly used for contrast.

Two caveats on the evidence base should be recorded at the outset. First, the field's publication rate accelerated sharply after 2024, and a substantial fraction of the most directly relevant work [9, 16, 21, 30] exists as preprints that have not completed peer review; these are cited as indicative of methodological direction rather than as settled results. Second, performance figures reported across studies are frequently not comparable, because they derive from different datasets, different label definitions and different evaluation protocols. Wherever numerical results are quoted in this chapter, the cohort and validation regime that produced them are stated alongside, and cross-study comparisons are avoided unless the underlying protocols genuinely align.

2.1.3 Structure of the Chapter

Sections 2.2 to 2.4 establish the clinical substrate: the disease, the imaging protocol used to observe it, and the grading systems that convert observation into ordinal labels. This material is not background decoration. The severity classes a model is trained to predict are defined entirely by these systems, and their documented unreliability propagates directly into every reported accuracy figure in the chapters that follow.

Section 2.5 traces the methodological history of the field, and Section 2.6 isolates anatomical localisation as the design principle that most consistently separates effective systems from ineffective ones. Sections 2.7 and 2.8 then address representation and context: the backbones available, and how spatial, sequential and cross-plane information is supplied to them. Section 2.9 treats class imbalance and the ordinal structure of the label space, and Section 2.10 the datasets and benchmarks—principally the RSNA LumbarDISC collection—on which the field now converges. Sections 2.11 and 2.12 consider validation practice and interpretability respectively. Section 2.13 synthesises the whole and states the research gap.

2.2 Clinical Background: Lumbar Degenerative Disease

2.2.1 Epidemiology and Socioeconomic Burden

Low back pain is the single largest contributor to years lived with disability worldwide, and lumbar degenerative change is among its most frequently implicated structural correlates [1, 31]. Lifetime prevalence is reported as high as 84%, and in the United States alone the combined direct medical and indirect productivity costs are estimated between $50 and $100 billion annually [10]. Lumbar spinal stenosis in particular is the most common indication for spinal surgery in patients over 65 [4, 32], and demographic ageing is expected to increase both its incidence and the imaging workload it generates [26].

The operational consequence for this thesis is one of volume. MRI referrals for low back pain have grown faster than the radiology workforce able to report them [1, 33], and each lumbar study requires the reader to assign severity at multiple anatomical levels across several sequences—an average of 14 to 19 minutes of expert time per study [1]. It is this combination of high volume, repetitive structure and documented inter-reader variability that makes the task a plausible target for automation, rather than any claim that machines perceive the underlying pathology more acutely than radiologists do.

2.2.2 The Degenerative Cascade

Degeneration of the lumbar spine is a progressive, multifactorial process driven by ageing, mechanical loading, genetic predisposition and lifestyle factors [34, 35]. It begins in the intervertebral disc, whose nucleus pulposus loses proteoglycan content and therefore water from approximately the third decade of life onward. The resulting dehydration reduces the disc's height, its capacity to distribute axial load, and its signal intensity on T2-weighted imaging—the last of these being precisely what the Pfirrmann scale measures [36].

Disc height loss initiates a compensatory cascade through the rest of the motion segment. The annulus fibrosus weakens and may permit bulging or frank herniation of nuclear material; the ligamentum flavum buckles and hypertrophies; the posterior facet joints, now bearing altered load, develop osteoarthritic change and osteophytosis [32, 37]. Changes in the adjacent vertebral endplates and marrow—the Modic phenotypes [38]—frequently accompany this process and have themselves been the target of automated detection [39, 40]. The clinical significance of the cascade is that these individually modest changes act cumulatively to encroach on the spaces occupied by neural tissue.

2.2.3 The Three Anatomical Forms of Stenosis

Encroachment is conventionally partitioned by anatomical compartment, and this partition is directly mirrored in the label structure of the dataset used in this thesis [10]:

1. **Spinal canal stenosis** is narrowing of the central canal, which contains the thecal sac and the cauda equina. It is best appreciated on axial T2-weighted images, where the relationship between cerebrospinal fluid and nerve rootlets is visible [4, 5]. Severe central stenosis characteristically produces neurogenic claudication.

2. **Neural foraminal narrowing** is constriction of the lateral foramina through which the exiting nerve roots pass. It is assessed on parasagittal images, where the epidural fat surrounding the root provides the contrast that makes compression visible—T1 weighting being particularly suited to this [6]. It produces dermatomal radicular symptoms.

3. **Subarticular stenosis**, also termed lateral recess stenosis, affects the zone between the medial edge of the facet and the foramen, where it compresses the traversing root before it exits at the level below [3, 41].

The distinction matters computationally as well as clinically, because the three compartments are not equally visible in any single plane. Foraminal narrowing is poorly assessed from axial images and subarticular stenosis poorly from sagittal ones, a constraint that Section 2.8 shows to have concrete consequences for model design. It is also the compartment that correlates with which sequence a model must be given, and thus sits at the centre of the multi-sequence question this thesis examines.

2.2.4 Clinical Presentation and the Treatment Decision

Severity grading is not an academic exercise; it partitions patients into distinct management pathways. Conservative management—physiotherapy, analgesia, epidural steroid injection—is appropriate for mild and most moderate disease, whereas severe compression with corresponding symptoms may warrant decompressive surgery or fusion [4, 42]. The evidence that morphological grade carries genuine prognostic weight is strongest for the Schizas classification, where grade C and D patients were substantially more likely to fail conservative treatment [4], and for the MSU herniation grades, where size 2 and 3 lesions identify microdiscectomy candidates [42].

Two qualifications temper this. First, the correlation between imaging findings and symptoms is imperfect: anatomical stenosis is demonstrable in a substantial proportion of asymptomatic individuals [3, 34], so imaging severity constrains rather than determines management. Second, and more consequentially for automated systems, the clinically decisive boundary is not evenly distributed across the ordinal scale. The distinction between *normal* and *mild* rarely changes management, whereas the distinction between *moderate* and *severe* frequently does. A model that distributes its errors uniformly across grade boundaries is therefore not clinically equivalent to one of the same overall accuracy that concentrates its errors at the harmless end—a point developed in Section 2.9.6.

2.3 MRI Acquisition: Sequences, Planes, and Diagnostic Yield

2.3.1 Why MRI is the Reference Modality

MRI is the reference standard for non-invasive assessment of the lumbar spine. Computed tomography and plain radiography depict osseous anatomy well but resolve soft tissue poorly, and both expose the patient to ionising radiation. MRI resolves the intervertebral disc, thecal sac, individual nerve roots and ligamentous structures with a contrast unavailable to the alternatives [28, 35], and it is the disposition of exactly these soft-tissue structures that defines every grading system reviewed in Section 2.4. MRI has correspondingly displaced CT myelography for stenosis assessment [43].

2.3.2 Complementary Information Across Sequences and Planes

Lumbar protocols are inherently multiplanar and multisequence, because the spine is a three-dimensional structure whose pathologies are not all visible in one projection. Three acquisitions dominate clinical practice and constitute the input available in the dataset used here [10].

Sagittal T2-weighted imaging renders fluid hyperintense, so cerebrospinal fluid in the thecal sac appears bright against darker ligament and bone. It is the sequence on which disc hydration—and therefore Pfirrmann grade—is judged [36], and on which the craniocaudal extent of canal compromise is surveyed. Short tau inversion recovery (STIR) suppresses the signal from fat and thereby exposes marrow oedema and inflammatory change, making it the sequence of choice for active Modic type 1 change [39].

Sagittal T1-weighted imaging renders fat hyperintense. Because the exiting nerve root within the neural foramen is normally surrounded by epidural fat, T1 provides the contrast against which obliteration of that fat—the basis of the Lee foraminal grading system—becomes visible [6].

Axial T2-weighted imaging provides the transverse cross-section through the disc level. It is the only view in which the morphology of the dural sac, the arrangement of the cauda equina rootlets and the lateral recesses can be directly assessed, and it therefore underpins both the Schizas and Lee central canal systems [4, 5] and the assessment of subarticular stenosis [41].

2.3.3 Matching Pathology to Sequence

The correspondence between pathology and sequence is not a matter of convention but of physics, and the empirical literature bears this out with unusual clarity. The most direct evidence comes from external validation of sagittal-only systems. Nigru et al. evaluated SpineNetV2 across eleven radiological features on 1,747 lumbosacral discs and found agreement above $\kappa = 0.7$ for most targets but distinctly lower agreement for foraminal stenosis and disc herniation, attributing the shortfall specifically to the absence of transverse anatomical context in sagittal input [44]. Hallinan et al. approached the same constraint from the opposite direction, deliberately assigning axial T2 to central canal and lateral recess grading while reserving sagittal T1 for the neural foramina [11].

This has a direct bearing on the experiments in this thesis. If a single-sequence model is to be evaluated, the choice of which sequence to retain is not arbitrary: sagittal T2 is the most defensible candidate because it is acquired in essentially every back-pain protocol and because it carries the signal for canal stenosis and disc degeneration, but the literature predicts that it should be weakest precisely on the foraminal and subarticular targets. Whether a sufficiently expressive architecture can compensate for that predicted weakness is an open question, and one this thesis is designed to answer.

2.3.4 Sources of Acquisition Variability

Models trained on one institution's images frequently degrade on another's, and the proximate causes lie in acquisition. Scanner manufacturer and field strength (typically 1.5 versus 3.0 T), slice thickness and spacing, in-plane resolution, echo and repetition times, the presence or absence of fat suppression, and patient positioning all vary between sites [29, 45]. Axial acquisitions vary further in whether slices form a continuous stack or are angled to lie parallel to each disc level—a difference the LumbarDISC inclusion criteria explicitly accommodate [10], and one that any model consuming axial data must tolerate.

Sáenz Gamboa et al. treat this variability as the research problem in its own right rather than as a nuisance, evaluating segmentation across multiple centres and acquisition parameters and finding that ensembles of U-Net variants were required to maintain performance [45]. Batra et al. make the complementary design argument: rather than assuming uniform histograms and a fixed slice count per study, their pipeline is built to accept a variable number of axial and sagittal slices, on the grounds that this reflects the conditions under which clinical data actually arrives [9]. Variability is thus not merely a threat to validity but a specification constraint, and Section 2.11.3 returns to how effectively the field has met it.

2.4 Clinical Grading Systems as the Origin of Ground Truth

Every supervised model reviewed in this chapter learns to reproduce labels generated by a human applying one of the systems described below. The properties of those systems—their definitions, their granularity and above all their reliability—therefore bound what any model trained on them can achieve. This section treats them as the measurement instruments they are.

2.4.1 Disc Degeneration: Pfirrmann and its Modified Extension

The Pfirrmann classification, introduced in 2001, remains the most widely adopted scheme for intervertebral disc degeneration and is applied to sagittal T2-weighted images [36]. It assigns one of five ordinal grades on the basis of four morphological criteria: homogeneity of the nucleus pulposus, distinction between nucleus and annulus, T2 signal intensity as a proxy for hydration, and disc height. Grade I denotes a homogeneous, bright, normally proportioned disc; grade V a collapsed disc of uniformly low signal.

Its virtues are simplicity and universality; its limitations are well documented. It is a subjective visual judgement, it is insensitive to change over short intervals, and it captures only intrinsic disc state, saying nothing about Modic change, high-intensity zones or the downstream compression of neural structures [34, 35]. It also suffers a ceiling effect in elderly cohorts, where most discs present as grade IV or V and clinically meaningful variation is compressed into the top of the scale. Griffith et al. addressed this with an eight-level modified scale supported by a 24-image reference panel, validated on 260 discs in 52 subjects of mean age 73; intra-observer agreement was excellent (weighted κ 0.79–0.91) and inter-observer agreement substantial (0.65–0.67), with most disagreement confined to a single grade [46].

Esposito et al. place these two systems in a wider context that is worth recording: a scoping review of 569 studies identified *93* distinct grading systems for lumbar disc degeneration—63 subjective, 25 quantitative—of which the Pfirrmann method accounted for just over half of all reported use [34]. The apparent standardisation of the field is therefore partly illusory, and comparisons of automated grading accuracy across studies must confirm that the same instrument was being reproduced.

2.4.2 Central Canal Stenosis: Schizas and Lee

Two morphological systems have largely displaced simple dural sac cross-sectional area (DSCA) thresholds for central canal assessment, on the grounds that area measurements correlate poorly with symptoms and overlap substantially between symptomatic and asymptomatic individuals [4].

The **Schizas classification** grades the morphology of the dural sac on axial T2 images by the ratio of cerebrospinal fluid to nerve rootlets. Grades A and B retain visible CSF—in A the rootlets occupy part of the sac, in B they fill it but remain individually discernible, giving a characteristic grainy appearance. Grade A is itself divided into four sub-grades by the position and extent of the rootlets: A1, lying dorsally and occupying less than half the sac; A2, dorsal and in contact with the dura in a horseshoe configuration; A3, dorsal but occupying more than half the sac; and A4, lying centrally and occupying the majority of it [4]. Grades C and D show no discernible rootlets and no CSF signal, distinguished from one another by whether posterior epidural fat is preserved (C) or obliterated (D) [4]. This seven-level structure is worth noting against the three-class scheme used in this thesis: a benchmark that collapses severity to Normal/Mild, Moderate and Severe is discarding distinctions the clinical instrument makes, which places an upper bound on how finely any model trained on it can discriminate. Applied to 95 subjects across surgical, conservatively managed and low-back-pain groups, intra- and inter-observer agreement were substantial and moderate respectively ($\kappa = 0.65$ and 0.44). Critically, area measurement over-diagnosed stenosis in 35 patients and under-diagnosed it in 12 relative to the morphological grade, and grades C and D predicted failure of conservative management [4].

The **Lee central canal system** offers a simpler four-grade alternative based on the degree of separation of the cauda equina rootlets on axial T2: grade 0 no stenosis; grade 1 obliteration of the anterior CSF space with all rootlets still separated; grade 2 some rootlets aggregated; grade 3 none separated [5]. Assessed by four musculoskeletal radiologists across 81 patients, it achieved inter-reader ICC of 0.730–0.953 and intra-reader κ of 0.863–0.900, and grades tracked cross-sectional area and anteroposterior diameter at nearly all levels—notably better agreement than the Schizas system, at the cost of finer morphological detail.

2.4.3 Neural Foraminal Stenosis: Lee and Sartoretti

The **Lee foraminal system** grades parasagittal images by the pattern of perineural fat obliteration and the morphology of the nerve root: grade 0 normal; grade 1 fat obliteration in two opposing directions; grade 2 obliteration in all four directions without morphological change to the root; grade 3 nerve root collapse or distortion [6]. Evaluated on 576 foramina in 96 patients from L3–L4 to L5–S1 by two radiologists, inter- and intra-observer agreement were near-perfect (κ approximately 0.80–1.0). This is the direct clinical precursor of the left and right foraminal narrowing targets in the benchmark used in this thesis. The Sartoretti classification offers a six-point sagittal alternative (grades A–F) indexed to progressive contact between the exiting root and each foraminal boundary.

The near-perfect agreement Lee et al. report deserves scepticism when generalised. It was obtained by two experienced readers applying explicit criteria to a curated series; the multi-reader, multi-institution agreement figures in Section 2.4.6 are considerably lower, and it is the latter that describe the conditions under which large training corpora are actually annotated.

2.4.4 Subarticular and Lateral Recess Stenosis

Subarticular stenosis is graded on axial T2 by the proportion of the lateral recess compromised: mild at up to one third, moderate between one and two thirds, severe beyond two thirds, with nerve root involvement described as touching, displacing or compressing [3, 41]. Compared with the disc and central canal systems this is a notably less crisply specified instrument, resting on a visually estimated ratio rather than a discrete morphological pattern. That imprecision shows in the reliability data: subarticular stenosis is consistently the least reproducible of the targets [3], and it is also the target on which the severity distribution in LumbarDISC is least skewed (Section 2.10.4). Its difficulty is thus intrinsic, not merely a consequence of scarce severe examples.

Because the lateral recess is bounded posteriorly by the facet joint, facet degeneration contributes directly to subarticular narrowing, and it is graded by its own instrument—the four-point Fujiwara scale for facet joint osteoarthritis, assessed on T1 and T2 images by joint space narrowing, osteophyte formation, articular process hypertrophy and subchondral erosion. The Fujiwara scale is not among the RSNA targets, but it is instructive here because it represents the reliability floor of this family of instruments: Nikpasand et al. report average human agreement of only $\kappa = 0.13$ for Fujiwara grading, against 0.68 for Pfirrmann on the same cohort, and their CNN correspondingly reached $\kappa = 0.18$ against 0.68 [37]. Where the human instrument is that unreliable, no model trained on it can be expected to do better, and the authors read their own result as evidence that facet scoring needs better criteria rather than a better network.

2.4.5 Disc Herniation: Nomenclature and the MSU Classification

Herniation is described qualitatively by the standardised nomenclature of bulge, protrusion, extrusion and sequestration [47], and quantitatively by the Michigan State University (MSU) classification, which grades size against a transverse line joining the medial edges of the facet joints on the axial slice of maximal herniation [42]. Grade 1 extends less than half the distance to that line, grade 2 more than half without crossing it, grade 3 completely beyond it. The MSU system supplies the reference standard for several automated herniation studies [13], and its size 1 versus size 2/3 boundary maps onto the conservative versus surgical decision.

2.4.6 Inter- and Intra-Observer Reliability

The most consequential study for present purposes is that of Lurie et al., who assessed standardised MRI interpretation of lumbar stenosis using 58 randomly selected studies from the SPORT trial read by four independent readers, with 20 re-read after at least a month [3]. Inter-reader agreement was substantial for central stenosis ($\kappa = 0.73$) but only moderate for foraminal stenosis (0.58) and nerve root impingement (0.51), and poorest for subarticular stenosis (0.49), with marked variation between reader pairs. Quantitative measurement fared no better in absolute terms: mean absolute inter-reader difference in thecal sac area was 128 mm^2, or 13%. Intra-reader agreement exceeded inter-reader agreement for every feature.

Two structural findings recur across the wider reliability literature. First, agreement is systematically worse for the lateral compartments than for the central canal—the same ordering that later appears in automated model performance. Second, disagreement is overwhelmingly confined to adjacent grades: Griffith et al. report 84% of intra-observer and 91% of inter-observer disagreement spanning a single grade [46], and the same pattern recurs in automated evaluation [12, 48]. Earlier consensus efforts confirm that the underlying definitions were themselves contested: the Delphi survey of Mamisch et al. [49] and the consensus conference reported by Andreisek et al. [41] were both convened precisely because radiologic criteria for stenosis varied between institutions.

2.4.7 Label Noise as a Performance Ceiling

The reliability figures above are not a peripheral caveat; they are a hard constraint on the interpretation of every result in this field. If four expert readers agree on subarticular severity at $\kappa = 0.49$, then a model trained on one reader's labels and evaluated against another's cannot exceed that agreement, and a reported accuracy approaching unity on such a target should prompt scrutiny of the evaluation protocol rather than admiration. The LumbarDISC dataset description states the point explicitly, noting that published inter-rater agreement on these grades ranges from poor to substantial depending on anatomical location [10].

The field has responded to this in two ways. The first is to make the reference standard more robust by construction—majority voting across a panel with adjudication of disagreements, as in Tumko et al. (seven radiologists) [43] and Su et al. (two clinicians with expert adjudication) [50]. The second is to reframe the evaluation question: rather than asking whether a model matches a single reader, asking whether its disagreement with readers falls inside the band of disagreement between readers. Won et al. furnish the cleanest example, reporting that the difference between their two experts was statistically significant while the difference between each expert and the model trained on that expert's labels was not [51]. Comparatively few studies adopt either safeguard, and Section 2.11.7 returns to this as a systemic weakness.

2.4.8 The Ordinal Structure of the Label Space

Severity labels are ordinal, not categorical: *normal*, *mild*, *moderate* and *severe* stand in a fixed order, and the distance between them is neither uniform nor clinically symmetric. Treating them as unordered classes under categorical cross-entropy—as the majority of the reviewed work does—discards that structure, penalising a severe-graded-as-normal error exactly as heavily as a moderate-graded-as-severe one.

Two empirical observations follow. First, errors concentrate overwhelmingly at adjacent grade boundaries. McSweeney et al. report that SpineNet disagreed with human raters on 20.83% of discs but that only 0.85% of discs differed by more than one grade [12]; Udomluck et al. observe the same concentration [48]; Ishimoto et al. find exact-grade agreement of 65.7% rising to 94.1% when collapsed to a severe-versus-rest dichotomy [52]. Much of the apparent error in ordinal grading is therefore boundary error rather than gross misclassification.

Second, and less expectedly, explicitly ordinal objectives have not reliably improved on this. Niemeyer et al. tested soft-kappa loss, ordinal cross-entropy and regression losses against plain cross-entropy for Pfirrmann grading on 1,599 patients and 7,948 discs, together with class pooling and class-weighted losses, and found that none improved overall classification performance; the default cross-entropy classifier reached $\kappa = 0.92$ with predictions within one grade of ground truth in 99.2% of cases [23]. Patil et al. likewise found performance strongest at the extremes of the Pfirrmann scale and on the binary low-versus-high distinction, with intermediate grades remaining the difficulty [53]. The ordinal structure of the problem is thus real and measurable, but the question of how to exploit it is genuinely open—a point that informs the loss functions adopted in Chapter 3.

2.5 Evolution of Automated Lumbar MRI Analysis

2.5.1 Handcrafted Features and Classical Machine Learning

Before deep learning, automated analysis of lumbar MRI depended on features designed by hand. Investigators specified mathematical descriptors of local appearance and passed them to a conventional classifier. Ghosh et al. combined Histogram of Oriented Gradients (HOG) descriptors with a support vector machine and a set of geometric heuristics, reporting 99% disc localisation accuracy across 53 clinical cases and 318 discs, and—unusually for the period—emitting a tight bounding box for each disc rather than a single interior point, so that a downstream system could operate on the region directly [54]. Athertya and Kumar took a comparable approach to vertebral degeneration, extracting local binary patterns, Hu moments and gray-level co-occurrence matrix (GLCM) statistics to classify Modic changes, endplate defects and focal changes, reaching 85.91% accuracy for the extent of Modic change on 100 patients [40]. Related efforts applied hybrid classical models to disc abnormality detection [55, 56] and semi-automatic computer-aided classification of degenerative disease [57], while Huber et al. pursued texture analysis with decision trees as a quantitative alternative to visual stenosis grading [58]. He et al. applied a synchronised superpixel representation to define consistent regions of interest across levels before classifying neural foraminal stenosis, an early attack on one of the exact conditions graded in this thesis [59].

These systems established feasibility but generalised poorly. Handcrafted descriptors are sensitive to scanner hardware, field strength, slice thickness and patient positioning, and the anatomical variability of the lumbar spine defeats fixed geometric assumptions. One finding from this era nevertheless retains value: Athertya and Kumar's feature-sensitivity analysis showed that GLCM entropy alone sufficed to detect focal changes, while endplate-defect classification depended almost entirely on the first two Hu moments [40]. Such explicit attribution of predictive power to individual features is something the deep models that displaced these methods recovered only later, and imperfectly, through the interpretability techniques of Section 2.12.

2.5.2 The First End-to-End Deep Learning Systems

Convolutional neural networks removed the need for explicit feature design by learning hierarchical representations directly from pixels [60]. In lumbar imaging the decisive contribution was SpineNet [61]. Trained on 2,009 patients and 12,018 disc volumes from sagittal T2 images, it predicted six radiological scores simultaneously—Pfirrmann grade, disc narrowing, upper and lower endplate defects, and upper and lower marrow changes—from a shared convolutional trunk that branched into task-specific heads. Two design decisions proved durable. First, multi-task learning let one architecture serve several targets, reducing the free parameters relative to training separate networks and exploiting the correlation between co-occurring pathologies. Second, the system required only disc-level class labels: no voxel-wise segmentation and no slice-level localisation, with the evidence hotspots emerging implicitly from classification training. Reported performance was state-of-the-art and near-human across all six scores.

The same group demonstrated shortly afterwards that longitudinal clinical scans could serve as free supervision. A Siamese network was pre-trained with a contrastive loss on whether two scans belonged to the same patient—information available from DICOM metadata without knowing the patient's identity—together with an auxiliary vertebral-level prediction task, then fine-tuned for degeneration grading. On 1,016 subjects, 423 with follow-up imaging, the pre-trained network outperformed training from scratch and reached equivalent performance from far fewer annotated examples [20]. This anticipated by several years the self-supervised pretraining strategies now routine in medical imaging, and it is the direct ancestor of the contrastive approach discussed in Section 2.9.5.

SpineNet's principal limitation was structural rather than architectural: it consumed preprocessed sagittal disc volumes, and could therefore not assess pathologies that require the transverse plane. SpineNetV2 extended coverage to eleven radiological features and moved to a multi-stage design, using a U-Net to detect and label vertebral bodies and discs before a ResNet classifier graded each [62]. The reliance on sagittal input nonetheless persisted, and Section 2.11.3 shows this to be the consistent locus of its external-validation failures.

2.5.3 From Binary Detection to Multi-Level, Multi-Pathology Grading

The task definition itself has evolved, and the trajectory matters more than any individual model. Early deep learning work asked binary questions: is stenosis present or absent [63]; is a herniation visible in this image [64]. Such formulations are tractable but clinically thin, because management depends on severity and on which level is affected.

Three shifts followed. The first was to *multi-class severity*: Won et al. graded stenosis status on 542 L4–5 axial images against two independent experts [51]; Niemeyer et al. predicted the full five-point Pfirrmann scale across 7,948 discs [23]. The second was to *multi-pathology* output, in which one model reports several findings at once: Lehnen et al. evaluated disc herniation, bulging, canal stenosis, nerve-root compression and spondylolisthesis across 888 lumbar segments from a single CNN [65]; Su et al. graded herniation, central canal stenosis and nerve-root compression from a shared ResNet-50 trunk with three classification heads [50]. The third was to *per-level, per-side* prediction, in which the unit of analysis is not the study or the image but the individual disc level and, for the lateral compartments, the individual side. Hallinan et al. exemplify the mature form, grading central canal, lateral recess and neural foraminal stenosis at each level with the sequence appropriate to each [11]. Single-stage detectors have since been adapted to deliver the same output more cheaply: Wang et al. modified a YOLOv5 network to detect regions of interest and grade central, lateral recess and foraminal stenosis simultaneously [66], and a related design grades Pfirrmann degeneration, herniation and high-intensity zones from sagittal and axial input in one pass [67]. Minin et al. pursued the same objective for Pfirrmann grading with a YOLOv8x model released openly, motivated by the inaccessibility of proprietary graders [68]. The progression through the YOLO generations—v3 for herniation detection [64], v5 for stenosis and degeneration grading [66, 67, 69], v8 for combined segmentation and keypoint localisation [68, 70]—tracks the field's gradual shift from detecting a lesion to characterising it.

This third formulation is precisely that of the benchmark used in this thesis, which requires twenty-five graded outputs per study: five conditions across five disc levels [10, 19]. The progression is therefore not merely historical. It explains why performance figures from the binary era cannot be compared with contemporary results, and why a model that classifies a whole image is solving a different and substantially easier problem than one that must attribute severity to a specific anatomical location.

2.5.4 Chronological Synthesis

Four phases are discernible. Until roughly 2016 the field was dominated by handcrafted features and classical classifiers, with localisation as the central problem [33, 54]. From 2017 to 2020, CNNs displaced these methods and multi-task grading became feasible [7, 61, 71]. From 2021 to 2023 the emphasis moved to clinical credibility: external validation [12], interpretability [32, 39], reader-assistance evaluation [18] and multi-centre testing [50, 72]. From 2024 onward, the release of a large public benchmark [10] redirected effort toward anatomically localised, multi-view severity grading, alongside the arrival of transformer and vision-language architectures [9, 16, 30, 73].

The through-line is a steady increase in the specificity of what is being predicted, accompanied—until recently—by a decrease in the rigour with which it is validated. Both systematic reviews of the period make the same complaint: Compte et al. found that algorithms performed markedly worse in replication or external validation than in their development studies [1], and Wang et al. concluded that despite pooled sensitivity of 0.84 and specificity of 0.87 across twelve studies and 15,044 patients, no model was reliable enough for clinical practice [2]. Section 2.11 takes up this tension directly.

2.6 Anatomical Localization as a Design Principle

2.6.1 Disc and Vertebra Detection, Labelling, and Level Assignment

Before severity can be attributed to L4–L5, the system must know which structure is L4–L5. Level assignment is thus a precondition of per-level grading, and it has been treated as a problem in its own right since the earliest work [54, 74]. Pan et al. illustrate the mature solution: a Faster R-CNN locates vertebral bodies L1–S1 on sagittal images, the geometric relationship between sagittal and axial acquisitions identifies the corresponding disc in each axial slice, and the disc region of interest is then extracted—each of the three stages reported at 100% accuracy, with the residual difficulty falling entirely on the subsequent classification [8]. Detection has, in short, largely been solved; grading has not.

2.6.2 Semantic and Instance Segmentation of Lumbar Structures

Dense segmentation provides a stronger anatomical substrate than bounding boxes. The U-Net architecture [75] and its descendants dominate, as a dedicated systematic review of deep-learning-assisted disc segmentation confirms [76]. Ghosh and Chaudhary segmented vertebrae, disc, dural sac and background jointly using cascaded random forests over image and sparsely sampled context features, achieving Dice 0.87 and 0.84 for dural sac and disc across 212 clinical cases [33]. Han et al. introduced adversarial training for multi-structure segmentation [71], Sáenz-Gamboa et al. established CNN semantic segmentation of the structural elements surrounding the spinal cord [77], and Al-Kafri et al. applied SegNet to boundary delineation for stenosis assessment across 515 symptomatic patients, contributing two purpose-built metrics—confidence and consistency—for assessing the quality of the ground truth itself rather than only the model [78].

Subsequent work has largely consisted of architectural refinement. Wang et al. added multilevel feature-map attention and residual blocks to an Attention U-Net [79]; Ahmed et al. targeted class imbalance between anatomical structures with a custom combined loss and modified upsampling [80]; Silveira et al. added an Inception module for multi-scale extraction and a dual-output mechanism, reporting mIoU 0.8974 and F1 0.9444 on SPIDER and—notably—outperforming TransUNet [81]. Möller et al. extended coverage to the whole spine with SPINEPS, segmenting fourteen structures including the posterior elements, and established a useful general result: segmenting semantically first and instance-wise second outperformed training directly on instance segmentation, beating an nnU-Net baseline [82] on every structure and metric [83]. Chen et al. merged convolutional and attention pathways in parallel with a novel position embedding, outperforming fifteen competing models [84].

2.6.3 Public Segmentation Resources and Benchmarks

Progress in this subfield has been unusually well served by shared data. The SPIDER dataset provides 447 sagittal T1 and T2 series from 218 patients across four hospitals, with reference segmentations of vertebrae, discs and spinal canal plus per-level radiological gradings, together with reference performance for a baseline algorithm and nnU-Net and a continuous public challenge on a sequestered test split [22]. Several of the segmentation models above are trained and benchmarked on it [53, 80, 81], which makes their results mutually comparable in a way that is rare elsewhere in this literature.

2.6.4 Quantitative Morphometry

Segmentation enables measurement, and measurement offers an escape from ordinal subjectivity. The dural sac cross-sectional area is the canonical example. Ghobrial and Roth compared U-Net, Attention U-Net and MultiResUNet for automated DSCA quantification on 683 scans with 50 held back for external validation; MultiResUNet performed best, with Pearson correlation 0.9917 and mean absolute error 23.70 mm^2, holding up externally at 99.95% accuracy and F1 0.9393 [85]. Zheng et al. pursued the same objective for disc degeneration, segmenting degeneration-related regions with a Swin-transformer network and computing signal-intensity and geometric features, then proposing a quantitative criterion to replace subjective Pfirrmann grading on the basis of strong correlation with grade across a large population [31]. Related work measures the lordosis angle from segmentation masks [86].

A caution is warranted. Ghobrial and Roth's model relies on T1 weighting alone and a modest dataset, as the authors acknowledge [85], and Schizas' original finding—that area measurement over- and under-diagnosed stenosis relative to morphological grade in a substantial minority of patients [4]—applies equally whether the area is traced by hand or by network. Automating a measurement does not repair a weak correlation between that measurement and the clinical state.

2.6.5 Evidence that Localisation Precedes Classification

The strongest recurring methodological finding in this literature is that explicit anatomical localisation before classification improves grading. The argument was made structurally by DeepSPINE, which chained U-Net vertebral segmentation and disc-level designation to a multi-task grading network across 4,075 patients and more than 22,000 disc-level annotations [7], and it has been corroborated repeatedly since.

Four lines of evidence converge. Bharadwaj et al. segmented dural sac and disc with V-Net, localised facet and foramen by geometric rule, and only then classified, reaching Dice 0.93 and 0.94 with foramen and facet localisation IoU 0.72 and 0.83 [32]. Šušteršič et al. showed that segmenting, cropping and enhancing the region of interest before classification produced 0.87 and 0.91 accuracy on axial and sagittal views for a multi-class herniation problem [87]. Kowlagi et al. made the comparative case most directly, arguing that a well-tuned three-stage pipeline of segmentation, localisation and classification allows convolutional networks to outperform transformer approaches on Pfirrmann grading in a population cohort [88]. Acharya and Kansakar tested the principle in its strongest form, hypothesising that anatomical focus rather than architecture is the binding constraint, and grading severity from a single localised region of interest per disc [21].

The implication for the present work is direct. If localisation dominates architecture in determining performance, then a comparison between backbones is only meaningful when both receive identically localised input—otherwise the comparison measures preprocessing rather than representation. This is a methodological requirement that Section 2.7.6 shows much of the existing comparative literature fails to satisfy.

2.6.6 Reduced-Annotation and Unsupervised Alternatives

Dense annotation is expensive, and several strands of work reduce the requirement. Kuang et al. eliminated manual labels entirely for vertebral segmentation: sub-optimal masks from a rule-based method were combined through a voting mechanism to supervise CNN training, with a regional strategy restricting attention to a specific vertebral region, validated across 243 volunteers [89]. SpineNet's evidence hotspots represent the complementary weakly supervised route, obtaining localisation from classification labels alone [61], while the SPIDER annotation protocol adopted a human-in-the-loop compromise, training on a small subset to produce initial masks that were then corrected and fed back [22]. Self-supervised pretraining on unlabelled or metadata-labelled data [20] completes the picture. Together these establish that the annotation burden, long cited as the principal obstacle to clinical deployment, is substantially negotiable.

2.7 Backbone Architectures for Severity Classification

2.7.1 Convolutional Networks

Convolutional networks exploit two properties well matched to imaging: local connectivity, since a filter responds to a neighbourhood rather than the whole field; and translation equivariance, since the same filter is applied at every position. Stacking convolutions builds a hierarchy in which early layers respond to edges and texture and later layers to composite anatomical structure.

ResNet addressed the degradation encountered when networks are made deeper, introducing residual blocks whose skip connections allow a layer's input to bypass subsequent convolutions and be added to their output, providing an unimpeded path for gradients and permitting stable training at fifty layers and beyond [60]. In lumbar imaging ResNet is ubiquitous: as the classification head in multi-task grading [50], as the detection backbone in staged pipelines [13], as the encoder in segmentation networks [9], and as the classifier in SpineNetV2 [62].

EfficientNet attacked a different problem—how to allocate a fixed computational budget—by scaling depth, width and input resolution together under fixed coefficients derived from a neural architecture search, rather than scaling one dimension in isolation [90]. The resulting models reach comparable accuracy at substantially lower parameter and FLOP cost, which matters for the deployment scenarios that motivate this thesis. EfficientNet appears as the classifier in Kowlagi et al.'s Pfirrmann pipeline [88] and in the final stage of M-SCAN [9]. **ConvNeXt** represents the most recent convolutional line, modernising the CNN with design choices borrowed from transformers while retaining convolution [91]; Udomluck et al. include ConvNeXt-Tiny in their feature extractor comparison and find it stable across classifiers [48].

The architectural limitation of convolution is a bounded receptive field. Deep stacks aggregate local responses into progressively broader ones, but the network never explicitly models a relationship between two distant regions—for instance, between the L1–L2 and L5–S1 levels, whose degenerative states are biomechanically coupled.

2.7.2 Vision Transformers and the Attention Bottleneck

Transformers dispense with convolution and rely on self-attention, in which every element of the input is weighted against every other regardless of distance [92]. The Vision Transformer applies this by partitioning an image into fixed patches, flattening them and treating the result as a sequence [93]. Global context is available from the first layer, and there is no locality prior to overcome—but also no locality prior to exploit, which is why vision transformers are markedly more data-hungry than convolutional networks of comparable capacity.

The practical obstacle is cost. Global self-attention scales quadratically with the number of patches, hence quadratically with image area. Clinical MRI demands high spatial resolution to resolve structures a few millimetres across, so a naive vision transformer over full-resolution lumbar images is computationally prohibitive.

2.7.3 The Swin Transformer

The Swin Transformer resolves this bottleneck through two mechanisms [94]. *Windowed self-attention* restricts attention to non-overlapping local windows, making cost linear rather than quadratic in image size; to restore cross-window communication, the window partition is *shifted* by half a window between consecutive layers, so information propagates across boundaries over successive blocks. *Hierarchical feature maps* are produced by patch-merging layers that progressively reduce spatial resolution while increasing channel depth, yielding a pyramid analogous to a CNN's and making the backbone usable for dense prediction as well as classification.

For lumbar grading the appeal is specific. Severity depends simultaneously on fine local detail—the texture of a nucleus pulposus, the obliteration of perineural fat—and on regional context, since a rootlet configuration is interpretable only relative to the surrounding dural sac and the levels above and below. A hierarchical, windowed attention mechanism is a plausible fit to that structure, which is the substantive reason for selecting it as the transformer representative in this thesis. Swin-derived architectures have been applied to spinal segmentation, and Zheng et al. incorporate Swin components in their quantitative degeneration network [31].

2.7.4 Hybrid CNN–Transformer Architectures

A third position holds that the two families are complementary. SymTC merges convolutional and transformer layers in a *parallel dual-path* arrangement rather than stacking them sequentially, and integrates a position embedding into the self-attention module to improve spatial awareness; benchmarked against fifteen existing models on both a private dataset and a released synthetic one, it performed best for both vertebral bones and discs [84]. Zheng et al. similarly combine convolutional and Swin components [31], and Yi et al. pair a 3D ResNet18 backbone with transformer components across three hospitals [72]. Baur et al. explore a different pairing, segmenting discs with a U-Net, reconstructing each as a 3D point cloud and grading it with a graph neural network; performance was uneven across levels—F1 0.71 at L2/3 against 0.46 overall—which the authors present as a foundation requiring refinement rather than a competitive result [17]. What hybrids indicate is where the field expects the answer to lie: not in the wholesale replacement of convolution, but in supplying it with a mechanism for long-range interaction.

2.7.5 Relational and Graph-Based Modelling

Attention supplies long-range interaction over a grid. A graph supplies it over an *arbitrary* structure, which matters when the entities of interest are not arranged in a grid at all. A graph $G = (V, E)$ places features on nodes and propagates them along edges by message passing, so that each node's representation is refined by its neighbours. Graph convolutional networks apply a fixed neighbourhood weighting; graph attention networks learn it; graph transformers extend attention over the node set, and *relation-aware* variants condition the attention on the type of each edge, so that a neighbour-of-one-kind is treated differently from a neighbour-of-another.

The case for graphs in medical imaging is that anatomy is relational in ways a rectangular image does not express, and the literature increasingly argues that graph construction should be *anatomically justified* rather than chosen for convenience. The lumbar spine is an unusually clean instance: it has an intrinsic chain topology over five levels, several conditions co-occurring at each level, and bilateral symmetry between left and right—three relationships that are clinically motivated rather than inferred from data.

Uptake in this specific domain remains minimal. Baur et al. are the only reviewed study to apply a graph network to lumbar grading, and their graph is constructed *within* a single disc—a point cloud over its reconstructed 3D surface—rather than *across* levels or conditions [17]. The graph therefore models disc shape, not spinal structure, and the levels remain independent predictions. Nguyen et al. move closer to the relational idea by injecting disc-level priors so that the model learns level-specific degeneration patterns [16]. Most recently, Chai et al. propose an anatomy-guided pipeline combining landmark parsing, multi-sequence ROI patches, quantitative anatomical biomarkers, and an inter-level Transformer to capture contextual dependencies across lumbar levels [15]. While Chai et al. model contextual dependence among lumbar-level feature tokens, the individual level–condition targets remain independent output predictions. No reviewed study constructs a heterogeneous graph whose nodes represent the level–condition and laterality targets themselves with typed edges for adjacency, co-occurrence, and bilateral symmetry.

The distinction matters for what can be claimed. Sharing a *trunk* across targets, as the multi-task systems of Section 2.5.3 do [50, 61], lets the encoder learn features useful to several predictions. It does not let a confident prediction at one target inform an uncertain one at another, because the heads remain independent. Message passing over a graph of targets is the mechanism that would allow it, and Section 2.13.3 takes up the consequences of its absence.

2.7.6 Published CNN-versus-Transformer Comparisons and Their Limits

Several studies bear directly on the architectural question, and their results do not favour transformers as strongly as the general computer vision literature might suggest.

Kowlagi et al. address it explicitly. Observing that recent gains had been attributed to transformer architectures, they show that a well-tuned three-stage convolutional pipeline outperforms the state-of-the-art transformer approaches on Pfirrmann grading, evaluated on the Northern Finland Birth Cohort 1966—a population cohort rather than a clinical convenience sample—with an ablation study and subgroup breakdown, and public code [88]. Their conclusion is that pipeline design and tuning outweigh backbone choice.

Lee et al. provide the closest published analogue to the present study, developing a CNN-based and a transformer-based model for lumbar stenosis classification on 564 MRIs with a 100-study external test set, labelled by four radiologists with expert consensus as reference and read by eight participants across four experience levels. Both models exceeded 94% detection recall; on dichotomised classification the CNN, transformer and human participants achieved kappas of 0.99/0.99/0.97–0.98 for central canal, 0.98/0.94/0.81–0.94 for lateral recesses and 0.98/0.95/0.91–0.95 for neural foramina. The CNN was superior overall, the transformer similar to superior against clinicians [95].

Udomluck et al. compared VGG19, ConvNeXt-Tiny and DINOv2 features with classical classifiers under external validation, finding conventional VGG19 features strongest (accuracy 0.9556 with logistic regression) and DINOv2 features generalising worst, dropping to 0.6222 with LightGBM [48]. Patil et al. reached a still more pointed conclusion through ablation: radiomic features carried essentially the entire signal for Pfirrmann grading, and frozen ImageNet-pretrained ResNet50 features added no measurable discriminative value [53]. Silveira et al. likewise found their enhanced U-Net outperformed TransUNet on segmentation [81].

Three methodological limitations recur across this evidence. First, the compared architectures are rarely given identical preprocessing, localisation and training budgets, so differences may reflect tuning effort rather than representational capacity—exactly the confound Kowlagi et al. identify [88]. Second, the input configuration is usually held fixed, leaving unexamined the interaction between architecture and available context, even though a transformer's advantage should plausibly be largest where global context must substitute for missing views. Third, several comparisons rely on frozen pretrained features rather than fine-tuned backbones [48, 53], which disadvantages transformers disproportionately given their weaker inductive biases. These three limitations concern how architectures have been *compared*; they are distinct from the five research gaps identified in Section 2.13, which concern what the field has not *attempted*.

2.7.7 Transfer Learning, Pretraining, and Self-Supervision

Medical datasets are small by the standards these architectures were designed for, so initialisation matters. ImageNet pretraining is near-universal, though Patil et al.'s ablation is a caution that its benefit is not automatic [53]. Domain-specific pretraining is better supported: Jamaludin et al. obtained equivalent performance from far fewer labels through longitudinal self-supervision [20], and Acharya and Kansakar used contrastive pretraining on disc-centred crops to reduce sensitivity to irrelevant appearance variation [21, 96]. Sequential transfer between related targets—initialising a subarticular model from spinal canal weights—has also been reported to help on smaller datasets, a strategy of particular relevance to transformers given their lack of convolutional inductive bias.

2.8 Exploiting Spatial Context: Slices, Volumes, and Views

2.8.1 2D, 2.5D, and 3D Input Representations

An MRI series is a volume, but severity is assigned per level, so a representation must be chosen. *2D* models classify individual slices and are cheap but discard through-plane context; since a disc spans several slices and a radiologist integrates across them, a single slice may not contain the evidence. *3D* models consume the volume directly [72] at considerable memory cost—Acharya and Kansakar note that a standard high-resolution series contains millions of voxels, making full-volume processing impractical for many architectures [21].

2.5D representations occupy the middle ground and have become the default in this domain. A small stack of adjacent slices is treated as channels or as a short sequence, capturing local through-plane context at roughly 2D cost. SpineNet's input was a stack of nine slices per disc [61]; M-SCAN describes its approach explicitly as 2.5D, selecting the three axial slices closest to each level and modelling them with recurrent layers [9]; the leading challenge solutions combined 2D CNNs with sequence models or attention-based multiple-instance learning over the slice dimension.

2.8.2 Cross-Sequence and Cross-Plane Fusion

Where multiple sequences are available, they must be combined. The simplest approach concatenates them as channels, which presumes spatial registration that sagittal and axial acquisitions do not satisfy. More considered strategies process each sequence separately and fuse at the feature level.

CrossSpine is the most explicit treatment. Nguyen et al. identify two failings of baselines—single-stream designs cannot integrate complementary information across sequences, and uniform treatment of all discs ignores level-specific anatomy—and address the first with a cross-sequence attention mechanism that fuses features at multiple spatial scales, letting each sequence attend to all others, followed by a weighting mechanism that prioritises the most diagnostically informative sequence. They report Macro F1 improved by more than 125%, Mean AUPRC by 99% and Mean AUROC by 36% over their baseline [16]. M-SCAN applies cross-attention to align sagittal and axial feature maps before classification [9], and Park et al. pursue transformer-based multimodal fusion on the same benchmark [97]. Zeng et al. extend fusion beyond imaging entirely, combining ConvNeXt image features with GPT-4o-generated textual descriptions encoded by RoBERTa through a Mamba fusion stage [73].

2.8.3 Geometric Correspondence Between Planes

Fusing sagittal and axial data requires knowing which axial slice corresponds to which sagittal level—a geometric problem solved through DICOM spatial metadata rather than learned. Pan et al. derive the correspondence from the geometric relationship between the two acquisitions, reporting 100% accuracy in identifying the disc in each axial slice [8]. M-SCAN projects 2D sagittal stenosis coordinates into 3D using patient orientation data from the DICOM headers, then selects the three nearest axial slices for each of the five levels [9].

This step is a hidden cost of multi-sequence modelling, and one that bears directly on the trade-off examined here. It adds a preprocessing dependency, a failure mode when metadata is absent or inconsistent, and an engineering burden that a single-sequence pipeline simply does not incur.

2.8.4 Multi-Sequence versus Single-Sequence Input

The clinical standard is unambiguous that all three sequences contribute [10, 11], and the strongest reported results on the RSNA benchmark use all three [9]. The countervailing consideration is cost: multi-sequence processing demands greater memory and compute, cross-plane registration, and longer inference, and the resulting infrastructure requirements exceed what many hospital environments provide.

The evidence on whether reduced input suffices is genuinely mixed, which is why the question remains open. On one side, external validation of sagittal-only SpineNetV2 found agreement dropping specifically for foraminal stenosis and herniation—the pathologies requiring transverse context [44]—and Wu et al. found ordinal Pfirrmann grading degrading in upper lumbar discs while binary detection held up [98]. On the other, Acharya and Kansakar grade severity from *sagittal T2 alone*, reaching 78.1% balanced accuracy with a severe-to-normal misclassification rate of 2.13%, on the argument that anatomical focus rather than input breadth is the binding constraint [21]. Kowlagi et al. likewise operate on sagittal input and outperform transformer baselines [88].

Two observations sharpen the question. First, the comparison has not been made under controlled conditions: the sagittal-only and multi-sequence results above come from different architectures, datasets, localisation strategies and evaluation protocols, so the contribution of sequence configuration cannot be isolated. Second, the expected penalty is target-dependent—severe for foraminal and subarticular grading, mild for central canal—so a single aggregate accuracy figure would obscure precisely the structure of interest. Both observations inform the design adopted in Chapter 3.

2.8.5 Robustness to Missing or Incomplete Sequences

Clinical data is incomplete: sequences are omitted, truncated or corrupted by artifact. The LumbarDISC inclusion criteria required all three sequences and excluded studies with substantial artifact [10], so a model trained on it has not been exposed to the degraded conditions of routine practice. A model that requires three sequences fails outright when one is absent, whereas a single-sequence model is unaffected—an argument for reduced input configurations that is about robustness rather than efficiency, and one that the literature on missing-modality learning addresses only in passing for this application.

2.9 Class Imbalance and Ordinal Severity Modelling

2.9.1 Severity Distributions in Lumbar MRI Datasets

Class imbalance in this domain is not an artefact of sampling but a property of the population: in any unselected clinical cohort, most disc levels are normal or mildly degenerate, and severe compression is uncommon. The RSNA LumbarDISC distributions make the scale explicit [10]. Of 13,474 spinal canal stenosis grades, 85.4% are normal or mild, 8.8% moderate and 5.9% severe. Of 26,919 neural foraminal grades, roughly 78% are normal or mild, 18% moderate and 4.5% severe. Subarticular grading is the least skewed of the three, with roughly 69% normal or mild, 19% moderate and 11% severe across 26,285 grades.

The same pattern recurs elsewhere. Liawrungrueang et al. report a Pfirrmann distribution of 220 grade I, 530 grade II, 170 grade III, 160 grade IV and just 20 grade V across 1,000 images [69]; Niemeyer et al. describe an approximately Gaussian grade distribution with grades 1 and 5 under-represented [23]. Two consequences follow. First, a model trained under unweighted cross-entropy can achieve roughly 85% accuracy on spinal canal stenosis by predicting *normal/mild* unconditionally, so aggregate accuracy is close to uninformative on this task. Second, the ordering of imbalance across targets is the inverse of the ordering of difficulty: subarticular stenosis has the most balanced label distribution yet the poorest human reliability [3], confirming that its difficulty is intrinsic rather than a consequence of scarce positive examples.

2.9.2 Data-Level Strategies

The most common data-level intervention is weighted random sampling, in which the probability of drawing a minority-class example is inflated so that each minibatch approaches a uniform severity distribution and gradients are not dominated by the majority class. Aggressive augmentation of minority classes complements this, since with few severe examples a network may memorise them rather than learn the morphology of severe compression. Reported techniques fall into three groups: intensity transformations, of which contrast-limited adaptive histogram equalisation (CLAHE) is the most common in this domain because it enhances local soft-tissue contrast without amplifying global noise; geometric transformations such as random scaling, affine rotation, horizontal flipping and elastic deformation; and sample-mixing methods such as MixUp and CutMix, which synthesise new training examples by blending or splicing pairs of images and their labels. Mixing methods warrant caution on ordinal targets, since interpolating between a normal and a severe disc produces a label with no clinical referent.

Tsai et al. provide the clearest single demonstration of augmentation's value in the small-data regime, treating it as the subject of the study rather than an implementation detail: using YOLOv3 with a DarkNet-53 backbone to detect disc herniation, mean average precision reached 92.4% at 550 images with augmentation, and augmentation was decisive in preventing both under- and over-fitting [64]. Ahmed et al. attack a related form of imbalance—between anatomical structures rather than severity classes—through preprocessing that rectifies class inconsistencies before training [80].

2.9.3 Loss-Level Strategies

Class-weighted cross-entropy multiplies each class's contribution to the loss by a scalar, forcing the optimiser to penalise minority-class errors more heavily. The RSNA challenge encoded this directly into its evaluation metric, assigning weights of 1, 2 and 4 to normal/mild, moderate and severe respectively, so that aligning the training objective with the evaluation criterion is a matter of adopting the same coefficients.

Focal loss goes further, introducing a modulating factor that down-weights well-classified examples dynamically and concentrates gradient on hard, borderline cases [99]. This is well matched to the present problem, where the abundant normal cases are mostly easy and the difficulty concentrates at the mild/moderate and moderate/severe boundaries—precisely where human readers also disagree. Acharya and Kansakar combine weighted focal loss with contrastive pretraining [21], and Islam Sk et al. propose an adaptive PID-Tversky loss that borrows feedback control to adjust training penalties dynamically toward under-segmented minority instances [30].

2.9.4 Ordinal-Aware Objectives and Whether They Help

Given the ordinal label structure of Section 2.4.8, it is natural to expect ordinal-aware objectives to outperform categorical ones. The evidence does not support this expectation as strongly as the intuition suggests.

Niemeyer et al. conducted the most systematic test available, evaluating soft kappa loss, ordinal cross-entropy and regression losses against standard cross-entropy for Pfirrmann grading on 1,599 patients and 7,948 discs, with extensive hyperparameter optimisation, and additionally trying class pooling and class-weighted losses to counter imbalance. None of these improved classification performance overall. The default cross-entropy classifier reached Cohen $\kappa = 0.92$, average sensitivity 90.2% and precision 92.5%, with predictions within one grade of ground truth in 99.2% of cases and a mean absolute error of 0.08 grades [23]. Patil et al. list ordinal regression among the supplementary analyses they ran, and still report that intermediate-grade discrimination remains the outstanding difficulty [53]. More recently, Trînc et al. presented a multi-level evaluation framework for LumbarDISC using condition-specific Vision Transformers, arguing that conventional accuracy is insufficient and proposing distance-aware ordinal penalties alongside patient-burden and anatomical correctness metrics [24].

Two readings are possible. Either the ordinal structure is already implicitly captured—a network trained with cross-entropy on visually ordered classes will naturally confuse adjacent grades more often than distant ones—or the specific formulations tested were inadequate. The literature does not currently distinguish these, which is why this thesis treats the choice of objective as an empirical question rather than a settled one.

2.9.5 Contrastive and Representation-Learning Approaches

A third strategy attacks imbalance in the representation rather than in the sampling or the loss. Contrastive learning pulls examples of similar severity together in the latent space and pushes dissimilar ones apart before any classifier is fitted [96], so the representation is shaped by the structure of the problem rather than by class frequency.

Acharya and Kansakar apply this directly, combining contrastive pretraining with disc-level fine-tuning over a single anatomically localised region of interest per disc, on the reasoning that this forces attention onto intrinsic disc features and reduces sensitivity to irrelevant appearance variation. An auxiliary regression task handles disc localisation and a weighted focal loss addresses the residual imbalance. The result—78.1% balanced accuracy with the severe-to-normal misclassification rate reduced to 2.13% relative to supervised training from scratch—is notable less for the headline accuracy than for what it optimises [21]. The lineage runs back to Jamaludin et al.'s longitudinal self-supervision [20].

2.9.6 The Clinical Asymmetry of Error

The most important point in this section is that not all errors are equal, and that standard metrics conceal the difference. Grading a severe stenosis as normal risks a missed diagnosis and delayed decompression; grading a normal disc as moderate risks an unnecessary review. Overall accuracy, macro-F1 and even Cohen's κ treat these identically.

Three studies take the asymmetry seriously. Acharya and Kansakar report the severe-to-normal misclassification rate as a headline result alongside balanced accuracy, arguing explicitly that conventional metrics do not capture the differential clinical risk [21]. Wu et al. characterise SpineNetv2's error profile as specificity-oriented, with false negatives exceeding false positives, and conclude that the system is suitable as a confirmatory second reader but that reliance on its negative findings is not recommended—a conclusion about deployment that follows from the shape of the errors rather than their number [98]. The RSNA weighted metric encodes the same judgement into the benchmark itself.

Ishimoto et al. supply the complementary observation: collapsing four grades to a severe-versus-rest dichotomy raised agreement from 65.7% to 94.1% with $\kappa = 0.75$ [52]. If the clinically decisive question is whether a level is severe, then performance on that question is considerably better than the multi-class figures suggest, and evaluation should report both. This chapter therefore recommends, and Chapter 3 adopts, reporting per-class recall for the severe class alongside aggregate measures.

2.10 Datasets and Benchmarks

2.10.1 Private and Single-Institution Cohorts

Most work before 2024 used private single-institution data, with cohort sizes spanning three orders of magnitude—from 75 exams [39] and 146 consecutive patients [65] through 200 studies [32] and 542 images [51] to 4,075 patients [7] and 15,254 axial images [50]. Aggregate performance correlates only loosely with cohort size, because larger private datasets are frequently drawn from a single scanner and protocol.

Two consequences follow, and both are structural rather than incidental. Results are not comparable across studies, since each is measured against a different label distribution and reference standard; and models cannot be independently re-evaluated, since the data cannot be shared. Both systematic reviews of the period identify this as the field's central methodological weakness [1, 29].

2.10.2 Population and Epidemiological Cohorts

A smaller body of work uses population cohorts, which differ from clinical convenience samples in a way that matters for generalisation: participants are not selected by referral, so the severity distribution reflects the population rather than the imaged-because-symptomatic subset. The Northern Finland Birth Cohort 1966, comprising lumbar MRI from participants imaged at age 46–47, is the most used [12, 88]; the Wakayama Spine Study [52] and the Hong Kong Disc Degeneration Population-Based Cohort [100] serve similar roles. Kowlagi et al. are explicit that evaluating on a population cohort rather than a clinical sample is a deliberate methodological choice, reflecting the transferability of results to downstream research [88].

2.10.3 Public Lumbar MRI Datasets

Public data arrived late to this field relative to CT, where large annotated vertebral collections have existed for years. SPIDER is the principal segmentation resource: 447 sagittal T1 and T2 series from 218 patients across four hospitals, with reference segmentations of vertebrae, discs and spinal canal, per-level radiological gradings, baseline and nnU-Net reference performance, and a continuous challenge on a sequestered split [22]. Its influence is visible in the number of subsequent methods benchmarked on it [53, 80, 81, 85]. Smaller public collections support classification work [37, 101, 102].

2.10.4 The RSNA 2024 Challenge and LumbarDISC

The RSNA Lumbar Degenerative Imaging Spine Classification dataset is the largest and most diverse expertly annotated lumbar spondylosis collection assembled, and it is the dataset used in this thesis [10]. It comprises MRI studies from 2,697 patients and 8,593 image series drawn from eight institutions across six countries and five continents—a geographic spread chosen deliberately so that models trained on it face genuine acquisition heterogeneity rather than a single site's conventions.

Inclusion required a sagittal T2-like series (conventional spin echo T2, STIR or Dixon), a sagittal T1 series and an axial T2 series covering at least three lumbar disc levels, from outpatients aged 18 or over imaged for degenerative disease. Studies with lumbosacral hardware, active non-degenerative pathology, severe scoliosis or substantial artifact were excluded [10]. Annotation was performed by expert volunteer neuroradiologists and musculoskeletal radiologists recruited through the ASNR [19]. The associated challenge [19] made the dataset the field's common benchmark and shifted attention decisively from whole-image classification toward anatomically localised, per-level severity grading.

2.10.5 Label Structure, Distribution, and the Weighted Metric

The label structure is the defining feature of the benchmark and warrants precise statement. Each study receives twenty-five graded outputs: five conditions —spinal canal stenosis, left and right neural foraminal narrowing, left and right subarticular stenosis—at each of five disc levels from L1/L2 to L5/S1. Each is assigned one of three ordinal classes: Normal/Mild, Moderate or Severe. Expert coordinate annotations accompany the grades, so anatomical localisation can be learned in a supervised manner rather than inferred [10, 19].

Three properties follow. The task is *not* generic three-class image classification, and results from studies that treat it as such are not comparable. The severity distribution is severely imbalanced and differently so for each condition (Section 2.9.1). And the challenge metric is a sample-weighted log loss with weights of 1, 2 and 4 across the three classes, which encodes the clinical asymmetry of Section 2.9.6 directly into the evaluation. A model tuned for unweighted accuracy is therefore optimising a different objective from the one the benchmark measures.

2.10.6 Challenge Solutions and the Practices They Standardised

The solutions converged on a common template, and that convergence is itself a finding. Nearly all adopted multi-stage pipelines in which localisation precedes classification: a detection model—commonly a U-Net, YOLO variant or Faster R-CNN with a feature pyramid—identifies each disc level on the sagittal series, and a second-stage classifier grades tightly cropped regions, typically using 2.5D representations that combine a 2D backbone with a sequence model or attention over adjacent slices.

M-SCAN exemplifies the approach and reports AUROC 0.971 on 1,975 studies, using a U-Net with ResNet50 encoder to locate stenosis points on sagittal images, projecting these into 3D via DICOM orientation metadata to select corresponding axial slices, and applying cross-attention to align sagittal and axial features [9]. Liu et al. pair MobileNetV2 with U-Net for efficiency, reporting 94.93% accuracy under five-fold cross-validation [103]; Park et al. apply transformer-based multimodal fusion [97]; and further challenge-derived work continues to appear [104, 105].

Two cautions attach to these figures. Most are cross-validation results on the public training split rather than external validation, so they measure fit to LumbarDISC rather than transportability beyond it. And the top solutions rely on large ensembles, which inflate reported accuracy relative to what a single deployable model achieves—a gap directly relevant to the efficiency question this thesis examines.

2.11 Evaluation, Validation, and Generalization

2.11.1 Metrics and Their Appropriateness

Reported metrics in this field are heterogeneous and frequently ill-matched to the task. Overall accuracy remains the most common headline figure despite being close to meaningless under the severity distributions of Section 2.9.1. Area under the ROC curve is widely reported [2, 9] but is defined for binary problems and must be extended by one-versus-rest averaging for three-class grading, which conceals which class is failing.

More appropriate choices appear in the better studies. Cohen's and Gwet's κ measure agreement above chance and are directly comparable with the human reliability figures of Section 2.4.6, which is the correct reference point [12, 18, 50]. Balanced accuracy corrects for prevalence [12, 21]. Mean absolute error in grades exploits the ordinal structure and distinguishes a one-grade slip from a two-grade error [23, 98]. Ishimoto et al. report both exact-grade agreement and the collapsed severe-versus-rest figure, making the difference visible rather than choosing between them [52].

The recommendation that follows is to report a small panel rather than a single number: a prevalence-corrected aggregate, an agreement statistic comparable to inter-reader values, per-class recall for the severe class, and the distribution of error magnitudes across grade boundaries.

2.11.2 Internal Validation and Optimistic Reporting

A recurring pattern is that internally validated performance is high and externally validated performance is materially lower. Several reported accuracies in the reviewed literature exceed 95% [69, 102, 103, 106], which sits uneasily beside inter-reader agreement of $\kappa = 0.49$ for subarticular stenosis [3]. Where a model appears to exceed the reproducibility of the process that generated its labels, the explanation usually lies in the evaluation protocol: single-institution data, cross-validation rather than held-out external testing, labels from a single reader, or a task collapsed to a binary decision.

The point is not that these studies are wrong on their own terms, but that their numbers do not transfer. Compte et al. state the general finding directly: algorithms did not perform as well in replication or external validation as in their development studies [1].

2.11.3 External Validation and Domain Shift

External validation on geographically and temporally separate cohorts is the strongest available evidence of clinical viability, and a small number of studies supply it.

McSweeney et al. evaluated SpineNet on 1,331 participants from the Northern Finland Birth Cohort 1966, a dataset separate from its UK training data in both geography and time. Balanced accuracy was 78% for disc degeneration and 86% for Modic changes, with Pfirrmann reliability against expert radiologists of Lin concordance 0.86 and Cohen $\kappa = 0.68$, and Modic detection at $\kappa = 0.74$; both were essentially unchanged in a low-back-pain subset. The informative detail is the error structure: 20.83% of discs were graded differently from human raters, but only 0.85% differed by more than one grade [12]. Nigru et al. evaluated SpineNetV2 across eleven features on 1,747 discs from 353 patients, finding agreement above 0.7 for most targets but distinctly lower for foraminal stenosis and herniation, attributed to the limits of sagittal input [44]. Wu et al. tested SpineNetv2 on 491 patients and 2,455 discs, reporting overall accuracy of 83.5–97.5% and superiority over a junior orthopaedic surgeon on canal stenosis, spondylolisthesis and bilateral foraminal stenosis [98]. Grob et al. contribute a further independent validation on 882 scans [107].

Studies that build external testing into their development are still uncommon but instructive. Tumko et al. evaluated on 150 studies graded by a panel of seven radiologists with majority voting and external adjudication [43]; Su et al. used a 1,273-image external set from a different hospital and observed accuracy falling by roughly 7–10 points relative to internal testing [50]; Zhang et al. found classification accuracy dropping from 87.70% to 74.23% externally [13]; Zeng et al. report an unusually small external drop, from 93.7% to 92.2% [73].

2.11.4 Detection Versus Fine-Grained Grading

A consistent and under-appreciated finding is that these two capabilities generalise differently. Zhang et al. provide the clearest illustration: moving to an external institution, detection degraded only modestly—mean IoU 0.82 to 0.70, precision 98.4% to 96.3%, sensitivity 99.4% to 97.8%—while classification accuracy fell from 87.70% to 74.23% [13].

Wu et al. report the same asymmetry in different terms: binary detection across five pathologies held up well while ordinal Pfirrmann grading remained the limiting factor, degrading further in older patients and upper lumbar discs [98]. Yilihamu et al. show the effect within a single study as task granularity increases, using a dual-branch YOLOv8 pipeline that pairs YOLOv8-seg for disc segmentation with YOLOv8-pose for canal key-point localisation: segmentation IoU remained above 97% externally, while classification precision fell from 81.21% to 74.50% for an 18-category problem yet held at 92.51% to 90.07% for four-category severity [70].

The implication is that where a system fails is predictable. Finding the anatomy is robust; judging how bad it looks is not. Reporting a single aggregate figure for a pipeline that performs both obscures this, and evaluation should separate the two stages.

2.11.5 Benchmarking Against Clinicians

Comparison against human readers is the most clinically meaningful benchmark, and the better studies construct it carefully. Won et al. provide the cleanest design: two experts each labelled the data, a model was trained on each expert's labels, and the key result was that the difference between the two experts was statistically significant while the difference between each expert and their corresponding model was not [51]. Tumko et al. compared against a seven-radiologist panel average and found the model matched or exceeded it on all three stenosis types [43]. Lee et al. benchmarked against eight participants spanning four experience levels [95], and Wu et al. against orthopaedic surgeons of differing seniority [98].

Two cautions apply. Comparisons against junior readers or against surgeons rather than radiologists set a lower bar than subspecialty radiological practice [98, 108]. And readers in these studies typically work from cropped images without clinical history, which is not how they read in practice—so such comparisons measure a narrower task than clinical interpretation.

2.11.6 Longitudinal Stability and Reproducibility

One-shot accuracy does not establish that a model is stable over repeat imaging, which matters for monitoring progression. Murto et al. supply the only longitudinal evidence in this collection, comparing SpineNet's Pfirrmann grading against radiologists in 19 volunteers imaged at age 37 and again at 51. Radiologist-to-SpineNet concordance was 0.54 at both timepoints, clearly below the 0.83 and 0.78 observed between radiologists, and SpineNet systematically assigned grade 1 to several discs and grade 5 to more discs than the readers did [109]. The cohort is small, but a systematic offset that is stable across time is exactly the failure mode that one-shot evaluation cannot detect. Cheung et al. approach the temporal dimension differently, predicting five-year progression from baseline imaging across 1,343 participants [100].

2.11.7 Reporting Quality and Recurring Weaknesses

The systematic reviews converge on a consistent diagnosis. Wang et al. pooled twelve studies covering 15,044 patients, obtaining pooled sensitivity 0.84 and specificity 0.87 with SROC AUC 0.92, but with heterogeneity of I^2 near 99% and four of twelve studies at high risk of bias; they concluded that no current model is reliable enough for clinical practice, citing the need for accepted reference standards, larger datasets, external validation and fuller reporting [2]. Compte et al. found no significant difference between algorithm families and identified the development-to-validation performance drop as the field's central problem [1]. Mendes et al. found that most of 56 segmentation studies relied on private datasets and lacked external validation [29]. Liawrungrueang et al. report metrics spanning 71.5% to 99% across twenty studies—a range so wide as to indicate that the underlying tasks are not comparable [28].

Four weaknesses recur: single-institution data without external testing; ground truth from a single reader with no adjudication; task definitions that vary between studies while sharing a label vocabulary; and metric selection that flatters imbalanced problems. The present work adopts the public LumbarDISC benchmark, its panel-derived labels and its weighted metric specifically to avoid the first, second and fourth of these.

2.12 Interpretability and Clinical Integration

2.12.1 Saliency, Evidence Hotspots, and Grad-CAM

The earliest interpretability contribution in this domain is SpineNet's evidence hotspots, which map output-layer gradients back to the input to produce a heatmap per radiological score, obtained without any localisation supervision [61]. Grad-CAM [110] has since become the default. Tabarestani et al. use it to confirm that their cascade attends to clinically relevant regions [106], and Silveira et al. report that attention concentrated on vertebral bodies and nucleus pulposus in spatial correspondence with manual segmentations [81].

The value of these methods should not be overstated. A saliency map shows where a network responded, not why, and confirming that attention falls on plausible anatomy is a weak check—it excludes gross reliance on background artefact but does not establish that the model has learned the morphological criteria the grading system specifies.

2.12.2 Inherently Interpretable Models

A stronger position is to build models whose reasoning is legible by construction. Bharadwaj et al. provide the most striking result in this literature: alongside a Big Transfer classifier they trained a decision-tree classifier operating on segmentation-derived features for central canal stenosis, and the interpretable model *outperformed* the black box, reaching $\kappa = 0.80$ against 0.54 and approaching inter-reader agreement of 0.74–0.86 [32]. Quantitative morphometry belongs to the same family: a measured dural sac area [85] or a quantitative degeneration criterion [31] is inspectable in a way a logit is not. Patil et al. make the related point that radiomic features carried the entire discriminative signal for Pfirrmann grading while deep features added nothing measurable, and that radiomics additionally yields interpretable shape and intensity descriptors usable as imaging biomarkers [53].

Together these findings undercut the assumption that interpretability must be purchased at the cost of accuracy—at least for targets whose clinical definition is already explicitly morphological.

2.12.3 Vision–Language Models and Report Generation

The most recent direction outputs prose rather than labels. Islam Sk et al. present an explainable vision–language framework that classifies severity, segments the anatomy and generates radiologist-style reports, built on biomedical foundation models, with a spatial patch cross-attention module that preserves the anatomical hierarchies global pooling discards and an adaptive PID-Tversky loss for class imbalance; they report 90.69% classification accuracy, macro-averaged Dice 0.9512 and CIDEr 92.80% [30]. Zeng et al. take a narrower approach, fusing ConvNeXt image features with GPT-4o-generated descriptions for foraminal stenosis classification across three centres and seven scanners [73]. These systems are early and their language metrics measure fluency rather than clinical correctness, but they indicate where the field is heading.

2.12.4 Reader-Assistance Studies and Workflow Impact

The most consequential evidence for clinical value comes not from model metrics but from measuring what changes when readers use the model. Lim et al. conducted the definitive study of this kind: eight radiologists with 2–13 years of experience read the same studies with and without deep-learning assistance, separated by a one-month washout, against a 32-year-veteran musculoskeletal radiologist as reference. Interpretation time per study fell from 124–274 seconds to 47–71 seconds, and interobserver agreement was superior or equivalent for every stenosis grading. The largest gain fell to general and in-training radiologists on four-class foraminal stenosis, where κ rose from 0.39 to 0.71 and 0.70 [18]. Gao et al. observed a comparable effect for Modic assessment, where AI assistance raised agreement between a junior radiologist and a senior neuroradiologist from $\kappa = 0.52$ to 0.58 [39].

The pattern in both is that assistance helps least-experienced readers most and standardises rather than replaces judgement. This reframes the objective: the target is not a model that outperforms the best radiologist, but one that raises the floor.

2.12.5 Uncertainty and Calibration

Uncertainty estimation is conspicuously underdeveloped in this literature. Where errors concentrate at grade boundaries and where the reference standard is itself unreliable, a calibrated confidence would let borderline cases be referred rather than silently graded—a natural fit for triage. Almost none of the reviewed work reports calibration. Patil et al. are the exception, applying isotonic calibration on a validation split and reporting Brier scores alongside discrimination metrics [53]. Since modern networks are systematically overconfident [25], an uncalibrated severity probability should not be interpreted as a clinical confidence, and this remains an open opportunity rather than a solved problem.

2.13 Synthesis and Research Gap

2.13.1 What the Literature Has Established

Five findings are supported well enough to be treated as settled.

Anatomical localisation precedes classification. From DeepSPINE [7] through the interpretable pipelines of Bharadwaj et al. [32] to essentially every leading RSNA challenge solution [9], systems that localise the disc level before grading it outperform those that classify whole images. Detection is close to solved, with several studies reporting near-perfect level assignment [8, 54].

Automated grading is competitive with human reliability, because human reliability is low. Inter-reader agreement is substantial for central canal stenosis but only moderate for foraminal and poor for subarticular grading [3]. Against that benchmark, models sit inside the band of human disagreement [12, 43, 51]. This is a real result, but it is a statement about the modesty of the reference standard as much as the capability of the models.

Errors concentrate at adjacent grade boundaries. Disagreement above one grade is rare—0.85% of discs in the largest external validation [12]—and collapsing to a severe-versus-rest decision raises agreement substantially [52]. The difficulty lies in boundary placement, not gross misclassification.

Detection generalises across institutions; fine-grained grading does not. The asymmetry is consistent and large [13, 70, 98].

Class imbalance must be addressed explicitly. With 85.4% of spinal canal grades normal or mild [10], unweighted objectives yield degenerate solutions, and weighted sampling, weighted or focal losses and contrastive pretraining are all established mitigations [21, 99].

2.13.2 What Remains Contested

Three questions remain genuinely open, and the disagreement is substantive rather than terminological.

Does architectural sophistication help? The general computer vision literature would predict transformers to outperform convolutional networks given sufficient data, but the lumbar evidence does not bear this out. Kowlagi et al. report a tuned CNN pipeline beating transformer approaches [88]; Lee et al. find a CNN superior and a transformer merely comparable [95]; Silveira et al. beat TransUNet with an enhanced U-Net [81]; Udomluck et al. find conventional VGG19 features generalising better than DINOv2 [48]; and Patil et al. find deep features adding nothing over radiomics [53]. Set against this, hybrid architectures report gains [31, 84], and cross-sequence attention reports large ones [16].

Does the ordinal structure of the labels admit exploitation? Errors demonstrably cluster at adjacent grades, yet the most systematic test of ordinal-aware objectives found none improved on plain cross-entropy [23].

How much imaging is actually required? Clinical practice and the strongest benchmark results use all three sequences [9, 10], while sagittal-only systems report competitive performance [21, 88] and sagittal-only systems also fail in a specific, predicted way on the lateral compartments [44].

These three disputes are noted here because any system built on this literature inherits them, and because they bound what can be claimed from a comparison of architectures or input configurations. They are not, however, the gaps this thesis sets out to close. Each concerns a choice *within* the prevailing formulation—which backbone, which loss, which sequences—while leaving that formulation itself unexamined. The five gaps that follow question the formulation itself: that the lumbar spine is a set of independent predictions; that anatomical correspondence between sequences is a preprocessing convenience rather than a training signal; that the diagnostic contribution of each sequence is a fixed design decision rather than something a model can learn; that a model validated once at one institution has been shown to generalise; and that severity errors are symmetric.

2.13.3 Gap 1: The Spine Is Modelled as Independent Predictions

The dominant formulation treats each level–condition pair as an independent classification. A model predicting severity at L4–L5 does so without reference to L3–L4, without reference to the other conditions at its own level, and without reference to the contralateral side. This is a modelling convenience, not a clinical or biomechanical claim, and the literature contains three distinct lines of evidence that it discards real structure.

First, *degeneration propagates between adjacent levels*. The degenerative cascade described in Section 2.2.2 is explicitly sequential: disc height loss at one level alters load distribution at its neighbours, driving ligamentum flavum hypertrophy and facet change there in turn [32, 37]. Levels are not independent samples.

Second, *conditions co-occur at a level for shared mechanical reasons*. The same disc bulge that narrows the central canal also encroaches on the lateral recess and the foramen, which is precisely why the multi-pathology systems of Section 2.5.3 share a trunk across targets [50, 65]. Those systems share *features*, however, not *predictions*: the heads remain independent, so a confident finding at one target cannot inform another.

Third, *laterality is anatomically symmetric*, and the RSNA label structure encodes left and right as separate targets that are nonetheless produced by one underlying morphology [10].

Relational modelling of this structure is almost absent from the reviewed literature. Baur et al. reconstruct each disc as a 3D point cloud and grade it with a graph neural network—but the graph is *within* a disc, over its surface geometry, not *across* levels or conditions, and the reported performance was uneven (F1 0.71 at L2/3 against 0.46 overall) [17]. Nguyen et al. inject disc-level priors [16], while Chai et al. introduce an inter-level Transformer to capture contextual dependencies across lumbar level tokens [15]. However, in all these formulations, the level–condition targets remain separate output predictions rather than communicating nodes. No study in this review constructs a heterogeneous graph whose nodes are the specific level–condition and laterality targets themselves and whose typed edges encode adjacent-level continuity, same-level condition co-occurrence, and bilateral symmetry.

This matters because the structure is not merely decorative. If neighbouring evidence genuinely constrains a prediction, then relational modelling should help most exactly where single-target evidence is weakest—the ambiguous mild/moderate boundary of Section 2.4.8, and the lateral compartments where human agreement itself is poorest [3].

2.13.4 Gap 2: Cross-Sequence Correspondence Is Available but Unused as a Training Signal

Section 2.8.3 established that DICOM patient-space metadata makes the correspondence between sequences exactly computable: a disc level identified on a sagittal series determines which axial slices depict the same anatomy. That correspondence is currently used only for *preprocessing*—to select slices or align crops [8, 9]. It has not been used as a *supervisory signal*.

The omission is conspicuous because the field is demonstrably short of labels and because generic pretraining performs poorly here. Patil et al. found through ablation that frozen ImageNet-pretrained ResNet50 features added *no measurable discriminative value* over radiomic features for Pfirrmann grading [53], and Udomluck et al. found DINOv2 features generalised worst of the extractors they compared, falling to 0.6222 accuracy externally [48]. Natural-image pretraining transfers weakly to this domain, yet it remains the default initialisation.

Where domain-specific self-supervision has been attempted, it has not exploited anatomy across sequences. Jamaludin et al. demonstrated the principle most directly, using *longitudinal* scans—the same patient at different timepoints, identifiable from DICOM metadata without knowing patient identity—as free supervision, and reported equivalent performance from far fewer annotated examples [20]. Acharya and Kansakar apply contrastive pretraining to disc-centred crops, but within a single sagittal sequence [21]. In both cases the positive pair is defined by patient or by augmentation, never by the fact that two visually dissimilar images depict the same anatomical structure.

The unexploited signal is therefore this: sagittal T1, sagittal T2 and axial T2 at the same spinal level in the same patient are three views of one anatomy, related by a geometric transform the scanner has already recorded. Treating them as positive pairs defines a pretraining objective grounded in anatomy rather than in image similarity, and one that requires no additional annotation whatsoever. No study in this review does so.

2.13.5 Gap 3: Sequence Contribution Is Assumed, Not Learned

Section 2.3.3 established that the correspondence between pathology and sequence is a matter of physics rather than convention: foraminal narrowing is assessed against perineural fat on T1, canal morphology against cerebrospinal fluid on axial T2. The literature applies this knowledge as a *fixed design decision*. Hallinan et al. assign axial T2 to central canal and lateral recess grading and sagittal T1 to the foramina, hard-coded in advance [11]. Fusion architectures combine sequences through mechanisms whose weighting is either fixed or learned globally rather than per condition [9, 97].

Nguyen et al. come closest, adding a weighting mechanism that prioritises the most diagnostically informative sequence and demonstrating that the technique is model-agnostic [16]. Even there the weighting is not conditioned on *which target is being predicted*, so the model cannot learn that the evidence for left foraminal narrowing at L4–L5 lives in a different sequence from the evidence for canal stenosis at the same level.

The cost of getting this wrong is documented rather than hypothetical. Nigru et al. found SpineNetV2's agreement dropping specifically for foraminal stenosis and disc herniation, and attributed it to the absence of transverse context in sagittal-only input [44]; Wu et al. observed the same pattern by target [98]. A model that cannot weight sequences by condition inherits these failures by construction.

A second and compounding limitation is brittleness. Section 2.8.5 noted that LumbarDISC required all three sequences for inclusion [10], so models developed on it have never been exposed to the incomplete or degraded acquisitions of routine practice. The prevailing response is to train a separate model per input configuration, which multiplies training cost and leaves no single deployable system. No reviewed study learns a condition-specific weighting that also degrades gracefully when a sequence is absent or corrupted.

2.13.6 Gap 4: Cross-Institutional Transfer Is Asserted, Not Measured

Section 2.11.3 documented a consistent finding: models degrade when moved between institutions, and detection degrades far less than fine-grained grading [13, 70, 98]. What the literature does not provide is a *quantified adaptation curve*—how much local annotation is required to recover performance at a new site.

The external validations reviewed here are one-shot: a model is transported, evaluated, and the drop reported [12, 44, 98, 107]. None asks the operationally decisive question, which is not *how much performance is lost* but *how cheaply it can be regained.* For a hospital deciding whether to adopt such a system, the difference between needing ten locally annotated studies and needing five hundred is the difference between feasible and not.

Two further limitations compound this. Every external validation in this review tests a model built elsewhere on a Western or East Asian cohort; no published evaluation covers a Middle Eastern population or the scanner protocols in routine use there. And parameter-efficient adaptation methods, now standard elsewhere in deep learning, appear nowhere in the reviewed lumbar literature despite being the natural tool when local data is scarce.

2.13.7 Gap 5: Ordinal Structure and Clinical Cost Remain Partially Unexploited

Section 2.4.8 established two facts that sit awkwardly together. Errors concentrate overwhelmingly at adjacent grade boundaries—only 0.85% of discs differed by more than one grade in the largest external validation [12]—so the ordinal structure is real and measurable. Yet the most systematic test of ordinal-aware objectives found that none of soft-kappa loss, ordinal cross-entropy or regression losses improved on plain cross-entropy [23].

The natural reading is that conventional ordinal formulations tested in isolation were inadequate rather than that ordinal structure is unusable. Recent work on LumbarDISC, including Trînc et al. [24], has evaluated distance-aware ordinal metrics and patient-burden scores, confirming that standard ordinal classification alone is no longer an unexplored standalone novelty claim. This gap is therefore narrower than the four preceding it, and the thesis treats it accordingly: as *supporting methodology* rather than a primary contribution.

Two dimensions nonetheless remain unexploited. Ordinal objectives have scarcely been coupled with *relational target consistency* across neighbouring predictions—which only becomes possible once Gap 1 is addressed. And the *asymmetry of clinical cost* has rarely been modelled at the loss level: grading a severe stenosis as normal is not equivalent to the reverse, yet standard models optimise symmetric objectives. Acharya and Kansakar are a notable exception in tracking severe-to-normal misclassifications [21], while Wu et al. characterise their model's error profile as specificity-oriented [98]. The RSNA benchmark encodes asymmetry in its evaluation metric, but no reviewed system optimises for it while also maintaining calibrated uncertainty suitable for selective prediction.

2.13.8 Research Questions and Objectives

The five gaps are related rather than independent, and they divide cleanly by the role each plays. Gaps 2 and 3 concern *how evidence is represented*: what supervisory signal shapes the features, and how the sequences carrying that evidence are weighted. Gap 1 concerns *how predictions communicate* once the evidence is represented. Gap 5 concerns *how the ordered output and its uncertainty are expressed.* Gap 4 asks whether any of these choices survive the move from a curated benchmark to real clinical data at an unseen institution.

All five bear on one underlying question—whether encoding *what is already known about lumbar anatomy* into the model produces grading that is more accurate, more clinically aligned and more portable than treating the problem as a collection of independent image classifications. Each gap yields one research question:

RQ1 Does explicitly modelling the lumbar spine as a relational structure—with nodes representing level–condition targets and edges encoding adjacency, within-level co-occurrence and bilateral symmetry—improve severity grading over architectures that predict each target independently, and is the improvement concentrated at the ambiguous grade boundaries and lateral compartments where independent prediction is weakest?

RQ2 Can anatomically defined cross-sequence correspondence serve as a self-supervised training signal—treating the same spinal level imaged in different sequences as a positive pair, established through DICOM geometry rather than image similarity—and does it outperform generic pretraining under the label scarcity characteristic of this domain?

RQ3 Can a single model learn a condition-specific weighting over available MRI sequences that both reflects the sequence–pathology correspondence of Section 2.3.3 and degrades gracefully when a sequence is absent, rather than requiring a separate model per input configuration?

RQ4 What is the zero-shot performance loss when a model trained on multinational benchmark data is applied to an unseen single-institution Middle Eastern cohort, and how many locally annotated studies are required under parameter-efficient adaptation to recover it?

RQ5 Do cost-sensitive asymmetric ordinal objectives reduce clinically consequential errors—specifically severe cases graded as normal or mild—relative to the symmetric categorical objectives that dominate the literature, and does relational consistency across neighbouring predictions coupled with calibrated uncertainty provide a reliable signal for selective prediction and human review?

The corresponding objectives are to implement a staged system in which each component can be evaluated in isolation: a shared localisation and ROI stage so that every subsequent comparison receives identical anatomical input; a cross-sequence self-supervised pretraining phase; an adaptive fusion mechanism trained under simulated sequence dropout; a relational graph over the level–condition targets; and ordinal, cost-aware output heads with calibrated uncertainty. Each is added as a measured increment rather than assumed, so that the contribution of every component is quantified rather than asserted.

2.13.9 Scope and Delimitations

Five boundaries are stated explicitly.

The work addresses *severity grading and identification*, taking localisation as a shared and largely solved precondition (Section 2.6.1) rather than as a contribution in itself.

Model development uses the *public RSNA benchmark*, and transfer is evaluated on a *single-institution Middle Eastern cohort*. Because that cohort's reports describe herniation morphology while the benchmark grades stenosis severity, the two label spaces overlap only partially; transfer evaluation is therefore scoped to spinal canal stenosis, where both provide comparable ground truth, with herniation morphology treated as a separate local task. Conclusions do not extend to populations excluded by the benchmark's own criteria—spinal hardware, non-degenerative pathology, severe scoliosis or substantial artifact.

The relational structure is defined *anatomically rather than learned*. Edges encode adjacency, co-occurrence and symmetry because those relationships are clinically motivated; discovering an optimal graph topology from data is a distinct problem and is not attempted.

Reference-standard reliability *bounds every result reported here* (Section 2.4.7). Where human inter-reader agreement is $\kappa \approx 0.49$ for a compartment, a model cannot be shown to exceed it, and figures approaching perfect agreement should be read as evidence of a protocol artefact rather than clinical readiness.

Finally, the work reports *model performance and computational cost, not clinical impact*. The reader-assistance literature of Section 2.12.4 [18, 39] demonstrates that workflow benefit requires a prospective reader study, which lies outside the present scope and is identified as the natural successor to this work.

2.14 Chapter Summary

This chapter reviewed the literature on automated assessment of lumbar degenerative disease from MRI, from the clinical grading systems that define the task through to the architectures that currently attempt it.

The clinical sections established that severity labels originate in morphological grading systems—Pfirrmann for disc degeneration [36], Schizas and Lee for central canal [4, 5], Lee for foraminal [6]—whose inter-reader reliability ranges from substantial to poor depending on the compartment [3]. This imposes a ceiling on achievable performance and makes agreement statistics, rather than accuracy, the appropriate measure. The labels are ordinal, errors concentrate at adjacent boundaries, and the clinical cost of an error depends on its direction.

The methodological sections traced the field from handcrafted features through SpineNet's multi-task convolutional grading [61] to contemporary multi-stage, multi-view pipelines [9]. The most robust design principle to emerge is that anatomical localisation should precede classification [7, 88]. The most robust empirical regularity is that detection transfers across institutions while fine-grained grading does not [13, 98].

What the field has not attempted is the common thread of the five gaps identified in Section 2.13. Almost without exception, these systems grade the lumbar spine as a set of independent predictions: each level, each condition and each side classified in isolation, under a symmetric objective, and validated at a single institution. Yet the same literature establishes that degeneration propagates between adjacent levels, that conditions co-occur at a level for shared mechanical reasons, that the errors which matter clinically are asymmetric, and that performance does not survive the journey to another hospital.

The proposition this thesis tests is that encoding those known properties of lumbar anatomy—relational structure between targets, ordinal and cost-sensitive severity, anatomically defined correspondence across sequences, and explicit measurement of what transfer costs and what it takes to recover—produces grading that is more accurate where it is currently weakest, better aligned with clinical consequence, and portable to institutions the model has never seen. Chapter 3 sets out the experimental design through which that proposition is tested, component by component, so that each contribution is measured rather than assumed.

Chapter 3

Methodology

3.1 Introduction and Methodological Position

Chapter 2 established that automated lumbar MRI grading has moved beyond the stage at which novelty can be claimed from replacing one image backbone with another. Anatomical localisation, multi-sequence regions of interest, inter-level context and ordinal grading have all been demonstrated in recent work. The methodological question in this thesis is therefore not whether a sufficiently large network can classify lumbar MRI. It is whether the *structure of the prediction problem* can be represented more faithfully: which disease is being graded, at which level and side, which MRI sequence contains the relevant evidence, how neighbouring targets are related, and how the resulting model behaves when moved to a hospital whose scanners, protocols, population and reporting conventions were not represented during development.

The study is designed as a staged, controlled experimental programme around three methodological contributions and one translational validation programme. The first contribution is an anatomically aligned cross-sequence self-supervised representation in which positive relationships are defined by DICOM patient-space correspondence rather than by generic image similarity. The second is a disease-conditioned sequence router that learns how much each available MRI sequence contributes to each anatomical target and is trained explicitly to remain operational when one or more sequences are absent or technically present but degraded. Missingness and corruption are treated as distinct failure modes: an absent sequence is masked explicitly, whereas a corrupted sequence remains observable but is accompanied by quality evidence that the router may learn to distrust. The third is a heterogeneous disease–anatomy graph whose nodes correspond to level–condition–laterality targets and whose typed edges encode anatomical adjacency, within-level relationships and bilateral symmetry. The translational programme then separates development from deployment: the model is trained on a public multinational benchmark, frozen, evaluated zero-shot on an independent Rizgary Teaching Hospital cohort, and only afterwards adapted with controlled amounts of local labelled data.

Several components are deliberately treated as *enabling technology* rather than as doctoral contributions. Anatomical localisation, segmentation, 2.5D region construction, conventional convolutional or transformer encoders, class-imbalance handling, ordinal losses and calibration are all necessary to make the comparison scientifically fair, but none is advertised as new by itself. This distinction follows directly from the novelty boundary established in Chapter 2: the thesis tests a new formulation of the structured task rather than presenting a collection of familiar modules as if their combination alone constituted originality.

The chapter is written as a protocol-grade methodology. Wherever an implementation choice has not yet been fixed by evidence, it is specified as a controlled comparison rather than invented retrospectively. Wherever an institutional fact remains unresolved—for example the final number of Rizgary studies surviving data reconciliation or the formal ethics reference— it is marked explicitly. This is intentional. A doctoral methodology should describe what is known, what is pre-specified and what will be decided by a transparent rule; it should not convert planning assumptions into apparently observed facts.

3.1.1 Research Questions

The five research questions derived in Chapter 2 are operationalised as follows.

RQ1 — Relational disease modelling. Does representing lumbar disease targets as a typed heterogeneous graph improve grading over independent target heads, an ordered five-level transformer and a homogeneous graph, and are any gains concentrated where single-target evidence is weakest?

RQ2 — Anatomically aligned self-supervision. Does cross-sequence pretraining based on DICOM-defined anatomical correspondence improve grading, label efficiency and transfer compared with ImageNet initialisation, generic medical pretraining and ordinary augmentation-based contrastive learning?

RQ3 — Disease-conditioned modality routing. Can one model learn target- and patient-dependent sequence weights while remaining robust to both missing and quality-degraded MRI sequences, without requiring a separately trained network for every combination of sagittal T1, sagittal T2/STIR and axial T2?

RQ4 — Cross-institutional transfer. What is the zero-shot degradation when a benchmark-trained model is applied to a previously unseen Middle Eastern clinical cohort, how much of that degradation is attributable to localisation versus grading/reference-standard shift, and how does performance recover as the amount of local annotation available for adaptation increases?

RQ5 — Clinically asymmetric error and uncertainty. Do ordinal and cost-sensitive objectives reduce clinically consequential distant errors, particularly Severe→Normal/Mild, and can calibrated uncertainty support selective prediction without overstating clinical safety?

The order is not arbitrary. RQ2 and RQ3 determine how evidence is represented; RQ1 determines how target predictions communicate; RQ5 determines how the ordered output and its uncertainty are represented; and RQ4 asks whether those choices survive the transition from a benchmark to real local clinical data.

3.1.2 Design Principles

Seven principles govern all experiments.

First, anatomy precedes grading. Every classifier and ablation receives the same anatomically localised input so that a comparison of representations is not confounded by one model being given a better crop than another. This follows the consistent evidence that localisation is a precondition of reliable per-level grading [7–9].

Second, DICOM geometry is preserved. The primary scientific representation is reconstructed from DICOM patient-space coordinates. PNG conversion may be used for visual inspection but is not the basis of cross-sequence alignment, because it discards the spatial metadata required to determine which axial slices correspond to a sagittal disc level.

Third, patient identity defines independence. No images, slices, levels, sides or sequences from one patient may cross between training, validation and test partitions. The unit of statistical independence for uncertainty intervals and resampling is the patient, not the image or the disc level.

Fourth, the external cohort is not a tuning set. The primary model is developed without Rizgary images. Zero-shot performance is measured before any local tuning. Few-shot adaptation then uses a separately defined local adaptation pool while evaluation remains on a fixed local test set.

Fifth, every claimed contribution requires an ablation. A complete model score does not establish that anatomical self-supervision, routing or graph topology contributed anything. Each component is added and removed under a controlled experiment, and a negative result is reported as such.

Sixth, clinical interpretation is bounded by the reference standard. The model is trained to reproduce radiological grades whose inter-reader reliability is imperfect. Performance is therefore interpreted against human agreement and not as evidence that the network has discovered a more objective biological truth.

Seventh, each major dependency receives a diagnostic control. The primary end-to-end result is not replaced by manually corrected data, but a pre-specified subset is used to isolate failures that arise upstream of the claimed contribution. In particular, external grading is repeated with verified anatomical localisation, graph behaviour is stress-tested on isolated single-level disease, cross-sequence pretraining is tested for loss of sequence-specific information, and modality routing is evaluated under image corruption as well as complete absence. These controls are diagnostic rather than additional novelty claims.

3.2 Overall Research Design

3.2.1 Study Type

The project is a retrospective, multi-dataset medical-imaging methods study with three linked stages: public benchmark development, controlled methodological ablation, and independent cross-institutional transfer. It does not prospectively alter patient care and does not use model output to make treatment decisions. The Rizgary component is an external retrospective evaluation of imaging and radiology reports already acquired in routine clinical care.

The core development dataset is the RSNA Lumbar Degenerative Imaging Spine Classification benchmark (LumbarDISC), which contains multi-sequence MRI and expert per-level severity labels [10, 19]. SPIDER is used as an anatomical localisation/segmentation resource where its licence and task definition permit [22]. The local Rizgary cohort is reserved for external transfer and domain-adaptation analysis rather than being mixed into the benchmark training data.

This separation is central to the inference the study is intended to support. A random split of one pooled dataset can estimate internal generalisation under that dataset's acquisition distribution. It cannot answer whether a model trained on international benchmark data transports to a different hospital. The latter requires the hospital to remain unseen during development.

3.2.2 Unit of Analysis

The primary unit of prediction on LumbarDISC is a *level–condition–laterality target*. At five disc levels (L1–L2, L2–L3, L3–L4, L4–L5 and L5–S1), the benchmark defines five targets: central canal stenosis, left and right neural foraminal narrowing, and left and right subarticular stenosis. Full benchmark scope therefore contains $5 \times 5 = 25$ targets per examination. Each target receives the ordered label Normal/Mild, Moderate or Severe [10].

The unit of *statistical independence*, however, remains the patient. Twenty-five outputs from the same examination are correlated measurements, not twenty-five independent observations. This distinction affects data splitting, confidence intervals, significance tests and error analysis throughout the chapter.

For the local Rizgary cohort, the target space is narrower. The reports contain disc morphology and central-canal descriptions but do not reproduce the entire RSNA schema reliably. The primary cross-institutional comparison is therefore restricted to the five central-canal targets, L1–L2 through L5–S1, for which a defensible overlap can be created. Bulge, protrusion and extrusion are analysed as a separate local morphology task rather than being relabelled as if they were RSNA stenosis grades. Foraminal and subarticular transfer are excluded unless the local reference standard is expanded by new expert grading.

3.2.3 Experimental Staging

The methodology is intentionally cumulative:

1. establish reproducible DICOM reconstruction, patient-level splits, localisation and an independent ROI baseline;
2. establish anatomically correct cross-sequence correspondence;
3. test disease-conditioned sequence routing and missing-modality training;
4. test anatomically aligned cross-sequence self-supervision;
5. test homogeneous and heterogeneous graph reasoning against independent target prediction;
6. test ordinal/cost-sensitive output heads and probability calibration;
7. freeze the selected public-data model and perform zero-shot external evaluation at Rizgary;
8. adapt with prespecified numbers of local cases and estimate the annotation–performance recovery curve.

The staging protects the thesis from a common failure mode of complex medical AI systems: if the final model underperforms, a monolithic experiment cannot reveal whether the failure arose from localisation, fusion, representation, graph reasoning, loss design or domain shift. Here each stage creates a measurable intermediate object and a direct baseline.

3.2.4 Pre-Specified Hypotheses

The principal hypotheses are directional but not expressed as arbitrary target accuracies.

H1 Anatomical cross-sequence self-supervision will improve label-efficient grading and cross-institutional robustness relative to generic pretraining, particularly when only a fraction of benchmark labels is used.

H2 Disease-conditioned routing with explicit availability masking, modality dropout and quality-aware gating will degrade less under missing- sequence and corruption experiments than fixed concatenation or ordinary cross-attention trained only on clean complete studies.

H3 A typed heterogeneous graph will outperform independent target heads and a homogeneous graph if the anatomical edge types carry useful relational information; performance under a shuffled-edge control should deteriorate if the topology itself matters.

H4 Zero-shot performance on Rizgary will be lower than internal benchmark performance because of scanner, protocol, population, localisation and reference-standard shift; the magnitude and pattern of that drop are treated as findings rather than failures of the project. A verified-ROI control is expected to separate the portion attributable to localisation from the residual grading-domain shift.

H5 Local adaptation will improve performance as annotation increases, but no predetermined percentage of "recovery" is required. The shape of the recovery curve and the location in the network at which limited adaptation is most effective are both treated as results.

H6 Cost-sensitive/ordinal objectives may reduce distant Severe→Normal/Mild errors even if they do not improve aggregate macro-F1; plain cross-entropy remains the mandatory baseline because previous lumbar work has shown that ordinal objectives do not automatically help [23].

3.3 Data Sources

3.3.1 Public Development Cohort: LumbarDISC

The primary development data are drawn from the publicly released RSNA 2024 Lumbar Spine Degenerative Classification dataset. The published LumbarDISC resource comprises 2,697 patients and 8,593 MRI series collected across eight institutions in six countries [10]. The locally held labelled competition training subset contains approximately 1,975 studies and is used for core development. The distinction between the published cohort size and the available labelled subset is retained explicitly throughout the thesis rather than treating the two numbers as interchangeable.

Eligible benchmark examinations contain sagittal T2-like imaging, sagittal T1 and axial T2, with expert severity grading at the five lumbar levels [10, 19]. Coordinate/localiser annotations are also available for relevant disease locations. These properties make the dataset appropriate for all three methodological contributions: it is multi-sequence, level-resolved, multi-target, geographically heterogeneous and sufficiently large to separate representation learning from local external validation.

The benchmark serves three roles:

1. supervised development of per-target grading;

2. construction of anatomically corresponding cross-sequence training pairs;

3. internal/site-aware robustness analysis before the model is exposed to Rizgary.

All dataset versions, downloaded files, checksums where feasible and the final patient split lists are archived in the reproducibility record. After local quality-control exclusions the LumbarDISC cohort comprised 1,974 studies carrying 48,657 annotated targets, partitioned at patient level into 1,381 development, 296 validation and 297 held-out studies.

3.3.2 Public Anatomical Resource: SPIDER

SPIDER provides manually delineated lumbar anatomy on sagittal MRI from multiple hospitals, including vertebrae, intervertebral discs and spinal canal [22]. It is used only for anatomical localisation, segmentation benchmarking or pretraining where permitted by its licence. It is not used to manufacture disease labels that are absent from the dataset.

The practical purpose is to avoid making localisation the doctoral novelty. Rather than spend the main experimental budget designing another segmentation network, the thesis will use or reproduce a strong established anatomical model and quantify its localisation error. If a SPIDER-pretrained model is used to initialise the localiser, the external disease-grading experiments remain separate because the disease targets originate in LumbarDISC.

3.3.3 Independent Local Cohort: Rizgary Teaching Hospital

The local cohort consists of lumbar MRI examinations and corresponding English radiology reports supplied by Rizgary Teaching Hospital. Current project inventory identifies 299 narrative reports and 294 candidate cases with matching multi-sequence imaging, while a wider raw DICOM audit contains additional studies. These counts are not treated as final until a one-to-one cohort reconciliation is complete.

The raw imaging inspected during protocol development confirms the sequence families required by the thesis—sagittal T1, sagittal T2 and axial T2—and preserves standard

DICOM spatial metadata. A sample examination was acquired on a Siemens Avanto 1.5-T system; scanner/vendor distribution across the full local cohort will be reported after complete metadata extraction rather than inferred from one case.

The local cohort is valuable precisely because it is not curated to reproduce the public benchmark. Its scanner, acquisition protocol, prevalence, reporting language and label completeness differ from LumbarDISC. Those differences are the domain shift the transfer experiment is intended to measure, not nuisances to be eliminated by silently harmonising labels or discarding difficult cases.

3.3.4 Local Radiology Reports and Historical Spreadsheet

The narrative radiology reports are the primary textual source for the local reference standard. The existing spreadsheet is treated as a secondary historical transcription aid rather than as unquestioned ground truth. It contains patient-level variables and comma-separated level references, but it does not constitute a complete level-resolved RSNA-style matrix and has known transcription inconsistencies. The methodology therefore reconstructs the local reference matrix from source reports and uses the spreadsheet only for cross-checking.

The canonical local matrix will preserve, at minimum:

- pseudonymous case identifier;
- lumbar level;
- central-canal stenosis severity where explicitly stated;
- disc bulge, protrusion and extrusion as separate morphology fields;
- foraminal findings and side only where explicitly stated;
- nerve-root pressure/impingement where reported;
- an explicit missing/not-reported state distinct from a negative finding.

This last distinction is essential. Absence of a phrase in a narrative report is not automatically equivalent to "normal". A field may be unreported because the radiologist considered it clinically irrelevant, because reporting practice varied, or because the report was not structured around that compartment. "Not reported" is therefore encoded separately and excluded from target- specific external evaluation unless a clinician has explicitly adjudicated the absence as normal.

3.3.5 Cohort Reconciliation

Before any local model evaluation, a case-flow table will reconcile the raw archive, reports and analysable imaging. For each candidate case the audit will record:

- whether a report exists;
- whether sagittal T1 exists and is usable;
- whether sagittal T2/STIR exists and is usable;
- whether axial T2 exists and is usable;
- whether DICOM geometry is internally consistent;
- whether a central-canal reference label can be derived;
- whether laterality-dependent labels are available;
- whether the case is duplicated or represents repeat imaging;
- the reason for any exclusion.

The final thesis will report the flow from raw studies to the fixed external test cohort, rather than presenting the current 299/294/raw-study counts without explanation. Repeat examinations from one patient, if identified, remain in the same partition to preserve patient independence.

3.4 Ethics, Governance and De-Identification

The Rizgary component uses retrospective clinical data. No MRI is acquired for the purposes of this study, no patient undergoes an additional intervention, and model output is not returned to the clinical record. Nevertheless, the data are sensitive medical information and are subject to institutional governance.

The final thesis will state the formal approval mechanism and reference: [**TO CONFIRM: hospital research committee/IRB approval number, date and responsible institution**]. It will also state the legal/ethical basis for retrospective use: [**TO CONFIRM: consent waiver, broad consent or other applicable basis**]. These items must be completed before the local study is described as ethically approved.

The raw local DICOM archive contains direct identifiers and therefore cannot be given to students or uploaded to external compute in its original state. De-identification is performed on a controlled copy before analysis. At minimum, the process removes direct patient identifiers, replaces case identifiers with study pseudonyms, strips or audits private tags, and handles dates according to the approved policy while preserving only the temporal information required for scientific analysis. The de-identification procedure must also check for burned-in identifiers where applicable. The re-identification key, if one must exist for hospital governance, is stored separately from the research repository and is not available to the model-training pipeline.

External/cloud processing of local images is prohibited unless explicitly allowed by hospital policy: [**TO CONFIRM: whether external cloud/GPU processing is permitted and under what conditions**]. Public benchmark data are handled according to their published licence and competition terms.

Only aggregate results are reported. Illustrative local images may be included in the thesis only after de-identification review and only if institutional permission permits publication. The local cohort is not assumed to be publicly shareable; any future data release would require separate approval and a more stringent disclosure review.

3.5 Reference Standard and Target Encoding

3.5.1 Benchmark Target Representation

For LumbarDISC, the target set is represented explicitly rather than collapsed into a generic three-class image label. Let the five lumbar levels be

$$\mathcal{L} = \{L1-L2, L2-L3, L3-L4, L4-L5, L5-S1\},$$

and let the five condition/laterality targets be

$$\mathcal{C} = \{\mathrm{Canal}, \mathrm{LF}, \mathrm{RF}, \mathrm{LSA}, \mathrm{RSA}\},$$

where LF/RF denote left/right neural foraminal narrowing and LSA/RSA left/right subarticular stenosis. One examination therefore has the target set

$$\mathcal{T} = \mathcal{L} \times \mathcal{C}, \qquad |\mathcal{T}| = 25.$$

For target $t \in \mathcal{T}$ the observed grade is

$$y_t \in \{0, 1, 2\},$$

corresponding to Normal/Mild, Moderate and Severe. Encoding the targets in this form is not merely convenient notation. The pair (l, c) is retained throughout data loading, feature extraction, graph construction and evaluation, which prevents a severity label from losing the anatomical meaning that generated it.

Where the benchmark provides an absent or unusable annotation, a target mask

$m_t \in \{0, 1\}$ is carried alongside the grade. Masked targets do not contribute to the supervised loss. Missing labels are never imputed from neighbouring levels for the main experiments, because doing so would introduce exactly the relational assumption the thesis is intended to test.

3.5.2 Local Reference Matrix

The Rizgary reference standard is constructed at the same anatomical level but not forced into the same disease schema. A canonical row represents one case–level pair. The minimum schema is:

Field	**Type**	**Interpretation**
Case ID	categorical	pseudonymous patient/study identifier
Level	categorical	L1–L2 through L5–S1
Canal stenosis	ordinal/missing	explicit report-derived severity, if stated
Bulge	binary/missing	circumferential/generalised bulge stated at level
Protrusion	binary/missing	protrusion stated at level
Extrusion	binary/missing	extrusion stated at level
Foraminal finding	categorical/missing	narrowing/pressure effect if stated
Laterality	left/right/bilateral/NR	side only where stated
Nerve-root effect	categorical/missing	pressure, displacement or compression
Other morphology	text/coded	migration, dehydration, height loss and related findings

The local matrix keeps *not reported* (NR) separate from *explicitly absent*. This rule is especially important for foraminal and subarticular findings, because local reporting completeness differs from the structured RSNA annotation scheme. A target is eligible for external evaluation only when the reference standard provides a defensible class label.

3.5.3 Report Extraction and Human Verification

Automated report extraction, if available from the parallel clinical-NLP work, may accelerate matrix construction but does not define the reference standard. Every label that enters Selar's external grading analysis is manually checked against the original report text. This requirement applies to positive findings, explicitly negative/normal statements and the anatomical level to which the statement refers; an NLP output is never accepted as ground truth merely because its parser confidence is high. Label noise of this kind is well documented in medical image analysis, where labels derived from clinical text or from a single reader carry error rates high enough to change measured performance and to reward models that learn the annotation process rather than the disease [111]. The verification procedure below is designed so that residual noise is bounded and reported rather than absorbed silently into the reference standard.

Verification is organised in three levels, which deliberately distinguish *textual fidelity* from *clinical harmonisation*.

1. **Complete source-text audit.** Every externally evaluated case–level label is traced to the exact report phrase and its level and laterality. Multi-level expressions such as "L2–3 and L4–5, L5–S1 shows circumferential disc bulge" are expanded into separate level records without changing their meaning. Ambiguous phrases are coded as unresolved rather than forced into a class.

2. **Harmonised central-canal regrading.** The preferred confirmatory external reference standard is generated by two qualified readers who independently review the MRI for the fixed Rizgary test set and assign the same three-grade target used for the cross-institutional comparison (Normal/Mild, Moderate, Severe) at L1–L2 through L5–S1 using a common grading sheet. Readers are blinded to model predictions and, where feasible, to the original routine report during the first pass. Disagreements are adjudicated by a more senior reader or a pre-specified consensus procedure. Weighted κ [112], exact agreement and the distribution of one-grade versus two-grade disagreements are reported.

   Adjudication is by explicit consensus rather than by statistical label fusion. Estimator-based fusion such as STAPLE [113] is designed to recover both a reference standard and per-rater performance from a collection of raters, and is not adopted for a two-reader design: with two observations the performance parameters are weakly identified, and a disagreement is better resolved by a senior reader who can inspect the image than by an expectation-maximisation estimate over a pair of labels.

3. **Resource-limited reliability fallback.** If dual regrading of the entire fixed external test set is not feasible, a stratified subset is regraded before any model result is inspected. The size and sampling scheme are fixed in advance to cover lumbar levels and severity categories. The harmonised subset is then reported as the strongest external reference analysis, while report-derived labels on the remaining cases are explicitly labelled a secondary real-world analysis rather than treated as equivalent to the LumbarDISC reference process.

The historical spreadsheet is never used to override a source report or an adjudicated regrade. Where the report-derived label and harmonised reader label differ, both are retained with provenance so that reference-standard shift can be quantified rather than hidden. This separation is important because a cross-institutional performance drop may arise from image-domain shift, reference-standard shift, or both.

3.5.4 Schema Alignment Between LumbarDISC and Rizgary

Cross-institutional evaluation requires equality of *task*, not merely similar words. The mapping is therefore conservative.

LumbarDISC target	Rizgary evidence	Transfer status
Central canal stenosis	level-resolved mild/moderate/severe statements where present	Primary external target
Left/right foraminal narrowing	pressure/narrowing often incompletely lateralised	Exclude unless laterality and grade are adequate or freshly re-read
Left/right subarticular stenosis	not routinely represented in current reports	Exclude unless new expert annotation is created
Disc bulge/protrusion/extrusion	directly described locally but not an RSNA stenosis target	Separate local morphology task

No local disc-herniation morphology is converted into a stenosis grade by heuristic rule. Likewise, "pressure on exit foramina" is not automatically treated as a left/right foraminal severity label unless the report or expert review supplies the information required by that target. This prevents apparent external validation from being created by label re-interpretation.

3.6 Data Partitioning and Leakage Control

3.6.1 Patient-Level Independence

All splits are made at patient/study level before slice extraction or data augmentation. For a patient p, every sagittal series, every axial series, every disc level, every side and every augmented crop belongs to one and only one partition. A programmatic assertion is run before training to confirm that the intersection of patient identifiers across partitions is empty.

This discipline is not a formality. A systematic review of 294 papers across seventeen fields found leakage of exactly this kind—the same subject present on both sides of a split—to be among the most common causes of irreproducible machine-learning results in applied science [114]. The risk is acute in this study because a single patient contributes many slices, several sequences and up to twenty-five targets, so an image-level split would place near-identical data in training and test while appearing well formed.

The split record is version-controlled as a list of pseudonymous IDs. Data loaders consume those fixed lists rather than performing a new random split each time the training script is executed.

3.6.2 Public Development, Validation and Held-Out Evaluation

Where the available public dataset exposes institution identifiers, a site-aware evaluation is preferred because it tests a more meaningful shift than a random split. LumbarDISC does not expose institution identifiers, so the partition was drawn at patient level under a fixed seed, written once and reloaded by every run; it was frozen before any model comparison. It provides:

- a development/training partition;
- a validation partition for hyperparameter choice, checkpoint selection and calibration;
- a held-out public evaluation partition not used for model selection.

If site identifiers permit leave-one-site-out or institution-held-out analysis, that analysis is reported in addition to the ordinary benchmark split. It is not a replacement for Rizgary, because the local hospital introduces a different reporting/reference-standard domain as well as a different imaging domain.

3.6.3 Fixed Local Test Set and Adaptation Pool

The local transfer study is protected against adaptation leakage by separating Rizgary into two parts *before* the public model is evaluated:

1. a **fixed local test set**, which is never used for adaptation, calibration or threshold selection; and
2. a **local adaptation pool**, from which the $N = 10, 25, 50, 100$ few-shot subsets are sampled.

The exact number allocated to each part will be determined after the audited class distribution is known. The split will be stratified by central-canal severity and lumbar level where feasible without producing clinically implausible or extremely small strata.

The fixed local test set is used for the zero-shot baseline and for every subsequent adapted model, making the recovery curves directly comparable.

For each adaptation size N, several independent stratified subsets are sampled from the adaptation pool. Results are averaged across these draws with variance reported, because one set of ten cases can be unrepresentative by chance. The test set itself never changes across adaptation sizes.

3.6.4 Prevention of Information Leakage During Self-Supervision

Self-supervised learning does not remove the need for split discipline. A model pretrained on the held-out test patients, even without labels, has already seen their anatomy and acquisition characteristics. Therefore anatomical self-supervision uses only the development partition during internal experiments. The held-out public test set and fixed Rizgary test set remain unseen in both supervised and self-supervised training.

The same restriction applies to intensity normalisation parameters and learned harmonisation models: any transformation whose parameters are estimated from data is fitted on the training/development partition and then applied unchanged to validation/test data.

3.7 DICOM Reconstruction and Quality Control

3.7.1 Series Identification

Each study is first organised by DICOM `SeriesInstanceUID`. Candidate series are classified into sagittal T1, sagittal T2/STIR and axial T2 using a combination of DICOM metadata and image-orientation checks. Series descriptions are informative but not trusted alone because naming conventions differ across institutions. The classifier therefore considers fields such as `SeriesDescription`, `ProtocolName`, sequence parameters where available, and the orientation derived from `ImageOrientationPatient`.

Ambiguous or duplicated candidate series are flagged for manual review rather than selected by filename order. The selected series identifiers are stored in a manifest so that every experiment uses the same source series.

3.7.2 Patient-Space Geometry

For each DICOM slice, the image plane is reconstructed from `ImagePositionPatient`, `ImageOrientationPatient` and `PixelSpacing`. Let $\mathbf{s}$ be the patient-space location of the upper-left pixel, $\mathbf{r}$ and $\mathbf{c}$ the row and column direction cosines, and Δ_r, Δ_c the corresponding pixel spacings. The physical coordinate of pixel (i, j) is

$$\mathbf{x}(i, j) = \mathbf{s} + i\,\Delta_r\,\mathbf{r} + j\,\Delta_c\,\mathbf{c}. \tag{3.1}$$

The slice normal is

$$\mathbf{n} = \mathbf{r} \times \mathbf{c}. \tag{3.2}$$

Slices are ordered by projection onto $\mathbf{n}$ rather than by filename. Spacing between consecutive slice origins is inspected for discontinuities. This procedure preserves patient coordinates across sagittal and axial acquisitions, allowing a level localised in one plane to be projected into the other.

For a sagittal target point $\mathbf{q}$ and candidate axial slice k with origin $\mathbf{s}_k$ and normal $\mathbf{n}_k$, the perpendicular distance to that slice plane is

$$d_k = \left| \mathbf{n}_k^\top (\mathbf{q} - \mathbf{s}_k) \right|. \tag{3.3}$$

Distance alone is not treated as sufficient evidence of anatomical correspondence. Lumbar axial stacks are frequently prescribed obliquely so that the slice plane follows the local disc space. Let $\mathbf{n}_l^{\mathrm{disc}}$ denote an estimated normal to the target disc plane, obtained from the local anatomical landmarks or endplate/disc geometry available from the localisation stage. The acquisition–disc angular discrepancy for slice k is

$$\theta_{k,l} = \arccos\left(\left| \mathbf{n}_k^\top \mathbf{n}_l^{\mathrm{disc}} \right| \right). \tag{3.4}$$

A geometry relevance score combines distance and angular compatibility,

$$s_{k,l} = -\frac{d_k^2}{2\sigma_d^2} - \lambda_\theta \, \theta_{k,l}, \tag{3.5}$$

with corresponding normalised slice weights

$$w_{k,l} = \frac{\exp(s_{k,l})}{\sum_{j \in \mathcal{K}_l} \exp(s_{j,l})}. \tag{3.6}$$

The reference implementation uses the geometry score $s_{k,l}$ in a ranking/selection step (Top$-K$ or argmax) to select the physical 2.5D slice stack from the candidate set $\mathcal{K}_l$. The continuous weights $w_{k,l}$ modulate feature aggregation in downstream modules and are *never* multiplied directly against raw MRI pixel intensities, avoiding any distortion of T2 tissue contrast. This is preferred to forcing every clinical axial stack into a globally orthogonal grid, because resampling an intentionally disc-aligned acquisition can introduce through-plane interpolation and blur small neural structures. Canonical resampling may still be evaluated as a secondary robustness comparison, but it is not the default alignment strategy.

The per-level value θ_l (for example the median or maximum discrepancy within the selected stack) is retained as a quality-control variable. External error analysis tests whether unusually oblique or poorly aligned acquisitions are associated with localisation failure or grading error. The overall method remains an implementation of physical correspondence rather than a learned registration shortcut and follows the geometric principle used by prior lumbar pipelines [8, 9].

3.7.3 Orientation Standardisation

Volumes are reoriented to a common anatomical convention for tensor processing while retaining the transform to original patient coordinates. Left/right must never be inferred after arbitrary array transposition; all laterality-dependent targets are checked against the DICOM orientation transform. This is particularly important for foraminal and subarticular labels, where a left/right inversion would create a scientifically silent but catastrophic error.

3.7.4 Voxel Spacing and Resampling

The acquisition spacing varies by institution, scanner and plane. Inputs are therefore resampled only after anatomical coordinates have been reconstructed. The target spacing is chosen from the training-set distribution so that it preserves structures relevant to grading without creating unnecessary memory cost. Sagittal and axial streams may use different in-plane resolutions because their native geometry and diagnostic roles differ.

Interpolation uses a continuous method (e.g. linear or higher-order interpolation) for MR intensities and nearest-neighbour interpolation for discrete masks or labels. Original spacing is retained in metadata for any measurement reported in millimetres. Crops are defined by a physical field of view of 60 mm resampled to 128 pixels, giving an effective in-plane spacing of 0.469 mm per pixel for every study regardless of its acquired resolution.

3.7.5 Intensity Processing

MRI intensities are not quantitatively standardised across scanners. The preprocessing therefore removes gross scale differences without imposing a global histogram learned from test data. A default pipeline is:

1. suppress non-finite values and obvious padding/background values;
2. define a foreground mask from the body/spine region;
3. clip extreme foreground intensities using training-derived robust percentiles;
4. perform per-volume or per-ROI robust z-score normalisation;
5. optionally compare with bias-field correction or histogram standardisation as a robustness ablation.

Any learned harmonisation is fitted only on the development data. The external Rizgary zero-shot result is reported both with the baseline preprocessing and, if a separate harmonisation experiment is conducted, with that method clearly labelled as an intervention rather than folded into the zero-shot definition.

3.7.6 Quality-Control Flags

Automated QC records the following per series:

- slice count and ordering consistency;
- missing or duplicated instance positions;
- orientation consistency within series;
- axial-to-disc acquisition angulation discrepancy θ_l where target landmarks are available;
- in-plane spacing and slice spacing;
- field of view and lumbar coverage;
- intensity dynamic range;
- severe motion or truncation where detectable;
- presence of all sequences required by the particular experiment.

Cases failing geometry checks are not silently forced into the pipeline. They are reviewed, excluded with a recorded reason, or assigned to a robustness analysis if the failure itself represents a realistic missing-data condition. A final inclusion/exclusion table reports every such decision. The overlays and the reader checklist were generated over 60 studies at all five levels, covering 1,474 targets; the reader adjudication that converts the checklist into the completed table is outstanding at the time of writing and is reported as such rather than presented as complete.

3.8 Anatomical Localisation and ROI Construction

3.8.1 Role of Localisation

Localisation exists to make the subsequent comparison fair. The thesis does not claim that identifying L1–L2 through L5–S1 is itself novel; Chapter 2 showed that this problem is mature and that strong systems already attain highly reliable level assignment [8]. The requirement here is that the localiser be accurate enough that representation and graph experiments are not dominated by wrong crops.

An established model is therefore preferred. Candidate implementations include a SPIDER-pretrained nnU-Net-style segmentation network [22, 82], a heatmap-regression detector, or another state-of-the-art anatomical parser available when implementation begins. The choice is made on validation localisation error and robustness, not on whether it is architecturally fashionable.

3.8.2 Outputs

The localiser produces, for each study:

- centres of L1–L2 through L5–S1 in patient coordinates;
- vertebral/disc/canal masks or landmarks where the selected model provides them;
- a confidence or quality flag for each detected level;
- the transform between voxel coordinates and DICOM patient space.

A failed or low-confidence localisation is retained as a distinct failure mode. It is not corrected manually for the primary automated pipeline without being reported, because doing so would inflate downstream grading performance by removing a real deployment error.

3.8.3 Localisation Evaluation

Where point annotations are available, centre localisation error is measured in millimetres:

$$e_l = \|\hat{\mathbf{q}}_l - \mathbf{q}_l\|_2. \tag{3.7}$$

Mean/median error and percentile distributions are reported by level. Percentage-of-correct-keypoint (PCK) or an equivalent tolerance-based measure is also reported at pre-specified physical thresholds. Where full masks are available, Dice and surface-distance metrics are reported. These results are separated from disease grading, consistent with the literature showing that detection/localisation and fine-grained grading generalise differently.

3.8.4 Localisation-Controlled External Analysis

Because localisation is an upstream dependency, a large external performance drop cannot automatically be attributed to the grading representation. A pre-specified stratified subset of the fixed Rizgary test set is therefore used for a diagnostic *verified-ROI control*. For this subset, qualified review confirms the lumbar level and corrects the target centre or crop only when the automated localisation is wrong. The primary external result remains the fully automated pipeline; manually verified ROIs are never substituted silently into that headline result.

The same frozen grading model is then evaluated twice on the control subset:

$$M_{\text{autoROI}} \qquad \text{and} \qquad M_{\text{verifiedROI}}, \tag{3.8}$$

and the localisation-attributable difference is reported as

$$\Delta M_{\text{loc}} = M_{\text{verifiedROI}} - M_{\text{autoROI}}. \tag{3.9}$$

A small ΔM_{loc} alongside a large public-to-Rizgary grading drop supports a representational/reference-standard domain-shift explanation. A large ΔM_{loc} instead shows that the enabling localiser is a material source of transport failure. The subset size and sampling scheme are fixed before the external grading results are inspected and are chosen to span lumbar levels, severity classes and, where possible, acquisition-quality conditions.

3.8.5 Disease-Specific 2.5D Regions of Interest

One rectangular crop is not assumed to be equally appropriate for every target. The crop definition is conditioned on anatomical compartment.

For central canal stenosis, the model receives sagittal T2/STIR context around the disc level together with corresponding axial T2 slices. For neural foraminal narrowing, the sagittal T1 stream receives greater parasagittal coverage and the ROI is side-aware, with sagittal T2 retained as complementary context. For subarticular stenosis, axial T2 supplies the dominant local evidence with sagittal context. These choices follow the clinical sequence– pathology correspondence established in Chapter 2 rather than being learned from arbitrary crop templates.

Each 2.5D stream contains a small ordered stack around the anatomically closest slice,

$$\mathcal{I}_l = \{I_{k-r}, \ldots, I_k, \ldots, I_{k+r}\},$$

where k is the level-centred slice and r is a small radius determined on the development set. A five-slice stack ($r = 2$) is the initial reference configuration because it captures through-plane context at modest memory cost; the final radius is reported after sensitivity testing rather than treated as a novel contribution.

Crops are defined in physical dimensions where possible, then resampled to the network input resolution. This avoids a fixed 100-pixel crop representing different anatomical widths on scanners with different pixel spacing.

3.8.6 ROI Quality Control

A validation subset is visually inspected using overlays of the detected level, crop boundary and corresponding axial slices. The inspection records:

- correct lumbar level;
- inclusion of the relevant canal/foramen/recess;
- correct left/right orientation;
- adequate axial coverage;
- crop truncation;
- correspondence failure caused by unusual axial angulation.

The purpose is not to manually curate the full dataset but to detect systematic geometry errors before they propagate into model training.

The executed inspection, the review sheets it produced and its inclusion and exclusion outcome are reported in Section 4.3, since they are findings rather than protocol.

3.9 Baseline Representation and Sequence Encoders

3.9.1 Independent ROI Classifier

The mandatory baseline (E0) is an independent ROI classifier. Each target is graded from its anatomically localised input without communication with other targets. This represents the prevailing formulation criticised in Chapter 2 and provides the reference against which relational reasoning is measured.

The baseline uses a well-characterised image encoder such as ResNet [60], EfficientNet [90], Swin [94], or another contemporary backbone selected before final experiments. The purpose is not to prove that one family is universally superior. A primary backbone is selected for the ablation ladder so that representation, routing and graph changes are measured under one stable feature extractor. A second backbone may be used in a model-agnosticity check for the strongest contribution.

3.9.2 Sequence-Specific Feature Extraction

Each available MRI sequence is processed by a sequence-specific encoder $E_m(\cdot)$,

$$\mathbf{f}_{p,l,m} = E_m(\mathcal{I}_{p,l,m}), \tag{3.10}$$

for patient p, level l and modality/sequence m. The encoders may share architecture while keeping separate normalisation statistics and weights, or may partially share weights if an explicit ablation shows that doing so is beneficial. Separate encoders are the default because T1, sagittal T2/STIR and axial T2 have different contrast and orientation statistics. In the standard taxonomy of multimodal learning, the problem posed by the MRI sequence stack is one of *alignment* and *fusion* rather than translation [115]: no sequence is to be generated from another, and the task is to place complementary views of one anatomy in a common frame and combine them. Naming the problem this way matters because it determines which prior work is a legitimate baseline.

To keep comparisons fair, the dimensionality of the output representation is held constant across fusion methods. Fixed concatenation, simple averaging and ordinary cross-attention all consume the same per-sequence features.

3.9.3 Controlled Backbone Selection

Backbone choice was selected on the validation partition before the central novelty experiments and then held fixed; a ResNet-18 was selected and used for every rung of the ladder. The candidate comparison used identical crops, augmentations, optimiser budget and patient split. If one backbone is selected because it gives the most stable validation performance rather than the absolute highest point estimate, that decision is recorded.

This design prevents the thesis from becoming a repeated architecture search. The main questions concern anatomical self-supervision, adaptive modality allocation and target-level relational reasoning. A backbone is a controlled feature extractor in those experiments.

3.9.4 Competitive Methodological Baselines

The internal E0–E8 ladder explains which proposed component contributes to a result, but it is not sufficient by itself to establish competitiveness with current lumbar MRI methods. Two additional reference implementations are therefore included where reproducibility permits.

1. **Multi-view cross-attention baseline.** A strong sagittal–axial fusion model following the design principle exemplified by M-SCAN is reproduced under the same patient split, ROI definitions and training budget used by the proposed system [9].

2. **Inter-level context baseline.** A model in which level-specific features communicate through an ordered Transformer, reflecting the recent anatomy-guided multi-level context literature, is trained under the same experimental conditions. This baseline tests whether the heterogeneous disease graph adds value beyond generic inter-level attention.

Published point estimates from these methods are reported only as contextual references when the original split or target definition differs. Claims of superiority require matched reimplementation or an otherwise genuinely comparable evaluation protocol; a higher number from a different split is not treated as a head-to-head result.

3.10 Core Contribution I: Anatomically Aligned Cross-Sequence Self-Supervision

3.10.1 Rationale

Conventional self-supervised vision learning defines related views by applying augmentations to one image, whether through contrastive objectives that contrast a positive against many negatives [96, 116] or through non-contrastive self-distillation that dispenses with negatives entirely [117, 118]. That is useful but does not exploit a distinctive property of multi-sequence MRI: sagittal T1, sagittal T2/STIR and axial T2 can look very different while representing the same underlying patient and spinal level. DICOM geometry supplies the correspondence directly.

The closest existing analogue is cross-modal contrastive alignment, in which paired observations in two modalities are pulled together in a shared embedding space [119]. There the pairing is supplied by human-authored text; here it is supplied by patient-space geometry, which is both cheaper to obtain and, for the anatomical question at issue, more precise. The proposed objective is therefore a known learning mechanism applied to a pairing signal that the lumbar MRI literature has treated as a preprocessing convenience rather than as supervision.

The proposed objective therefore treats anatomical identity, rather than visual similarity, as the primary source of positive pairs. For patient p, level l, and two sequences m_a and m_b, embeddings $\mathbf{z}_{p,l,m_a}$ and $\mathbf{z}_{p,l,m_b}$ are encouraged to encode a shared anatomical representation even though acquisition plane and contrast may differ.

This idea is evaluated as a representation-learning hypothesis, not assumed to be beneficial. The literature already shows that generic and longitudinal self-supervision can reduce label dependence [20, 96], and that self-supervised pretraining improves both accuracy and label efficiency in medical imaging specifically, including a multi-instance variant that treats several images of the same pathology as positives [120]; the experiment here asks whether explicitly anatomical cross-sequence correspondence is a stronger signal for lumbar grading.

3.10.2 Projection Head

Each sequence feature is mapped through a projection head P_m:

$$\mathbf{z}_{p,l,m} = \frac{P_m(\mathbf{f}_{p,l,m})}{\|P_m(\mathbf{f}_{p,l,m})\|_2}. \tag{3.11}$$

The projection space is used only to express the anatomical self-supervised constraint; the encoder features before projection are retained for routing and downstream grading. The projection head is discarded or decoupled after the pretraining stage unless an explicitly labelled joint-training ablation retains it. This separation reduces the risk that an objective designed to align anatomy across sequences will force the final disease representation itself to be sequence-invariant.

3.10.3 Positive-Pair Construction

The strongest positive relation is:

$$(p, l, m_a) \sim (p, l, m_b), \qquad m_a \neq m_b,$$

meaning the same patient and same anatomical level in two different sequences. Correspondence is established through the DICOM reconstruction and level localiser, not through nearest image appearance. To prevent artifact-degraded sequences from corrupting the contrastive representation, positive pairs are dynamically weighted by sequence quality scores $q_{p,m}$ (Section 3.7.6), excluding pairs where $q_{p,m} < \tau_{\text{quality}}$.

An optional softer relation may be tested for adjacent levels from the same patient, because neighbouring levels share biomechanical context. It is not included in the primary objective by default. Treating L3–L4 and L4–L5 as equivalent positives would risk erasing genuine level-specific pathology, so any adjacent-level term is isolated as a separate ablation.

Different patients at the same level are not defined as positives in the main objective, because common anatomical level does not imply common disease state. They may be used in a secondary supervised-contrastive variant once labels are available, but this is conceptually distinct from the proposed anatomical self-supervision.

3.10.4 Contrastive Objective

For anchor embedding $\mathbf{z}_i$ and its cross-sequence positive $\mathbf{z}_{i^+}$, the reference InfoNCE-style loss is

$$\mathcal{L}_{\text{ACSSL}}(i) = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i^+})/\tau)}{\sum_{j \in \mathcal{B} \setminus \{i\}} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \tag{3.12}$$

where sim is cosine similarity, τ is a temperature parameter and $\mathcal{B}$ is the contrastive batch. When more than one anatomically valid positive exists, the numerator is extended to the set of positives rather than choosing one arbitrarily.

False negatives are a concern because two different patients may have very similar normal anatomy. The implementation will therefore compare ordinary InfoNCE with a variant that does not force all non-positive samples to be hard negatives, such as a multi-positive objective or a non-contrastive consistency objective of the kind that removes negative pairs altogether [117]. The latter is attractive here precisely because the false-negative problem is structural rather than incidental: normal lumbar anatomy is genuinely similar between patients, so a correct model should not be penalised for representing it similarly. The novelty claim is tied to the *anatomical pairing*, not to a particular brand of contrastive loss.

3.10.5 Label-Efficiency Experiment

The primary test of RQ2 uses fixed fractions of the labelled development set: 10%, 25%, 50% and 100%. For each fraction:

1. the encoder is initialised from the candidate pretraining strategy;
2. the same supervised grading head is fitted;
3. patient splits and target distribution are held fixed;
4. several random labelled subsets are evaluated where computationally feasible;
5. macro-F1, weighted κ, severe-class recall and calibration are compared.

The competing initialisations are:

- training from random initialisation;
- ImageNet initialisation;
- generic medical/volumetric pretraining where a suitable public model is available [120];
- ordinary augmentation-based self-supervision;
- the proposed anatomically aligned cross-sequence self-supervision.

A successful result is not defined as beating every baseline at every label fraction. The useful scientific question is where anatomical pairing helps, where it does not, and whether any benefit is largest under label scarcity or domain shift.

3.10.6 Sequence-Specific Information Preservation Test

Anatomical alignment can become harmful if the encoder learns invariance by erasing contrast-specific information. The use of a projection head reduces this risk but does not eliminate it, because gradients from $\mathcal{L}_{\text{ACSSL}}$ still update the sequence encoders. A dedicated preservation experiment therefore tests whether pretraining retains information that is diagnostically useful only in particular acquisitions.

For each major sequence, a single-sequence grading probe is trained from the same encoder with and without ACSSL pretraining while the probe architecture and labelled data are held fixed. Performance is compared separately on targets whose expected evidence is sequence-specific, including foraminal targets for sagittal T1 and canal/subarticular targets for axial T2. The experiment is not intended to prove a causal hierarchy among sequences. Its purpose is to detect a failure mode in which improved cross-sequence alignment is purchased by loss of sequence-unique diagnostic signal.

A representation is therefore considered beneficial only if any gain in label efficiency or transfer is not accompanied by a reproducible deterioration on these sequence-specific probes beyond the uncertainty of the comparison.

3.10.7 Transfer Test of the Representation

The public-data model used for zero-shot Rizgary evaluation is trained once with and once without anatomical self-supervision while the downstream architecture is held constant. If the anatomically pretrained model loses less performance externally despite comparable internal performance, that would support the claim that the representation is less tied to dataset-specific appearance. If both models degrade equally, the self-supervised contribution is limited to label efficiency rather than domain robustness.

3.11 Core Contribution II: Disease-Conditioned Adaptive Sequence Routing

3.11.1 Problem Formulation

A conventional multi-sequence model applies the same fusion rule to every target. That assumption is clinically implausible because the sequences are not equally informative for each anatomical compartment. Sagittal T1 is particularly useful for foraminal fat obliteration, sagittal T2/STIR for disc/canal context, and axial T2 for canal and lateral recess morphology. The router turns that known heterogeneity into a learnable, testable mechanism.

For patient p, target $t = (l, c)$ and available modalities $\mathcal{M}_p$, let $\mathbf{f}_{p,l,m}$ be the feature from modality m. The fused target feature is

$$\mathbf{u}_{p,t} = \sum_{m \in \mathcal{M}_p} g_{p,t,m}\, \mathbf{f}_{p,l,m}, \tag{3.13}$$

where the non-negative routing weights satisfy

$$\sum_{m \in \mathcal{M}_p} g_{p,t,m} = 1. \tag{3.14}$$

The gate is conditioned on target identity, normalised patient features and sequence-quality evidence. To avoid a high feature norm acting as an accidental proxy for importance, the image representation is first normalised,

$$\tilde{\mathbf{f}}_{p,l,m} = \mathrm{LayerNorm}(\mathbf{f}_{p,l,m}), \tag{3.15}$$

and the routing weights are

$$g_{p,t,m} = \mathrm{softmax}_{m \in \mathcal{M}_p} \left(G[\tilde{\mathbf{f}}_{p,l,m}, \mathbf{e}_c, \mathbf{e}_l, \mathbf{a}_p, \mathbf{q}_{p,m}] \right), \tag{3.16}$$

where $\mathbf{e}_c$ and $\mathbf{e}_l$ are learnable condition and level embeddings, $\mathbf{a}_p$ denotes optional acquisition/context metadata available without leaking the target label, and $\mathbf{q}_{p,m}$ is a compact quality representation derived from non-diagnostic acquisition/QC evidence such as motion score, truncation/coverage flags, spacing irregularity or other pre-specified image-quality indicators. The router remains patient-specific because image features are retained; the quality vector supplements rather than replaces them.

The gating mechanism itself has clear precedent. Squeeze-and-excitation recalibrates channel responses through a learned, content-dependent gate [121], and sparsely-gated mixture-of-experts layers route each input to a subset of expert subnetworks through a trainable gating network [122]. Neither is adopted unchanged. What is proposed here is that the gate be conditioned on *target identity*, so that one study is fused differently for a foraminal target than for a canal target at the same level. The novelty claim is the conditioning variable and its clinical justification, not the existence of a gate.

The initial implementation uses a lightweight MLP or target-conditioned attention gate. A more complex mixture-of-experts router is considered only if the simpler mechanism is insufficient. Should that route be taken, the load-balancing failure reported for sparsely-gated experts applies directly [122]: without an explicit balancing term the gate can collapse onto one input and the remaining experts stop receiving gradient. The degenerate case is diagnostically important here, because a router that always selects sagittal T2 would be functionally identical to a fixed fusion rule while presenting as a learned one. Gate entropy and per-target weight distributions are therefore monitored during training and reported, and a collapsed router is recorded as a negative result rather than reported as successful routing.

3.11.2 Explicit Availability Mask

Missing sequences are represented by an availability mask $a_{p,m} \in \{0, 1\}$. Logits for unavailable modalities are masked before the softmax:

$$g_{p,t,m} = 0 \quad \text{if} \quad a_{p,m} = 0. \tag{3.17}$$

The remaining weights are renormalised over the modalities that actually exist. A missing sequence is therefore not represented by an all-zero image whose meaning the network must infer. Availability is explicit in the model.

This follows the hetero-modal principle, in which each modality is embedded independently and fusion is computed over whatever modalities are present, so that a single trained model accepts any subset without retraining [123]. The masked-softmax renormalisation used here is a simpler realisation of that idea, chosen because the routing weights must remain interpretable as an allocation over available evidence, which moment-based fusion would obscure.

3.11.3 Modality Dropout

During training, complete studies are stochastically converted into incomplete ones by removing one or more sequence streams. Modality-wise dropout is an established route to test-time robustness against missing modalities that still permits cross-modal correlations to be learned during training [124]. It is adopted here because incomplete studies are the norm rather than the exception in the local cohort, so tolerance of absence is a deployment requirement and not merely a stress test. The dropout distribution includes:

- all three sequences;
- sagittal T1 + sagittal T2;
- sagittal T2 + axial T2;
- sagittal T1 + axial T2 where clinically meaningful;
- each single-sequence configuration as an edge case.

The sampling probabilities are tuned so that the model sees enough complete studies to learn complementary fusion while also receiving substantial exposure to incomplete inputs. Each available sequence was dropped independently with probability 0.2 during training on the rungs that use it (E3 onward), and never at evaluation. That probability was fixed for the whole campaign rather than tuned per rung, so modality dropout is held constant everywhere it appears and cannot differ between the arms of any comparison that includes it. No schedule was annealed and no per-sequence probability was used.

No external test case is artificially repaired by copying another sequence into a missing slot. The model's purpose is to remain functional under genuine absence, not to conceal it.

3.11.4 Corrupted-Modality Robustness

A modality can be present yet clinically unreliable. This is distinct from missing-modality learning: an availability mask can suppress a sequence that is absent, but it cannot by itself protect the router from a motion-corrupted or truncated series that remains technically available.

During supervised routing experiments, controlled corruption is therefore applied to one sequence stream at a time using transformations that approximate plausible acquisition failures without manufacturing new pathology. Candidate corruptions include motion-like displacement/ghosting approximations, increased noise, bias-field/intensity distortion, partial field-of-view truncation and controlled slice loss. The exact corruption family and severity levels are pre-specified from the QC characteristics observed in the development data and are reported explicitly.

For each corruption, the analysis measures both prediction degradation and the change in routing allocation. If sequence m is corrupted, a robust router is expected, where clinically possible, to reduce $g_{p,t,m}$ relative to the clean input while redistributing weight to the remaining evidence. This expectation is evaluated rather than hard-coded. A model that preserves accuracy only by assigning high weight to a visibly corrupted sequence is treated as unstable even if aggregate performance appears acceptable.

Three controls are compared: routing without quality features, routing with $\mathbf{q}_{p,m}$ but no corruption training, and quality-aware routing exposed to controlled corruption during supervised training. The corruption experiment is a robustness test of RQ3 and not a separate novelty claim.

3.11.5 Routing Consistency

An optional consistency term encourages the prediction from an incomplete modality set to remain close to the complete-study prediction when the removed modality is non-essential:

$$\mathcal{L}_{\text{miss}} = D(p_\theta(y \mid x_{\text{full}}), p_\theta(y \mid x_{\text{subset}}))\,, \tag{3.18}$$

where D is a divergence such as KL divergence or Jensen–Shannon divergence. This term is not applied blindly: for targets known to depend strongly on a specific sequence, forcing invariance to its removal could suppress legitimate diagnostic information. The term is therefore ablated by target and compared with modality-dropout training alone.

3.11.6 Routing Baselines

RQ3 is evaluated against four progressively stronger controls:

1. fixed feature concatenation;
2. simple equal-weight averaging;
3. ordinary cross-attention over available sequence features;
4. disease-conditioned routing without modality dropout;
5. disease-conditioned routing with modality dropout.

For each method, encoder capacity, ROI inputs and training budget are matched as closely as possible. Missing-sequence robustness is evaluated on the same patients under the same artificial absence pattern.

3.11.7 Interpreting Routing Weights

The gate weights are model allocations, not causal estimates of sequence importance. A high axial T2 weight does not prove that axial T2 caused a correct decision. Clinical plausibility is assessed in two complementary ways.

First, aggregate routing weights are summarised by target. The expected pattern is that foraminal targets allocate more weight to sagittal T1 and subarticular/ canal targets more to axial T2, but this expectation is not built into the gate as a hard prior.

Second, controlled input ablation tests whether removing the sequence with high routing weight causes a greater performance loss than removing a low-weight sequence. Agreement between learned allocation and intervention strengthens the interpretation; disagreement is reported rather than rationalised post hoc.

3.12 Core Contribution III: Heterogeneous Disease–Anatomy Graph

3.12.1 Graph Construction

The proposed graph represents *prediction targets*, not pixels and not a 3D disc surface. At full LumbarDISC scope, the node set is

$$V = \{v_{l,c} : l \in \mathcal{L},\, c \in \mathcal{C}\}, \qquad |V| = 25. \tag{3.19}$$

Each node begins with the routed visual representation for that target together with embeddings that identify level and condition:

$$\mathbf{h}_{l,c}^{(0)} = [\mathbf{u}_{l,c} \| \mathbf{e}_l \| \mathbf{e}_c]. \tag{3.20}$$

The graph topology is defined anatomically before training. It is not learned from correlations in the test data.

The formalism is that of graph neural networks operating by neighbourhood feature propagation [125]. The setting is strictly inductive: each patient presents a graph whose node features the model has never seen, so what is learned is an aggregation function rather than an embedding tied to one fixed graph [126]. Node set and edge topology are constant across patients; node features and evidence masks are not.

3.12.2 Node Evidence Availability Versus Label Availability

The graph distinguishes two forms of "missingness" that must not be conflated. A node may lack a ground-truth label in a particular cohort while still having valid image evidence, or it may lack usable visual evidence because a required sequence, field of view or localisation is missing. These cases are handled differently.

On LumbarDISC, a missing target label sets the supervised-loss mask $m_t = 0$ but does not automatically remove the node from the forward graph if the image inputs required to construct its representation are available. At inference, absence of a local Rizgary label likewise does not imply absence of the anatomical structure: the model is permitted to compute auxiliary foraminal or subarticular representations even when those outputs cannot be scored locally. Suppressing such nodes solely because their labels are unavailable would change the trained graph at external deployment and would confuse missing supervision with missing anatomy.

By contrast, when the visual evidence itself is invalid—for example the axial coverage does not include a target, localisation fails, or a required sequence is absent—an evidence mask $e_{p,t} \in \{0, 1\}$ is propagated to the graph. Edges carrying messages from nodes with $e_{p,t} = 0$ are masked before the attention softmax. This prevents genuinely unsupported "ghost" representations from contaminating neighbouring nodes while preserving information from unlabelled but observable anatomy.

3.12.3 Typed Edge Families

Three primary edge types are used.

Adjacent-level edges connect the same target type across neighbouring lumbar levels, for example

$$v_{L3-L4,\text{Canal}} \leftrightarrow v_{L4-L5,\text{Canal}}.$$

These encode the ordered chain of lumbar motion segments.

Same-level cross-condition edges connect anatomical compartments that coexist at one level, for example central canal to left/right foraminal and subarticular targets. The edge expresses that the targets share local morphology, not that one grade deterministically implies another.

Bilateral edges connect left and right counterparts of the same condition, for example

$$v_{L4-L5,\mathrm{LF}} \leftrightarrow v_{L4-L5,\mathrm{RF}}.$$

This represents bilateral anatomical symmetry without forcing equal grades.

Additional cross-type edges are added only if clinically justified in the pre-specified topology. A denser graph is not automatically a better graph; the study tests whether explicit relation types add value over homogeneous message passing. Distinguishing edge types by giving each relation its own transformation is the defining feature of relational graph convolution [127]. The claim under test is therefore not that typed relations are a new idea, but that *these particular anatomical relations*—the motion-segment chain, same-level compartment coexistence and bilateral symmetry—carry information that undifferentiated message passing discards.

3.12.4 Relation-Aware Message Passing

For layer q, node i receives messages from neighbours $j \in \mathcal{N}(i)$ through relation-specific projections [127]. Each relation $r \in \mathcal{R}$ carries its own transform $W_r^{(q)}$, and messages are averaged over the neighbourhood:

$$\mathbf{m}_i^{(q)} = \frac{1}{|\mathcal{N}(i)|} \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} W_r^{(q)} \mathbf{h}_j^{(q)}, \tag{3.21}$$

where the normaliser is the total in-degree of node i across all relations, so a node is not made louder simply by participating in more edge families. Neighbours without visual evidence contribute nothing, because their hidden states are held at zero by the masking described below.

What is deliberately absent, and why it is recorded. Aggregation here is uniform over neighbours. It is *not* attention-weighted: there is no W_Q, no W_K^r, no per-edge α_{ijr}, and no graph-transformer variant. An earlier version of this protocol specified a relation-aware attention formulation, and a graph-transformer alternative with positional encodings, on the reasoning that type-dependent attention suits heterogeneous graphs [128, 129]. **Neither was built, and no result in this thesis comes from either.** The specification above is what the executed model computes, stated so that the methodology and the implementation cannot be read as describing different architectures. Attention over this graph remains an untested option, and is listed as such in the future work of

Section 5.7 rather than presented as part of the method. To reduce the risk that a focal abnormality is propagated indiscriminately into a normal adjacent target, local evidence and relational evidence are combined through a learnable residual gate,

$$\boldsymbol{\gamma}_i^{(q)} = \sigma\Big(W_\gamma[\mathbf{h}_i^{(q)} \| \mathbf{m}_i^{(q)}] + b_\gamma\Big)\,, \tag{3.22}$$

$$\mathbf{h}_i^{(q+1)} = e_{p,i} \cdot \mathrm{ReLU}\Big(\mathrm{LN}\Big(W_{\mathrm{self}}^{(q)}\mathbf{h}_i^{(q)} + \boldsymbol{\gamma}_i^{(q)} \odot \mathbf{m}_i^{(q)}\Big)\Big)\,, \tag{3.23}$$

The self term is a learned transform $W_{\mathrm{self}}^{(q)}$ rather than the raw hidden state, so a node's own evidence is re-projected at each layer on the same footing as the messages it receives. Nodes lacking visual evidence ($e_{p,i} = 0$) bypass message passing and LayerNorm updates entirely, hard-masking their hidden states to zero ($\mathbf{h}_i^{(q+1)} = \mathbf{0}$) to prevent LayerNorm noise amplification and feature drift across graph layers. The gate allows a node with strong local visual evidence to retain its own state when neighbouring messages are inconsistent, while still permitting relational evidence to contribute when it is useful. The ungated residual formulation remains a mandatory ablation; otherwise any gain cannot be attributed specifically to the proposed control of neighbour influence. The relation-specific transforms W_r allow an adjacent-level neighbour to influence a node differently from its contralateral counterpart. Relation-specific parameters grow linearly with the number of edge families, a known source of overfitting in relational graph models. With three families over twenty-five nodes the risk is modest, but the basis-decomposition remedy [127] is available should the pre-specified topology later be widened.

No transformer or attention variant was implemented, so the alternatives canvassed at protocol stage — positional encodings for lumbar level identity [130], or structural attention biases derived from graph distance [131] — were never exercised and are not part of what this thesis tests. They are noted because the spine's short fixed path structure makes them a natural next step, not because they contributed anything reported here. The contribution is the explicit typed target graph and its controlled evaluation, not a requirement to use the largest graph architecture.

3.12.5 Graph Baselines

The graph contribution is tested against:

1. **Independent heads:** no message passing between targets.

2. **Ordered level transformer:** features communicate across the five lumbar levels but do not form 25 typed condition/laterality nodes.

3. **Homogeneous graph:** the same 25 nodes and adjacency pattern but one undifferentiated edge type [125, 128].

4. **Heterogeneous graph:** typed adjacent-level, cross-condition and bilateral relations [127, 129].

5. **Random/shuffled-edge control:** graph capacity retained while anatomical edges are permuted.

The shuffled-edge control is essential. If a graph with arbitrary topology performs as well as the anatomical graph, the study has shown that extra message-passing capacity helps, not that the encoded anatomy matters.

3.12.6 Edge-Family Ablation

The full graph is further decomposed:

- remove adjacent-level edges;
- remove same-level cross-condition edges;
- remove bilateral edges;
- retain only one family at a time.

Results are reported by target family. It is plausible, for example, that bilateral edges help lateral-compartment grading but add little to central-canal classification. A single average graph improvement would conceal this structure.

3.12.7 Isolated-Lesion Specificity Stress Test

A graph can improve average grading while still introducing a clinically undesirable failure mode: a severe focal lesion at one level may elevate the predicted severity of an adjacent level that is visually normal. This "information contagion" is evaluated directly rather than inferred from aggregate specificity.

The held-out public evaluation set is searched, without altering labels, for cases in which one target is Severe while the same target at one or both adjacent levels is Normal/Mild. For these naturally occurring isolated-lesion cases, the independent-head, homogeneous-graph, ungated heterogeneous-graph and gated heterogeneous-graph models are compared on the adjacent normal levels. The primary stress-test outcomes are the adjacent-level false-positive rate, the change in predicted Severe probability after graph refinement, and the fraction of cases in which the graph changes a correct local prediction into an incorrect higher grade.

A graph contribution is not considered clinically coherent if an aggregate gain is accompanied by a material, reproducible increase in adjacent-level false-positive contamination. This stress test is therefore part of the acceptance criterion for retaining relational message passing in the final integrated system. To safeguard against gate collapse caused by heavy class imbalance (where residual gating parameters $\boldsymbol{\gamma}$ lazily suppress relational propagation to zero), gate activation distributions $(\mu_\gamma, \sigma_\gamma)$ are tracked across training iterations, with an auxiliary entropy-maximization loss penalty applied if average gating falls below an operational threshold.

3.12.8 External Graph-Scope Sensitivity

Rizgary provides reliable evaluation labels primarily for central-canal stenosis, whereas the public graph is trained over all 25 targets. Absence of a local label for a foraminal or subarticular node does not make that node inherently invalid, but the possibility that auxiliary unvalidated nodes alter central-canal predictions is tested explicitly.

For the external central-canal analysis, three inference configurations are compared where implementation permits: (i) the full trained 25-node graph using all visually available node representations; (ii) a central-canal-only graph restricted to the five canal nodes using a separately trained or appropriately ablated model; and (iii) the full graph with only nodes lacking valid visual evidence masked. The comparison is interpreted as a sensitivity analysis of external graph scope, not as a post-hoc search for whichever configuration scores highest.

3.12.9 Graph Consistency and Uncertainty

Graph information is not used to hard-code clinically deterministic rules. Severe stenosis at L4–L5 does not require moderate stenosis at L3–L4, and the model is allowed to preserve isolated pathology. Relational message passing is therefore soft.

For uncertainty analysis, node disagreement can be measured between an independent-head prediction and the graph-refined prediction, or between neighbour-conditioned and local evidence. This disagreement is explored as an auxiliary signal for selective prediction, but it is not equated with calibrated clinical uncertainty unless validated against error.

3.13 Supporting Method: Ordinal and Cost-Sensitive Grading

3.13.1 Categorical Baseline

Standard three-class cross-entropy is retained as the mandatory reference:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_t m_t \sum_{k=0}^{2} \mathbb{I}(y_t = k) \log p_{t,k}. \tag{3.24}$$

Class weighting or balanced sampling is applied in separate controlled variants. This baseline is necessary because previous lumbar work found that explicitly ordinal objectives did not consistently outperform cross-entropy [23].

3.13.2 Ordinal Threshold Formulation

An ordinal head represents a three-grade target as two ordered threshold questions:

$$P(y > 0 \mid \mathbf{h}), \qquad P(y > 1 \mid \mathbf{h}).$$

A CORAL/CORN-style implementation may be used provided the ordering constraint is respected. The exact ordinal formulation is selected before the final ablation and compared directly with categorical cross-entropy under identical features.

Performance is evaluated not only by exact class agreement but also by error distance:

$$d_t = |\hat{y}_t - y_t|. \tag{3.25}$$

The proportions with $d_t = 0$, $d_t = 1$ and $d_t = 2$ are reported separately.

3.13.3 Clinically Asymmetric Cost

A cost matrix C allows distant under-grading to receive a larger training penalty:

$$C = \begin{bmatrix} 0 & c_{01} & c_{02} \\ c_{10} & 0 & c_{12} \\ c_{20} & c_{21} & 0 \end{bmatrix}, \tag{3.26}$$

with the clinically motivated condition

$$c_{20} > c_{21},$$

so that Severe→Normal/Mild is more costly than Severe→Moderate. The exact numerical values are not invented by the model developer. They are either aligned with a defensible benchmark/clinical weighting or selected through a sensitivity analysis across a small pre-specified family of matrices. Specifically, baseline values for C are initialized to mirror the RSNA challenge evaluation weights ($1 : 2 : 4$ ratio for Normal/Mild, Moderate, and Severe grades), directly aligning the training penalty with the benchmark's clinical risk metric.

When using an ordinal threshold head (Section 3.13.2), the 3-class probability distribution $p_{t,k}$ is explicitly derived from cumulative threshold probabilities before computing expected cost:

$$p_{t,0} = 1 - P(y > 0 \mid \mathbf{h}_t), \qquad p_{t,1} = P(y > 0 \mid \mathbf{h}_t) - P(y > 1 \mid \mathbf{h}_t), \qquad p_{t,2} = P(y > 1 \mid \mathbf{h}_t).$$

This derived distribution feeds the expected cost formulation:

$$\mathcal{L}_{\text{cost}} = \sum_{t} m_t \sum_{k=0}^{2} C_{y_t,k}\, p_{t,k}. \tag{3.27}$$

The cost term is evaluated alongside, not instead of, standard calibration and discrimination metrics. A model is not considered better merely because it optimises the chosen matrix.

3.13.4 Under-Grading Versus Over-Grading Trade-off

Cost sensitivity is not accepted as beneficial merely because it lowers the Severe→Normal/Mild rate. A sufficiently large penalty on under-grading can shift the decision boundary toward Severe and create an unacceptable false-positive burden. The cost-sensitive experiments therefore report both sides of the trade-off for each pre-specified cost matrix and each candidate value of λ_{cost}.

The minimum reporting set comprises Severe-class sensitivity, specificity, positive predictive value, the proportion of all targets predicted Severe, the Severe→Normal/Mild rate and the Normal/Mild→Severe rate, together with macro-F1, weighted κ and calibration. A cost-sensitive configuration is retained only if the reduction in clinically consequential under-grading is not explained by indiscriminate over-prediction of Severe.

Decision-curve analysis [132] is not a mandatory part of this imaging-grading study because the thesis does not observe a validated treatment or referral action threshold, and net benefit is defined only once such a threshold exists. If a clinically agreed binary action and threshold probability are established in a separately approved extension, decision-curve analysis may be reported there. Until then, the thesis reports the empirical sensitivity–false-positive trade-off without presenting radiological grades as treatment utility.

3.13.5 Class Imbalance

Because severe grades are uncommon in LumbarDISC, the study compares a small set of established strategies rather than stacking all of them simultaneously:

- unweighted cross-entropy;
- inverse-frequency or effective-number class weighting;
- weighted sampling;
- focal loss [99];
- the cost-sensitive/ordinal variant above.

MixUp/CutMix are not the default for ordinal grading because mixed labels may have no clinical interpretation. Geometric augmentation is constrained so that left/right labels are updated consistently; horizontal flipping is prohibited for laterality-dependent targets unless labels and orientation are transformed together.

3.14 Supporting Method: Calibration, Uncertainty and Selective Prediction

3.14.1 Probability Calibration

Raw neural-network probabilities are not treated as calibrated confidence. Modern networks are systematically overconfident, and added capacity tends to worsen calibration even where it improves accuracy [25]. Temperature scaling is fitted on the validation set:

$$p_k^{(T)} = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}, \tag{3.28}$$

where $T > 0$ is selected without access to the test labels. Calibration is then evaluated with Expected Calibration Error, the Brier score [133] and reliability diagrams.

3.14.2 Predictive Uncertainty

The primary feasible uncertainty methods are Monte Carlo dropout, which approximates Bayesian inference by retaining dropout at test time and so adds no training cost [134], and a small ensemble of independently initialised models, which is typically better calibrated and degrades more gracefully under distribution shift [135]. The second property is the reason an ensemble is considered despite its cost, since the external Rizgary evaluation is itself a distribution shift and is where overconfidence would be most damaging. If S stochastic predictions are available, the predictive mean is

$$\bar{\mathbf{p}} = \frac{1}{S}\sum_{s=1}^{S} \mathbf{p}^{(s)}, \tag{3.29}$$

and uncertainty can be summarised by predictive entropy,

$$H(\bar{\mathbf{p}}) = -\sum_k \bar{p}_k \log \bar{p}_k, \tag{3.30}$$

together with between-member variance. The study reports whether uncertainty is higher on incorrect predictions, distant grade errors, low-agreement targets and external-domain cases.

3.14.3 Selective Prediction

A selective system abstains when uncertainty exceeds threshold δ:

$$U(x) > \delta \Rightarrow \text{refer/abstain}.$$

Rather than choosing one arbitrary threshold, the complete risk–coverage curve is reported: as increasingly uncertain cases are withheld, what fraction of the cohort remains covered and what error remains among the retained predictions? The area under, or summary points from, this curve can compare methods.

Conformal prediction offers a stronger guarantee than a heuristic threshold, converting any model into one that emits prediction sets with finite-sample coverage under exchangeability alone [136]. It is therefore reported for the internal held-out evaluation, where exchangeability is defensible. It is *not* claimed to hold on the Rizgary test set: that cohort is drawn from a different institution, scanner population and reporting process, which is precisely the violation of exchangeability under which the coverage guarantee fails. Where conformal sets are computed externally they are presented as a descriptive comparison against internally calibrated thresholds, with the broken assumption stated rather than left for a reader to notice.

The terminology "refer" describes model behaviour only. It does not claim a validated clinical workflow; a prospective reader study would be required to show that abstention improves patient care.

3.15 Composite Training Objective

In the primary phased pipeline, self-supervised pretraining (Phase 2) minimizes $\mathcal{L}_{\text{ACSSL}}$ independently. During supervised fine-tuning (Phase 3), the active objective is

$$\mathcal{L}_{\text{supervised\_total}} = \mathcal{L}_{\text{grade}} + \lambda_{\text{miss}}\mathcal{L}_{\text{miss}} + \lambda_{\text{cost}}\mathcal{L}_{\text{cost}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \tag{3.31}$$

where $\mathcal{L}_{\text{grade}}$ is categorical or ordinal supervised loss, $\mathcal{L}_{\text{miss}}$ is optional missing-modality consistency, $\mathcal{L}_{\text{cost}}$ is the optional clinical cost term and $\mathcal{L}_{\text{reg}}$ collects standard regularisation. Monolithic multi-task training incorporating $\lambda_{\text{ssl}}\mathcal{L}_{\text{ACSSL}}$ into the joint sum is evaluated as an optional ablation.

Graph message passing is normally part of the forward model rather than a separate loss. If an explicit graph-consistency regulariser is introduced, it receives its own coefficient and ablation.

Every non-zero λ is treated as a hyperparameter whose contribution must be demonstrated. A loss term is removed if it adds no reproducible value. The final thesis reports the selected values and the search range: The executed settings were: AdamW; a cosine schedule over the full budget with a 5% linear warm-up; load-balance weight $\lambda_{\text{bal}} = 0.01$ on the routing gate; cost-sensitive weight $\lambda_{\text{cost}} = 0.5$ at E7 and 0 elsewhere. Calibration temperature is not a training hyper-parameter: it is fitted post hoc on the validation partition, and the fitted values ranged from 1.13 to 2.02 across rungs.

3.16 Optimisation and Training Protocol

3.16.1 Training Phases

Training is separated into conceptually distinct phases:

1. **Anatomical/localisation training or loading.** The localiser is trained/reproduced and then fixed for the grading experiments unless a joint-training ablation is explicitly conducted.

2. **Self-supervised pretraining.** Sequence encoders are trained with anatomical cross-sequence pairs using development patients only. The primary ACSSL phase uses studies for which the sequence pair required by the positive relation is present and passes QC. Modality dropout is disabled in this phase so that the pretraining objective is not changed by stochastic removal of its own positive pair. An incomplete-modality SSL variant, if investigated, is treated as a separate exploratory experiment with a dynamically defined set of valid positives.

3. **Supervised grading.** Encoders are fine-tuned with the routing and graph modules on LumbarDISC labels. Modality dropout and controlled corruption are introduced here, after anatomical pretraining, so that missing-modality robustness is learned without altering the definition of ACSSL.

4. **Calibration.** Temperature/uncertainty parameters are fitted on validation data after model selection.

5. **External evaluation.** The selected model is frozen and evaluated on the fixed Rizgary test set.

6. **Few-shot adaptation.** New model copies are adapted using only the local adaptation pool and re-evaluated on the same fixed local test set.

3.16.2 Optimiser and Schedules

AdamW [137] is the default optimiser for transformer/graph components; SGD or AdamW may be compared for CNN baselines if required. AdamW is preferred over Adam [138] because weight decay is decoupled from the adaptive step size, so regularisation strength can be tuned without confounding the effective learning rate—a distinction that matters when one schedule must serve encoder, router and graph modules of different scales. Learning rate is selected on the validation set within a pre-specified range, with warm-up and cosine decay [139] or a plateau scheduler used consistently across the main ablation.

Every rung of the ladder is trained for the same fixed epoch budget, and early stopping is deliberately not used. This is a departure from common practice and is made for comparability. If training halted on a validation criterion, the budget would become data-dependent: one configuration might run the full schedule while another halted at a third of it, and the difference between two rungs would then confound architecture with training length—the very confound the ladder in Section 3.17 exists to remove. A truncated run also never completes its cosine annealing, so rungs would receive different amounts of learning-rate decay and would not be compared at an equivalent point on their schedules.

Fixing the budget does not mean testing whatever the final epoch produced. Model selection is retained and is separate from stopping: validation macro-F1 is evaluated after every epoch, the best-scoring checkpoint is retained, and that checkpoint—not the final one—is restored before any held-out evaluation, as required by Section 3.16.4. The selection criterion is prevalence-robust by construction; raw accuracy is not used, because a model predicting the majority grade for every target would score highly on it while ordering no severity at all.

The cost of a fixed budget is compute, since a configuration that converges early continues to train. That cost is accepted. The epoch budget is chosen on the development and validation partitions before the confirmatory comparison is run, and is held constant across the ladder thereafter.

The final report records:

- optimiser and version;
- initial learning rate and scheduler;
- batch size / effective batch size;
- fixed epoch budget, and the epoch selected by validation macro-F1;
- weight decay and dropout;
- input resolution and slice-stack radius;
- precision mode (FP32/BF16/FP16);
- random seeds;
- hardware and software versions;

- wall-clock and GPU time.

Mixed-precision training [140] is used where it does not change results, because volumetric multi-sequence batches are memory-bound on the hardware available to this programme. At least one configuration is repeated in single precision to confirm that reduced precision has not altered a comparison on which a contribution claim rests.

No final hyperparameter value is fabricated in this protocol chapter; each is Every rung was trained for a fixed budget of 50 epochs with no early stopping, so the schedule is identical across the ladder. Model selection recorded the epoch of best validation macro-F1 and restored that checkpoint before the held-out test pass; selection therefore never observes the test partition.

3.16.3 Augmentation

Training augmentation is designed to simulate plausible MRI variability without altering disease anatomy. Candidate transforms include modest intensity scaling, gamma/contrast change, bias-field simulation, noise, small translation/rotation and limited elastic deformation. Aggressive transformations that distort the canal or foraminal morphology beyond plausible acquisition variation are excluded.

Laterality-aware augmentation updates all side-specific labels. If a horizontal flip changes anatomical left/right, left and right target labels and graph node identity are swapped in the same transformation. Augmentation is disabled for validation and test data except in a separately labelled test-time-augmentation experiment.

3.16.4 Model Selection

Only validation data are used to choose architecture depth, routing hyperparameters, loss weights, the retained checkpoint and calibration. The held-out public test and fixed Rizgary test sets are not inspected for iterative model selection. If many exploratory experiments are performed, only the pre-specified final comparison is used for confirmatory statistical claims; exploratory results are labelled as such.

3.17 Experimental Ladder and Ablation Programme

The main experiment is not one comparison between “AMOG-Net” and a baseline. It is a ladder in which each additional assumption is isolated.

ID	Configuration	Question isolated
E0	Independent ROI classifier	How well can anatomically localised targets be graded without relational or adaptive fusion?
E1	+ DICOM-aligned multi-sequence ROIs	What is gained by correct cross-plane correspondence under fixed fusion?
E2	+ disease-conditioned sequence routing	Does target-specific routing outperform fixed fusion?
E3	+ modality dropout / quality-aware corruption training	Does explicit training for missing and degraded sequences improve robustness?
E4	+ anatomical cross-sequence SSL	Does anatomy-defined pretraining improve representation and label efficiency?
E5	+ homogeneous target graph	Does generic message passing help?
E6	+ typed heterogeneous disease–anatomy graph with gated residual update	Do clinically defined edge types add value without contaminating strong local evidence?
E7	+ ordinal/cost-sensitive/calibrated heads	Can clinically consequential errors and probability reliability be improved?
E8	Selected complete system + external transfer	Do the selected components survive institutional shift and adapt efficiently?

The ordering may be adjusted for engineering practicality, but the comparisons themselves remain. In particular, E6 must be compared with both E5 and a shuffled-edge control. Otherwise any improvement can be attributed to additional parameters rather than anatomical topology.

3.17.1 Contribution-Specific Ablations

Three ablation families correspond directly to the doctoral contributions.

Anatomical SSL ablation compares no SSL, generic image SSL and DICOM-aligned cross-sequence SSL under identical downstream training. It is repeated at reduced label fractions and accompanied by the sequence-specific preservation probes of Section 3.10.6.

Routing ablation compares fixed fusion, cross-attention, target- conditioned routing, target-conditioned routing with modality dropout, and quality-aware routing under controlled corruption. Performance is reported under complete, missing-sequence and corrupted-sequence input.

Graph ablation compares independent heads, an ordered five-level transformer, homogeneous graph, ungated heterogeneous graph, gated heterogeneous graph and random/shuffled edges. Edge-family removal further identifies whether adjacency, bilateral symmetry or same-level cross-condition interaction carries the signal, while isolated-lesion cases test whether any gain is purchased by adjacent-level contamination.

3.17.2 Negative Results

A component that fails to improve the pre-specified outcome is not silently removed from the research record. The thesis reports it, including confidence intervals and target-specific effects. This is particularly important for ordinal objectives and graph modelling, because both are theoretically attractive and therefore vulnerable to confirmation bias.

A component enters the final integrated system only if it adds a reproducible benefit, improves an outcome that the thesis has defined as clinically relevant, or yields a meaningful robustness/calibration advantage despite similar aggregate discrimination.

3.18 External Validation at Rizgary

3.18.1 Definition of Zero-Shot Transfer

Zero-shot transfer is defined strictly. The selected model, including encoder, router, graph, classifier and calibration strategy developed on public data, is frozen before any Rizgary test label is used. The local images are passed through the same preprocessing and localisation pipeline, with only non-learned DICOM reconstruction and pre-specified per-volume normalisation allowed.

No local fine-tuning, threshold optimisation or calibration is permitted in the zero-shot result. If a local intensity harmonisation method is fitted on Rizgary data, that constitutes a separate domain-adaptation experiment and is labelled accordingly.

The primary zero-shot analysis is restricted to the central-canal targets. Where the harmonised reader regrading described in Section 3.5.3 is available, that adjudicated three-grade matrix defines the confirmatory external reference standard. The manually audited routine-report matrix is retained as a secondary real-world reference analysis. The principal comparison is therefore not the full 25-target benchmark score versus a five-target local score; it is public held-out versus local external performance on the same central-canal grading task, with the reference-standard provenance stated explicitly.

3.18.2 Quantifying Domain Shift

Domain shift between institutions is a recognised and extensively catalogued failure mode in medical image analysis, with adaptation methods spanning supervised, semi-supervised and unsupervised settings [14]. This study measures the shift before attempting to correct it, because its size and structure are themselves the finding sought by RQ4.

Performance degradation is reported as both absolute and relative change:

$$\Delta M = M_{\text{Rizgary}} - M_{\text{Public}}, \tag{3.32}$$

$$\Delta M_{\%} = 100\, \frac{M_{\text{Rizgary}} - M_{\text{Public}}}{M_{\text{Public}}}, \tag{3.33}$$

for each primary metric M. The thesis does not interpret a large negative ΔM as failed research; the size and structure of the drop are among the main outcomes of RQ4.

The shift is decomposed by:

- lumbar level;
- severity class;
- available sequence configuration;
- scanner/vendor/field strength where the local cohort contains enough variation;
- slice thickness/in-plane resolution;
- axial-to-disc angulation discrepancy and geometry quality;
- localisation success and verified-ROI correction status;
- report/reference-standard confidence and harmonised-reader status;
- image-quality/corruption indicators where available.

This analysis separates representational failure from data-quality failure where possible. For example, if grading declines but localisation remains stable, the pattern supports the detection-versus-grading asymmetry documented in Chapter 2. The interpretation is explicitly decomposed into at least three potential contributors: localisation shift, image/representation shift and reference-standard shift. The verified-ROI control estimates the first; the harmonised-reader analysis reduces the third; the residual difference is the closest available estimate of grading-domain transport failure, although the three sources cannot be made perfectly orthogonal in retrospective data.

3.18.3 Local Error Review

A blinded error audit is performed after the zero-shot metrics are locked. Incorrect local predictions are grouped into:

- wrong level/localisation;
- adjacent-grade disagreement;

- distant Severe$\leftrightarrow$Normal/Mild disagreement;
- apparent label/report ambiguity;
- missing or atypical sequence;
- image-quality/artifact issue;
- plausible model error despite adequate evidence.

Where reader time permits, a subset of model/report disagreements is re-read without exposing the model prediction. The purpose is not to alter the test labels opportunistically but to estimate how much apparent model error may reflect reference-standard variability.

3.19 Few-Shot Domain Adaptation

3.19.1 Adaptation Sizes

The adaptation pool is sampled at

$$N \in \{10, 25, 50, 100\}$$

patient-level labelled cases, subject to the final number of eligible local studies. Each N is sampled multiple times, stratified where feasible by central-canal severity. The same fixed local test set is used after every adaptation.

The experiment therefore estimates a function

$$M(N) = \text{test performance after adaptation with } N \text{ local cases}, \tag{3.34}$$

rather than a single "fine-tuned" number.

3.19.2 Adaptation Strategies

Few-shot adaptation is not treated as a single operation, because the dominant source of institutional shift may reside in the image representation, the modality router, the relational module or the final decision boundary. A pre-specified family of progressively less constrained strategies is therefore compared:

1. **No adaptation.** The frozen zero-shot baseline.
2. **Non-learned/simple harmonisation.** Intensity preprocessing is adjusted using only the adaptation pool where a defensible method exists.
3. **Head-only adaptation.** Only the final grading heads are updated.

4. **Router + head adaptation.** Sequence encoders and graph are frozen; routing and output layers are adapted.

5. **Graph + head adaptation.** Sequence encoders are frozen while relation-aware graph parameters and output heads are adapted.

6. **Late-encoder PEFT + head.** Low-rank [141] or bottleneck-adapter [142] parameters are inserted into a limited set of late encoder blocks while the remainder of the image representation stays fixed.

7. **Constrained multi-module PEFT.** A small parameter budget is distributed across late encoder, router and graph components according to a pre-specified configuration selected on development/domain-shift validation rather than the fixed Rizgary test set.

8. **Full fine-tuning.** All or most parameters are updated as a high-capacity reference only where N and compute make this defensible.

To evaluate the local morphology transfer task (disc bulge, protrusion, extrusion) under extreme sample constraints ($N \leq 25$), multi-label linear probing with independent sigmoid heads on frozen ACSSL representations is evaluated as a standalone baseline, accommodating non-mutually-exclusive herniation co-occurrence and comparing its transfer performance against full fine-tuning to directly test Hypothesis H1.

This design avoids assuming in advance that graph-only or encoder-only adaptation is correct. The trainable parameter count is reported for every strategy, and performance is interpreted jointly with that count. The exact adapter/LoRA implementation is supporting methodology taken directly from the parameter-efficient transfer literature [141, 142], not a doctoral novelty; the scientific question is where a limited local adaptation budget is most effective and how that answer changes with N.

3.19.3 Annotation-Efficiency Summary

For each method, the recovery curve reports macro-F1, weighted κ, severe recall and calibration as a function of N. A useful summary is the area under the performance-versus-log-annotation curve, but raw points with confidence intervals remain primary because the curve may be non-linear.

No arbitrary threshold such as "95% recovery" defines success. If 100 local cases recover little performance, that is a meaningful negative finding about transportability. If 10 cases recover most of the loss, that is evidence for annotation-efficient adaptation.

3.19.4 Avoiding Adaptation Overfit

Few-shot training is particularly vulnerable to overfitting and hyperparameter selection bias, especially at $N = 10$. Repeatedly splitting ten cases into even smaller train/validation folds can itself create unstable model selection. The study therefore separates *method selection* from *few-shot estimation.*

- PEFT architecture, rank/adapter size, learning-rate range, maximum epoch budget and stopping rule are selected before the final $N = 10, 25, 50, 100$ recovery experiment, using public development/site-shift experiments or a separate local adaptation-development procedure that never touches the fixed Rizgary test set.
- Each N is repeated across several independent stratified adaptation samples; variance across these samples is reported because the identity of ten cases can dominate a point estimate.
- For the smallest N, the pre-specified stopping rule is used rather than inventing a two- or three-patient validation split. Where the adaptation pool is large enough, internal validation or cross-validation may be used for larger N as a secondary safeguard.
- The fixed external test set is never used to choose epoch, adapter rank, module placement, threshold, calibration or any other adaptation hyperparameter.
- Training-set loss and generalisation gap are reported for the tiny-N settings so that apparent recovery caused by memorisation is visible.
- PEFT is compared with full fine-tuning rather than assumed superior merely because fewer parameters are trainable.

3.20 Evaluation Metrics

No single metric is treated as sufficient because the task is imbalanced, ordinal and multi-target.

3.20.1 Macro F1 and Balanced Accuracy

For class k, precision and recall are

$$P_k = \frac{TP_k}{TP_k + FP_k}, \tag{3.35}$$

$$R_k = \frac{TP_k}{TP_k + FN_k}, \tag{3.36}$$

and

$$F1_k = 2\frac{P_k R_k}{P_k + R_k}. \tag{3.37}$$

Macro-F1 is the unweighted class mean,

$$F1_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} F1_k, \tag{3.38}$$

so the severe class cannot be hidden by the Normal/Mild majority.

Balanced accuracy is

$$BA = \frac{1}{K}\sum_{k=1}^{K} R_k. \tag{3.39}$$

These are primary discrimination metrics.

3.20.2 Quadratic Weighted Kappa

Quadratic weighted κ [112] is reported because the labels are ordered and the literature supplies human agreement values in κ. The weighting is not a technical detail but a clinical requirement: an unweighted κ would penalise a Normal/Mild-to-Moderate confusion exactly as heavily as a Severe-to-Normal/Mild one, collapsing the very distinction the grading task exists to preserve. For observed confusion matrix O and expected matrix E, with weight

$$w_{ij} = \frac{(i-j)^2}{(K-1)^2},$$

the statistic is

$$\kappa_w = 1 - \frac{\sum_{ij} w_{ij} O_{ij}}{\sum_{ij} w_{ij} E_{ij}}. \tag{3.40}$$

Exact agreement and within-one-grade agreement are also reported so that a high κ is not the only view of ordinal error.

3.20.3 Severe-Class Outcomes

The clinically decisive class is reported separately:

- Severe recall/sensitivity;
- Severe precision;
- Severe-versus-rest AUROC and AUPRC where appropriate;

- Severe→Normal/Mild error rate;
- Severe→Moderate error rate.

The separation distinguishes distant under-grading from adjacent boundary error.

3.20.4 AUROC and AUPRC

For three-class outcomes, AUROC is calculated one-versus-rest and macro-averaged. AUPRC is reported alongside AUROC for rare Severe classes because it is more sensitive to prevalence and false positives. Curves are constructed at patient- target level, while confidence intervals respect patient clustering.

3.20.5 Calibration

Calibration evaluation includes:

- Brier score [133];
- Expected Calibration Error (ECE);
- classwise reliability diagrams;
- negative log likelihood where useful;
- risk–coverage curves under selective prediction.

Calibration is reported both internally and externally because a model can retain discrimination while becoming overconfident after domain shift.

3.20.6 Localisation Metrics

Localisation is evaluated separately using:

- mean, median and 95th-percentile Euclidean error in millimetres;
- PCK at physically meaningful tolerances;
- Dice coefficient for available masks;
- surface-distance metrics where mask quality supports them;
- complete localisation-failure rate per study.

3.20.7 Efficiency Metrics

For each model configuration the thesis reports:

- trainable and total parameter count;
- FLOPs/MACs where they can be estimated consistently;
- peak VRAM;
- training wall time and GPU hours;
- inference time per study and per target;
- storage size of weights.

Efficiency comparisons use the same hardware and batch/inference conditions. Timing from different machines is not compared as if it were an architectural property.

3.21 Statistical Analysis

3.21.1 Patient-Level Bootstrap Confidence Intervals

Confidence intervals for primary metrics are obtained by bootstrap resampling *patients*, not individual targets. On each replicate, a patient is sampled with replacement and all of that patient's target predictions enter together, preserving within-study dependence. To prevent division-by-zero errors when evaluating minority classes or small subgroups (where unstratified resampling might draw zero positive cases), bootstrap sampling is stratified at the patient level by target severity class, ensuring every resample contains at least one positive instance of minority classes.

The default is a sufficiently large number of replicates to stabilise the 2.5th and 97.5th percentiles; the final number is 2,000 replicates. Ninety-five per cent percentile or bias-corrected intervals are reported consistently.

3.21.2 Paired Model Comparisons

Ablation models are evaluated on the same patients, so comparisons are paired. For scalar patient-level losses or correctness indicators, a paired permutation test or Wilcoxon signed-rank test is used where distributional assumptions are not justified. AUCs are compared with an appropriate paired procedure such as DeLong's test [143] when its assumptions are satisfied. The paired form is required rather than optional here: ablation models are evaluated on the same patients, so their AUCs are correlated and an unpaired comparison would misstate the variance of the difference. Bootstrap differences are used for metrics without a simple analytic test.

The primary quantity is the paired difference

$$\Delta M = M_A - M_B$$

with a 95% confidence interval. Point estimates alone are not used to claim one component is superior.

3.21.3 Multiple Comparisons

The ablation programme contains many target-level analyses. A limited set of primary comparisons is therefore declared in advance:

1. anatomical SSL versus generic/self-supervised baseline;
2. routed+dropout fusion versus ordinary cross-attention;
3. heterogeneous graph versus independent heads and homogeneous graph;
4. zero-shot public versus local performance;
5. PEFT versus no adaptation across annotation sizes.

Target-by-target and subgroup analyses are secondary/exploratory. Where many formal hypotheses are tested, false-discovery-rate control is applied. The thesis distinguishes confirmatory comparisons from exploratory patterns instead of presenting every $p < 0.05$ as independent evidence.

3.21.4 Repeated Training Seeds

Deep-learning optimisation is stochastic. Final ablation comparisons are therefore repeated across multiple fixed seeds where computationally feasible. The thesis reports mean and standard deviation/interval across seeds in addition to the patient-level test uncertainty. A one-off training run is insufficient to attribute a small improvement to a method.

3.21.5 Subgroup Analysis

Subgroup analysis is conducted only where sample size permits and includes:

- lumbar level;
- severity;
- sex/age bands where metadata and governance allow;

- scanner/vendor/field strength;
- acquisition resolution or slice thickness;
- presence/absence of individual sequences.

To prevent reporting unstable point estimates on small strata, quantitative metrics (weighted κ, AUROC) are computed only for subgroups containing a predefined minimum sample size ($N \geq 30$); strata with $N < 30$ are reported exclusively via descriptive counts and raw confusion matrices.

Sparse cells are reported descriptively rather than subjected to unstable significance testing. The subgroup analysis is intended to expose failure modes, not to make causal claims about demographic biology.

3.22 Robustness Experiments

3.22.1 Missing-Sequence Matrix

The selected public-data models are evaluated under every supported sequence configuration. For three sequences, the core matrix includes complete input, three two-sequence combinations and three single-sequence configurations where the required anatomical target remains observable.

Performance is reported per target family rather than only in aggregate. A model can remain strong on central canal grading while collapsing on foraminal narrowing when sagittal T1 is removed; an overall mean would obscure the clinical meaning.

3.22.2 Geometry Perturbation

The reliance on DICOM coordinates creates a specific failure mode. Controlled perturbations therefore test sensitivity to small localisation/geometry errors: axial slice selection shifted by one or two slices, modest ROI-centre offsets, simulated spacing variation, and perturbation of the axial-to-disc angular compatibility term. Results are stratified by the recorded acquisition-angle discrepancy θ_l. The purpose is not to train on corrupted geometry by default, but to measure how brittle the system is to the imperfect localisation and oblique acquisition expected in practice.

3.22.3 Image-Quality and Acquisition Perturbation

Where feasible, controlled perturbations approximating scanner and acquisition failure are applied to the public held-out set: contrast scaling, noise, bias field, resolution degradation, motion-like corruption, slice loss and partial truncation. These do not replace real external validation but help identify which changes reproduce the pattern observed at Rizgary. For router experiments, the change in modality weight before and after corruption is reported alongside the change in prediction, so robustness is not inferred from accuracy alone.

3.22.4 Graph Perturbation

The graph is challenged by:

- random edge deletion;
- shuffled edge types;
- random rewiring with matched edge count;
- removal of one relation family.

A robust anatomical graph should outperform these controls without depending on one fragile edge. If random topology performs equivalently, the anatomical interpretation is rejected.

3.23 Interpretability and Failure Analysis

3.23.1 Local Visual Evidence

Grad-CAM or equivalent saliency may be used as a gross sanity check that the encoder responds within the relevant ROI, but saliency is not presented as a proof of reasoning. The stronger explanations in this thesis arise from structural quantities that are part of the model itself:

- which sequence received routing weight for a specific target;
- which graph relation contributed to target refinement;
- how uncertainty changed before and after relational reasoning;
- how prediction changed under controlled removal of a sequence or edge.

These are still model diagnostics, not causal explanations, but they are more directly linked to the proposed mechanisms than a heatmap alone.

3.23.2 Case-Level Audit Template

A fixed template is used for qualitative audit of representative correct and incorrect cases:

- ground-truth target and grade;
- predicted probability distribution and calibrated grade;
- localisation overlay;
- available MRI sequences;
- routing weights;
- dominant incoming graph relations/attention where extractable;
- uncertainty;
- nearest error category;
- whether the discrepancy is within one grade.

Cases are sampled from predefined categories rather than cherry-picked from the best-looking outputs.

3.24 Reproducibility and Software Engineering

3.24.1 Environment

The implementation uses a reproducible Python/PyTorch medical-imaging environment with pydicom for DICOM access. The executed configuration was Python 3.14.2, PyTorch 2.11.0 (CUDA 12.8), NumPy 2.3.5 and pandas 3.0.5, on a single NVIDIA RTX 5090 with 31.8 GB of memory. A CUDA 12.8 build is required rather than incidental: the card is Blackwell-class (compute capability 12.0) and earlier CUDA builds carry no kernels for it. Environment specification is stored in a lockfile or container recipe.

3.24.2 Experiment Configuration

Each run is configured from a machine-readable file containing:

- dataset manifest and split version;
- selected sequence IDs;
- preprocessing and resampling;

- ROI dimensions;
- encoder architecture;
- SSL/routing/graph settings;
- loss coefficients;
- optimiser and scheduler;
- seed;
- epoch budget and checkpoint-selection rule.

The configuration is copied into the run directory with metrics and model weights so that a table in the thesis can be traced to an executable experiment.

3.24.3 Data Provenance

Raw data are never edited in place. A manifest records the path/identifier and processing status of every study. Derived crops can be regenerated from the manifest and DICOM source, and each derived sample retains its pseudonymous patient, level, condition, side, sequence and source-series identifier.

The local reference matrix is versioned independently of model output. If a label is corrected after clinical review, the change is documented and the affected experiments are rerun rather than silently updating test truth.

3.24.4 Code and Model Release

Code for the public-data methodology will be released where institutional and dataset licences permit. Pretrained weights are released if the training dataset terms allow redistribution. Local Rizgary images and report-derived individual-level data are not included unless a separate hospital approval authorises public release.

The repository will include:

- preprocessing and geometry reconstruction;
- split-generation/verification code;
- localisation/ROI pipeline;
- SSL, routing and graph modules;
- training and evaluation scripts;
- configuration files for primary experiments;
- instructions for reproducing public benchmark results.

3.25 Methodological Safeguards and Threats to Validity

3.25.1 Reference-Standard Shift

LumbarDISC uses structured expert grades; Rizgary originates from routine narrative reports. A performance drop can therefore reflect both imaging-domain shift and reference-standard shift. The thesis mitigates this by restricting transfer to defensible overlap, manually verifying every evaluated local label against source text and, for the strongest external analysis, constructing a harmonised central-canal reference through independent reader regrading with adjudication where feasible. Report-derived and harmonised-reader results are kept separate rather than averaged into one ambiguous reference standard.

3.25.2 Single-Centre Local External Validation

Rizgary is a genuinely independent institution but still only one local centre. External performance therefore demonstrates transport to *that* setting, not universal Middle Eastern generalisation. The thesis states this boundary explicitly. A future multi-hospital regional study is required before making a population-wide deployment claim.

3.25.3 Model Complexity

The full architecture contains several interacting components. Complexity can produce apparent gains merely through parameter count and tuning effort. The ablation ladder, matched backbone, homogeneous graph and shuffled-edge controls are specifically designed to separate mechanism from capacity.

3.25.4 Localisation Error

The graph and routing modules cannot recover evidence absent from a bad crop. Localisation is therefore measured and failures are reported separately. A verified-ROI control on a pre-specified Rizgary subset repeats grading after manual confirmation/correction of target localisation, allowing the external performance drop to be decomposed into an upstream localisation component and a residual grading/reference-standard component.

3.25.5 Graph Evidence and Unlabelled External Nodes

Rizgary cannot score every node used by the 25-target public graph. Missing local ground truth is not treated as missing anatomy, and nodes with valid visual evidence remain available to the trained graph. Truly unsupported nodes caused by absent sequences, failed localisation or inadequate coverage are masked. A full-graph versus canal-only sensitivity analysis tests whether unvalidated auxiliary nodes materially alter central-canal predictions.

3.25.6 Corrupted but Present Modalities

Availability masks solve absence but not poor quality. A motion-corrupted or truncated sequence can still enter the router and receive weight. Quality-aware gating, controlled corruption training and corruption-specific evaluation are therefore included, and routing weights are never interpreted as reliable when the model has not been tested under such degradations.

3.25.7 Class Imbalance

Severe grades are uncommon, so accuracy can be misleading and few-shot local subsets may contain few Severe examples. The primary metrics are macro-F1, balanced accuracy, weighted κ and per-class recall, with patient-level confidence intervals. Adaptation subsets are stratified where feasible and their severity counts are always reported.

3.25.8 Hyperparameter Multiplicity

A sufficiently large search can overfit the validation set even without touching test data. The study limits the number of tunable components in each stage, uses a fixed primary backbone for the core ablation, records the search space and freezes choices before external testing.

3.25.9 Clinical Utility

This methodology evaluates technical grading, calibration and portability. It does not demonstrate that radiologists become faster, safer or more accurate when using the model. Such a claim requires a prospective reader-assistance or impact study with independent clinical approval. The thesis therefore uses the language of "decision-support potential" rather than clinical benefit.

3.26 Mapping Research Questions to Evidence

RQ	Primary comparison	Primary outcomes	Failure interpretation
RQ1	independent/level transformer/homogeneous graph vs ungated and gated typed heterogeneous graph	macro-F1, κ, severe recall, edge ablations, isolated-lesion adjacent-level false positives	graph topology carries little useful signal or relational messages contaminate strong local evidence
RQ2	generic pretraining vs anatomical cross-sequence SSL at 10/25/50/100% labels	macro-F1, κ, label efficiency, external drop, sequence-specific preservation probes	anatomy-defined positives do not improve transferable representation or erase modality-specific signal
RQ3	fixed fusion/cross-attention vs routing, modality dropout and quality-aware routing	missing- and corrupted-sequence degradation, target-specific performance, routing-weight response, calibration	fixed fusion is sufficient or learned routing is unstable under absence/corruption
RQ4	public held-out vs fixed Rizgary test; automated vs verified ROIs; report-derived vs harmonised labels; module-wise PEFT vs no/full tuning	zero-shot Δ, localisation-controlled ΔM_{loc}, recovery curve, calibration shift	transport failure is dominated by localisation, reference-standard shift or representation shift; adaptation may remain inefficient
RQ5	CE vs ordinal/cost-aware objectives; calibrated vs uncalibrated prediction	Severe under-grading and over-grading rates, Severe PPV/specificity, grade distance, Brier/ECE, risk–coverage	lower under-grading is offset by unacceptable Severe false positives or ordinal/cost formulation adds no advantage

This table defines what evidence can answer each question. It also makes the thesis robust to partial negative results: RQ1 can be answered if the graph does not help, and RQ2 can be answered if generic pretraining is equal or superior. The contribution is the controlled experiment and the resulting knowledge, not a requirement that every proposed mechanism win.

3.27 Planned Reporting Standard

The final manuscript/thesis reporting will follow the current applicable medical-AI and diagnostic/prediction reporting checklists at the time of submission. Four apply to different parts of this work and are not interchangeable. TRIPOD+AI [144] governs the development and validation of the prediction model. CLAIM [145] governs the medical-imaging specifics, including acquisition, ground-truth definition and model description. STARD [146] governs any analysis reported as diagnostic accuracy against a reference standard, including the harmonised reader study. STROBE [147] governs the description of the retrospective Rizgary cohort itself. At minimum the report will include:

- complete cohort flow diagrams and inclusion/exclusion reasons;
- reference-standard source, annotator qualifications and agreement where available;
- exact patient-level split logic and IDs reproducible within the approved environment;
- all primary metrics with confidence intervals;
- per-class Severe performance and ordinal error distances;
- calibration and external-domain performance;
- ablations for every claimed novel component;
- repeated-seed variability;
- compute/hardware cost;
- explicit limitations and negative results;
- a statement of what was not tested clinically.

The reporting checklist itself will be versioned with the manuscript because CLAIM/ STARD/TRIPOD+AI guidance may be updated during the candidature.

3.28 Chapter Summary

This chapter translates the research gap of Chapter 2 into a controlled experimental design. The study begins from a structured definition of lumbar disease: twenty-five level–condition–laterality targets on the public benchmark rather than a generic image-level class. DICOM patient-space geometry is preserved so that sagittal and axial evidence can be aligned physically; localisation and disease-specific 2.5D ROIs are treated as shared enabling technology; and all data partitions are defined at patient level.

Three methodological contributions are then tested independently. Anatomically aligned cross-sequence self-supervision asks whether the same spinal level viewed under different MRI sequences provides a stronger representation-learning signal than generic augmentation. Disease-conditioned routing asks whether one model can allocate different sequence evidence to different targets while remaining functional when modalities are absent or degraded, with quality-aware gating tested under controlled corruption. The heterogeneous disease–anatomy graph asks whether explicit typed relations among adjacent levels, co-occurring compartments and bilateral targets improve grading beyond independent heads, an inter-level transformer or a homogeneous graph without inducing adjacent-level information contamination. Anatomical SSL is additionally tested for preservation of sequence-specific diagnostic information. Ordinal, cost-sensitive and uncertainty-aware outputs remain supporting methods and are retained only if their contribution is measurable and does not simply exchange severe under-grading for indiscriminate over-grading.

The translational design is deliberately separated from model development. Rizgary is not mixed into public training. A fixed local test set establishes zero-shot domain shift, while a separate adaptation pool supplies controlled few-shot subsets. External interpretation is strengthened by a verified-ROI control that isolates localisation error and by a harmonised central-canal reader reference where feasible, while the report-derived matrix is retained as a distinct real-world reference analysis. Few-shot adaptation compares where a restricted update budget is applied—head, router, graph or late encoder—rather than assuming one PEFT location is universally optimal. The primary local comparison remains restricted to central-canal stenosis; local herniation morphology is analysed separately rather than being forced into the RSNA schema.

Finally, the methodology defines success as an answer to the research questions, not as attainment of a predetermined accuracy. A graph that adds nothing, an ordinal loss that fails to beat cross-entropy, or a large cross-institutional performance drop are all informative if demonstrated under the same controlled protocol. This principle is what turns the architecture from an engineering assembly into a doctoral experiment: every claimed mechanism is falsifiable, every comparison is tied to a research question, and the external cohort is used to test portability rather than to decorate an internal result.

Chapter 4

Results

4.1 Overview

This chapter reports the executed experimental programme. It is organised around the research questions of Section 1.7 rather than around the architecture, because the questions are what were fixed prospectively and the architecture is only the instrument used to answer them.

Three statements summarise the chapter and are supported in the sections that follow.

1. **The complete system outperforms the single-sequence baseline.** Quadratic weighted kappa rises from 0.7270 to 0.7447, a paired difference of +0.0177 (95% CI $[+0.0097, +0.0260]$) that holds on all seven seeds and survives false discovery rate correction. Two further comparisons also survive: typed heterogeneous edges and the ordinal, cost-sensitive head.

2. **Most of the proposed mechanisms do not account for that gain.** Anatomically aligned self-supervision (RQ2) and disease-conditioned routing (RQ3) produce effects of +0.0030 and +0.0053 QWK, bounded by the paired intervals at 1.32% and 1.55% of baseline performance respectively. Relational structure (RQ1) helps, but the anatomical content of that structure does not: a typed graph beats a homogeneous one on every seed, while an anatomically correct topology does not beat a degree-preserving random shuffle and is bounded at 1.05% of baseline.

3. **There is a mechanism for the second statement, not merely a p-value.** Gradient-weighted class activation mapping shows that the convolutional encoder already concentrates 2.4× the chance share of its attention on the annotated target before any structural prior is added. The localisation problem the priors were designed to assist is largely solved upstream of them.

Section 1.9 anticipated exactly this possibility in stating that “if local evidence alone is sufficient, more complex mechanisms should not be retained". The results are read here in that spirit.

4.2 Experimental Execution

4.2.1 What Was Run

The ablation ladder of Section 3.17 was executed in full: ten configurations × seven training seeds (42–48), fifty epochs each, seventy runs in total, completed in 955.8 minutes on a single RTX 5090. No run failed.

What "LumbarDISC" refers to in this thesis. The name denotes the released lumbar degenerative classification dataset. The experiments here use a subset of it: the 1,974 studies for which all three required sequences could be located and for which the DICOM geometry needed to place anatomically corresponding crops was recoverable, contributing 48,657 annotated targets. Every number in this chapter refers to that subset, never to the release as a whole. The distinction matters for anyone comparing these results with published figures on the same dataset, since a different inclusion rule yields a different sample and the two are not interchangeable. The selection is a property of the pipeline rather than of the patients, is applied before any partition is drawn, and is identical across every rung of the ladder, so it cannot favour one configuration over another.

The campaign was extended from three seeds to seven after the three-seed analysis left two comparisons close to the corrected significance threshold and the power analysis of Section 4.5.5 indicated roughly ten seeds would resolve them. The extension is reported because it changed two verdicts, and Section 4.5.8 states what changed and in which direction.

The ten configurations comprise the eight internal ladder rungs E0–E7 and two controls attached to E6: `E6_shuffled`, in which edge endpoints are permuted while node and edge counts are preserved, and `E6_ungated`, in which the gated residual is replaced by an ungated one.

The ladder specified in Section 3.17 runs to E8, the selected complete system under external transfer. **E8 was not executed.** It is the operational form of RQ4 and is blocked by the two obstacles reported in Section 4.9; every result in this chapter is therefore internal to LumbarDISC, and no rung of the executed ladder speaks to institutional shift.

4.2.2 Data and Partitioning

All results are computed on the frozen patient-level test partition: 297 patients contributing 7,310 scored targets. The partition is written once and reloaded by every run, so all comparisons are paired at the target level. Partition membership is derived from a fixed module constant and is independent of the training seed by construction, which is verified behaviourally rather than by inspection.

4.2.3 Integrity Controls

Three controls guard the reported numbers.

A label-shuffled negative control. A copy of E0 trained on permuted labels must not exceed chance. It does not, confirming that the evaluation pathway cannot manufacture apparent skill from a broken model.

A run-configuration fingerprint. Every result file records the configuration that produced it. A run whose configuration has changed is re-executed rather than reused. This was added after a stale twelve-epoch result was silently reused in place of a fifty-epoch one.

A behavioural test suite. One hundred and twenty-two tests exercise the components against the claims made about them in Chapter 3, including the ordinal head's cumulative transform, evidence masking in the graph, split persistence, and the across-seed inference introduced in Section 4.5. All pass.

4.3 ROI Quality Control

Section 3.8.6 commits to visual inspection of a validation subset using overlays of the detected level, crop boundary and corresponding axial slices, recording correct lumbar level, inclusion of the relevant canal/foramen/recess, correct left/right orientation, adequate axial coverage, crop truncation, and correspondence failure caused by unusual axial angulation. It further commits to a final inclusion/exclusion table.

This section reports that inspection. It is placed before the results because one of its findings determines whether a later result can be interpreted at all.

4.3.1 Why the Correspondence Had to Be Verified

Core Contribution I rests entirely on DICOM-defined anatomical correspondence: ACSSL's positive pairs are anatomically aligned rather than augmentation-derived. If that alignment were wrong, the pretext task would have been matching misaligned patches, and the null result for RQ2 (Section 4.7) would be an artefact of the data pipeline rather than a finding about self-supervision.

No data-grounded evidence existed either way. The correspondence code records that an earlier implementation never read `ImageOrientationPatient` and substituted textbook direction cosines selected by a substring match on the series description, and that its reported round-trip error of 0.00000000 mm was vacuous: mapping a point through a matrix and back through its inverse returns the original point whatever the matrix contains.

4.3.2 An External Validation

The check used here compares two *independent human annotations* through the transform, so it cannot succeed by algebraic identity. LumbarDISC annotates different conditions of the same level on different sequences: central canal on sagittal T2, subarticular on axial T2. The sagittal canal annotation for a level is therefore projected into the axial series and asked whether it selects the slice that carries that level's own axial annotations.

Table 4.1: Cross-annotation correspondence check. Offset is in whole slices from the nearest independently annotated target at the same level.

Partition	Projections	0	1	2	> 2	Within 1
Test (297 studies)	1,429	69%	24%	2%	6%	93%
Dev (1,676 studies)	8,113	70%	24%	2%	5%	**94%**
All	9,542	70%	24%	2%	5%	93.6%

Median offset is 0.0 mm in every partition and the 90th percentile is 4.0 mm, one slice thickness. The dev row is the relevant one for Contribution I, since ACSSL pretrained on that partition.

4.3.3 The Residual Failures Are Not Systematic

Whether the remaining 5–6% undermines the verdict depends on whether failure is concentrated and whether it is level-dependent.

Of 1,973 studies, 1,734 (88%) have no failing projection at all, and the nine studies that fail on every projection account for only 8% of all outliers. Failure is spread thin rather than confined to a broken subpopulation.

More decisively, outliers are evenly distributed across levels: 19.7%, 20.6%, 19.9%, 19.1% and 20.8% for L1–L2 through L5–S1, against an overall distribution of 18.8–20.6%. A systematically incorrect affine transform would not produce that profile; it would fail worst at L5–S1, where segmental angulation is greatest. A flat level profile is the signature of annotation noise and genuine acquisition variation.

The nine complete failures, with median offsets of 16–67 mm, are recorded as the "correspondence failure caused by unusual axial angulation" category and form the first entries of the inclusion/exclusion table.

4.3.4 Left/Right Orientation

Orientation is measurable rather than impressionistic. Across 2,881 axial annotations, left subarticular targets sit at +8.99 mm and right subarticular targets at −8.14 mm relative to each slice's own horizontal centre, where positive is right of centre. Radiological convention renders patient-left on the image right, so left targets appearing at +9 mm is correct. Separation is 17.12 mm and departure from a perfect mirror is 0.85 mm.

This also constrains the lateralisation observation reported in Section 4.11: since the annotations are symmetric to within 0.85 mm, the asymmetry observed in the attention maps is not inherited from the data.

4.3.5 Review Sheets and Reproducibility

One sheet per study and level, five condition rows, two panels per row: the annotated plane with a solid marker and the same three-dimensional point projected into the orthogonal plane with a dashed marker. Markers are 10 mm in physical radius, sized per slice from `PixelSpacing` so they are comparable across scanners. Colour encodes the reference grade and a red border marks a crop truncated at the image edge.

Each panel states its *native* pixel size. A 100 mm field of view is 213 px at 0.469 mm spacing but 107 px at 0.938 mm, and an upsampled panel that does not declare what it was upsampled from implies detail it does not have.

The current set is 300 sheets over 60 studies — 20% of the test partition — at all five levels, covering 1,474 targets, with 0.5% truncated crops and a median cross-plane distance of 1.1 mm, well inside the 3.5–4.8 mm slice thickness, so the projected point lies within the displayed slice rather than being extrapolated to it.

Studies are drawn at random from the frozen partition under a fixed seed, so the subset is reproducible and cannot be reselected after a result is seen. The generating command is stored beside the figures, and a manifest records every sheet with its study, level, target count, partition, seed and geometry parameters.

The accompanying checklist carries one row per target with the six Chapter 3 fields left blank for a reader. Completing it produces the inclusion/exclusion table; at the time of writing the reader pass is outstanding, and this is stated rather than presented as complete.

Figures 4.1 and 4.2 show one sheet each, at the same level, from opposite ends of the correspondence distribution. The successful case is drawn at the *median* of that distribution rather than at its best available value, because a quality-control figure selected for its appearance demonstrates nothing about the cohort.

4.4 The Ablation Ladder

4.4.1 Aggregate Performance

Table 4.2 reports each configuration averaged over the seven seeds.

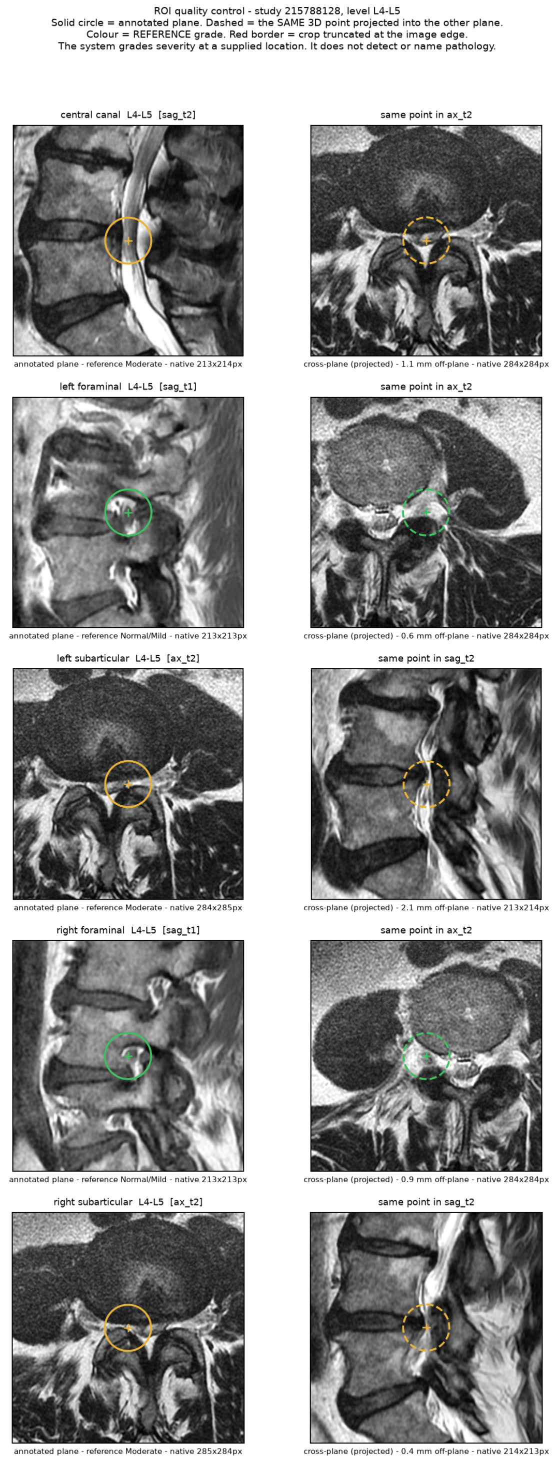

Figure 4.1: Review sheet for study 215788128 at L4–L5, a case at the median of the cross-plane distribution (1.09 mm). Left column: the plane on which each condition was annotated. Right column: the same three-dimensional point projected into the orthogonal plane. Residuals range from 0.4 to 2.1 mm and the anatomy agrees between planes. **The markers show the reference annotation supplied with the dataset, not a model output.** The system is handed a crop centred on an annotated coordinate and grades severity there; it neither searches a study for disease nor names a pathology, and LumbarDISC labels stenosis severity rather than its cause.

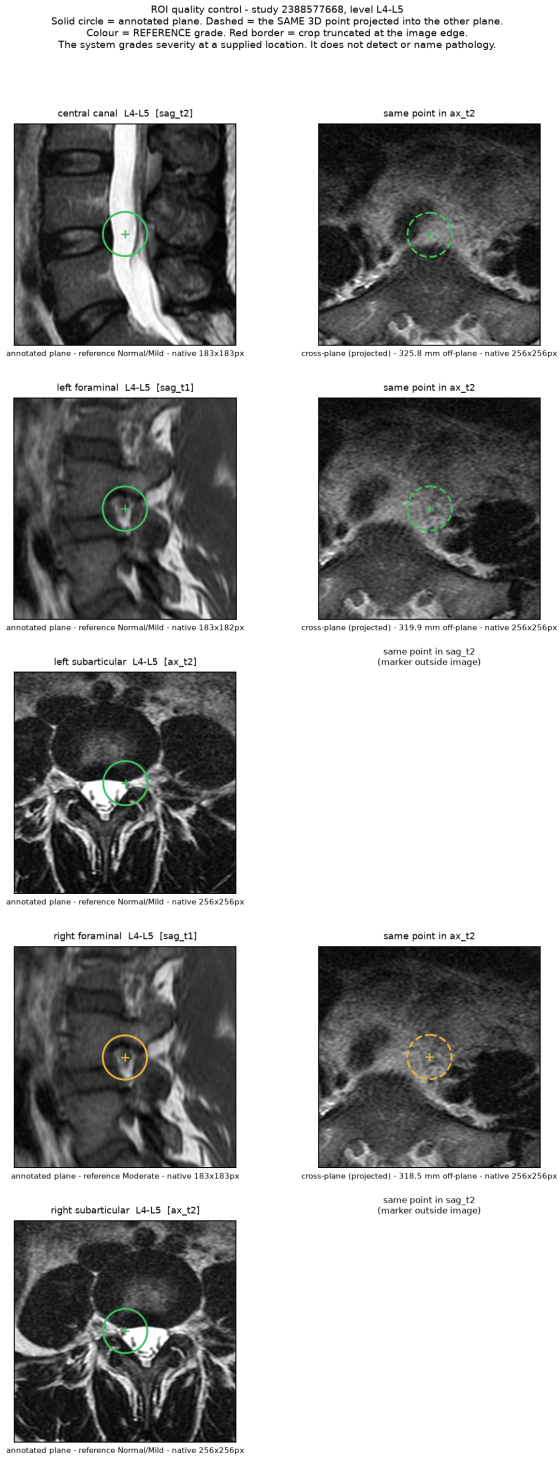

Figure 4.2: The same level in study 2388577668, the worst of the nine studies that fail on every projection. The sagittal annotations are correctly placed, but the axial series lies 318–326 mm from the annotated level, so the projected point falls in abdominal soft tissue and two targets fall outside the image altogether. The transform is behaving correctly: the acquisition does not cover the level. This is the "correspondence failure caused by unusual axial angulation" category of Section 3.8.6 in its most extreme form.

Table 4.2: Ablation ladder on the held-out test partition, mean ± standard deviation over seven training seeds.

Configuration	QWK	Macro-F1	Balanced acc.
E0 single sequence	0.7270 ± 0.0108	0.693 ± 0.009	69.2%
E1 multi-sequence	0.7246 ± 0.0075	0.694 ± 0.004	68.9%
E2 + disease routing	0.7298 ± 0.0123	0.701 ± 0.009	69.6%
E3 + modality dropout	0.7273 ± 0.0125	0.700 ± 0.009	70.1%
E4 + ACSSL	0.7303 ± 0.0030	0.697 ± 0.008	68.3%
E5 + homogeneous graph	0.7272 ± 0.0091	0.698 ± 0.008	69.1%
E6 + typed graph	0.7364 ± 0.0073	0.706 ± 0.010	70.7%
E6 shuffled edges	0.7344 ± 0.0052	0.704 ± 0.007	69.9%
E6 ungated residual	0.7350 ± 0.0167	0.700 ± 0.014	70.6%
E7 + ordinal, cost-sensitive	$\mathbf{0.7447 \pm 0.0051}$	$\mathbf{0.712 \pm 0.005}$	**71.0%**

4.4.2 Where the Gain Actually Arrives

Reporting only the endpoints would conceal the shape of the ladder. Table 4.3 decomposes the +0.0177 into its seven successive steps, and Figure 4.3 shows the same data with the individual seeds drawn in.

Table 4.3: Step-by-step decomposition from E0 to E7 over seven seeds. "Seeds +" counts the seeds on which the step improved QWK.

Step	Δ QWK	Seeds +	Cumulative
Multi-sequence input	−0.0024	4/7	−0.0024
Disease-conditioned routing	+0.0053	5/7	+0.0028
Modality dropout	−0.0026	4/7	+0.0003
Anatomical cross-sequence SSL	+0.0030	4/7	+0.0033
Homogeneous graph	−0.0031	2/7	+0.0002
Typed heterogeneous edges	+0.0093	**7/7**	+0.0094
Ordinal and cost-sensitive head	+0.0082	**7/7**	**+0.0177**

Through the first five steps the accumulated system goes nowhere. It moves within ±0.003 QWK of the baseline it began from and ends those five steps +0.0002 ahead, and not one of them improves on more than five of seven seeds. **The entire gain arrives in the final two steps, and those two are the only steps in the ladder that improve on every seed.** This is stated plainly because it is precisely the structure an ablation study exists to expose, and because a reader who reconstructs it independently from Table 4.2 would be entitled to distrust anything else in this chapter.

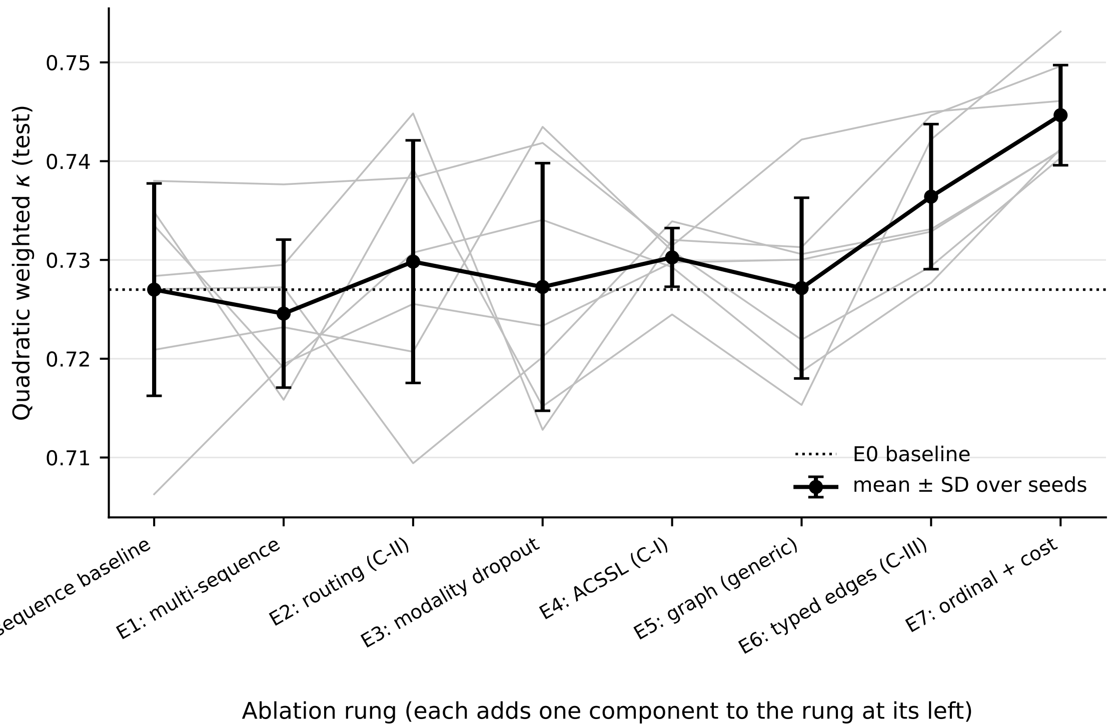

Figure 4.3: The ablation ladder on the held-out test partition. Heavy line: mean over seven seeds with between-seed standard deviation. Faint lines: the seven individual seeds. The seed traces are the point of the figure. Every rung from E0 to E5 sits within one standard deviation of the single-sequence baseline, and individual seeds cross one another repeatedly along the way, so no intermediate rung is distinguishable from the baseline by eye or by test. The separation appears only at E6 and E7. Generated by `implementation/make_figures.py` from the same run records as Table 4.2.

4.5 Statistical Inference

4.5.1 A Correction to the Planned Analysis

Section 3.21 specifies a paired patient-clustered bootstrap with false discovery rate control. Executed literally, that procedure produced a table that contradicted itself: cross-sequence self-supervision appeared as $+0.0270$ with $p = 0.000$ on seed 42 and as -0.0234 with $p = 0.000$ on seed 43. Both intervals excluded zero; their signs were opposite.

The cause is that the bootstrap resamples *patients* with the trained model held fixed. Its interval therefore covers test-set sampling and excludes training stochasticity. On this problem that omission is decisive, because the seed-to-seed standard deviation is larger than most of the effects under test. Reported per seed, the analysis would have permitted any claim to be supported by selecting a seed.

Two changes were made, and neither alters what is compared.

1. *Seeds are averaged inside each bootstrap replicate.* One patient resample is drawn and applied to all seeds, the difference computed per seed, and the seeds averaged before the replicate is recorded. The resulting interval is for the mean effect over training runs. The between-seed standard deviation is reported alongside it rather than absorbed into it.

2. *The false discovery rate family contains one test per comparison and metric.* Three seeds are three views of one comparison, not three independent tests; including them separately tripled the apparent multiplicity.

Per-seed intervals are retained in the appendix as descriptive quantities carrying no significance claim.

4.5.2 Primary Comparisons

Table 4.4 reports the pre-specified comparisons under the corrected procedure. Three survive correction, where the three-seed analysis produced one.

4.5.3 What Constitutes the Confirmatory Family

Multiplicity control is only meaningful if the family it corrects over is stated, so it is stated here rather than left to be inferred from the code.

The primary endpoint is quadratic weighted kappa. It is the ordinal agreement measure the grading task is defined by, it was named as the endpoint before the campaign was executed, and it is the only metric entering the correction. *The confirmatory family is the nine paired contrasts of Table 4.4*, each of which isolates one rung of the ladder or one of its matched controls.

Table 4.4: Pre-specified comparisons, QWK, seven seeds. Difference averaged over seeds with the patient resample shared across them; FDR controlled across these rows.

Comparison	Tests	Δ	95% CI	Seeds +	p_{FDR}	Sig
E7 vs E0	full system	+0.0177	[+0.0097, +0.0260]	7/7	**< 0.001**	**yes**
E6 vs E5	typed edges (RQ1)	+0.0093	[+0.0027, +0.0156]	7/7	**0.009**	**yes**
E7 vs E6	ordinal head (RQ5)	+0.0082	[+0.0025, +0.0143]	7/7	**0.009**	**yes**
E2 vs E1	routing (RQ3)	+0.0053	[−0.0002, +0.0112]	5/7	0.131	no
E4 vs E3	ACSSL (RQ2)	+0.0030	[−0.0038, +0.0096]	4/7	0.575	no
E6 vs E6_shuf	anatomy (RQ1)	+0.0020	[−0.0038, +0.0076]	4/7	0.671	no
E6 vs E6_ung	gating	+0.0014	[−0.0041, +0.0064]	3/7	0.713	no
E5 vs E0	message passing	+0.0002	[−0.0073, +0.0074]	5/7	0.967	no
E3 vs E2	modality dropout	−0.0026	[−0.0091, +0.0034]	4/7	0.575	no

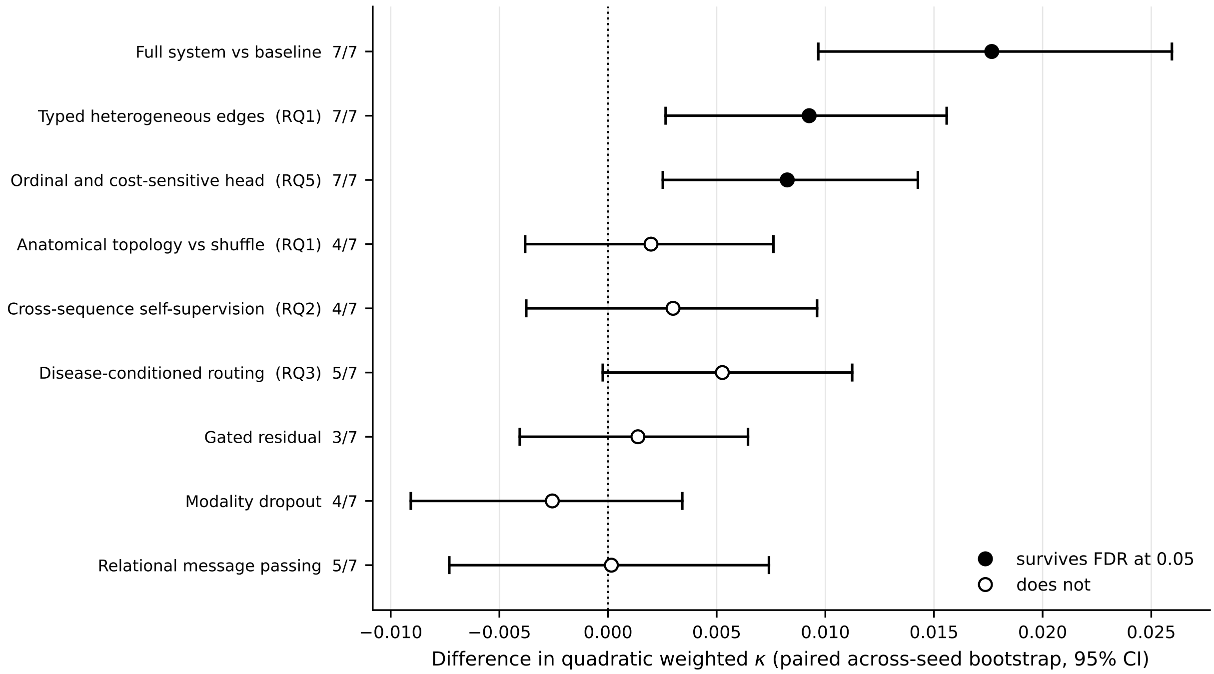

Figure 4.4: The pre-specified comparison family of Table 4.4, plotted on a common axis. Filled markers survive false discovery rate control at 0.05; hollow markers do not. The fraction beside each label is the number of seeds favouring the first arm. Plotting the nulls at the same scale as the positive results is deliberate: it shows that the six unconfirmed mechanisms are not merely unproven but tightly bounded, with intervals whose upper limits lie close to zero rather than admitting a large effect the study was too small to detect. Generated by `implementation/make_figures.py`.

Everything else in this chapter is secondary and is reported without correction: macro-F1 and balanced accuracy, the per-condition and per-level breakdowns, the clinical error structure of Section 4.5.9 and Section 4.5.10, the calibration analysis, the attribution study, and the post hoc variance observation of Section 4.7. These support interpretation; none is used to establish a claim, and no verdict in this chapter rests on one.

The analysis code computes tests on macro-F1 as well as on kappa. Those rows exist and are retained for inspection, but they are not part of the confirmatory family and are not corrected together with it, because a family assembled after seeing which metric behaved more favourably would not be a family at all.

The conclusions do not depend on the correction procedure. Benjamini-Hochberg controls the false discovery rate; Holm-Bonferroni controls the stricter family-wise error rate. Table 4.5 reports both.

Table 4.5: The nine primary QWK contrasts under Benjamini-Hochberg and under Holm-Bonferroni. Generated by `implementation/make_figures.py` from the same p-values as Table 4.4.

Comparison	p	p_{BH}	p_{Holm}	BH	Holm
E7 vs E0	0.0000	0.000	0.000	**yes**	**yes**
E6 vs E5	0.0020	0.009	0.016	**yes**	**yes**
E7 vs E6	0.0020	0.009	0.016	**yes**	**yes**
E2 vs E1	0.0580	0.131	0.348	no	no
E4 vs E3	0.3640	0.575	1.000	no	no
E3 vs E2	0.3830	0.575	1.000	no	no
E6 vs E6_shuffled	0.4950	0.671	1.000	no	no
E6 vs E6_ungated	0.6340	0.713	1.000	no	no
E5 vs E0	0.9670	0.967	1.000	no	no

The two procedures agree on all nine comparisons. The same three survive and the same six do not, and the three that survive do so under the more conservative procedure with adjusted p-values of 0.000, 0.016 and 0.016. The verdicts of this chapter are therefore not an artefact of choosing the more permissive correction.

4.5.4 An Exact Sign Test on the Unanimous Comparisons

Three comparisons favour the upper arm on all seven seeds. That unanimity is worth a statistic of its own, because it answers a question the patient-level bootstrap does not: whether the result depends on one fortunate training run.

Under a null in which the sign of each paired seed difference is a fair coin, seven agreeing signs give an exact two-sided binomial probability of $2 \times 0.5^7 = 0.0156$. This applies to E7 vs E0, E6 vs E5 and E7 vs E6.

The test is deliberately weak. It ignores the magnitude of each difference, treats the seeds as the only source of variation, and takes no account of patient-level sampling, so it cannot replace the paired bootstrap of Section 4.5 and is reported alongside it rather than in place of it. What it adds is that the direction of these three effects is not an artefact of a single seed.

4.5.5 Effect Size and Statistical Power

A p-value confounds effect size with sample size, and a null result is uninterpretable without knowing what the study could have detected. Table 4.6 therefore reports Cohen's d on the paired across-seed differences alongside the number of training seeds that would be required to detect an effect of the observed size at 80% power.

Table 4.6: Effect size and detectability over seven seeds. Cohen's d is computed on the paired across-seed differences. "Seeds required" is for 80% power at the observed effect size and between-seed variance.

Effect	Δ QWK	Cohen's d	Seeds required
Ordinal and cost-sensitive head (RQ5)	+0.0082	**1.97**	3
Full system vs baseline	+0.0177	**1.91**	3
Typed heterogeneous edges (RQ1)	+0.0093	**1.06**	7
Disease-conditioned routing (RQ3)	+0.0053	0.39	52
Anatomical topology vs shuffle (RQ1)	+0.0020	0.20	205
Modality dropout	−0.0026	−0.13	442
Cross-sequence self-supervision (RQ2)	+0.0030	0.24	134
Gated residual	+0.0014	0.07	1,615
Relational message passing	+0.0002	0.01	65,489

The table separates into two groups with nothing in between. Three effects are large by any convention, d between 1.06 and 1.97, and all three are the comparisons that survive correction. Everything else is $d < 0.4$. That gap matters: a study genuinely starved of power would show a spread of intermediate effects rather than a clean division, and would not resolve any comparison at three seeds.

The ladder is therefore not an instrument that reports nothing everywhere. Two effects are estimated to require approximately three seeds and a third approximately seven, under the observed effect sizes and variances, and the campaign ran seven. Where it reports nothing, that is a statement about the effect rather than about the instrument — subject to the caveat above, that these requirements are themselves estimated from seven seeds.

The largest standardised effect in the study is the ordinal and cost-sensitive head at $d = 1.97$, exceeding the full-system comparison it forms part of. Its between-seed standard deviation is 0.0042, the smallest of any comparison, so its modest raw difference is also the most reliably reproduced.

These figures supersede a three-seed version of the same table, and the differences are instructive. Three-seed variance estimates are themselves noisy: routing moved from an estimated 3,467 seeds to 52, and cross-sequence self-supervision from 216 to 134. The ordering is stable and the two groups are unchanged, but individual requirements should be read as order-of-magnitude guidance rather than as precise targets.

This table is not evidence that a mechanism does nothing. Both the effect and the variance entering each estimate are themselves measured on seven seeds, so the seed requirement inherits their uncertainty and is a description of this sample rather than a property of the mechanism. The argument it supports is only this: the observed effect is small relative to the observed between-seed variance, and a conventional experiment would require implausibly many training runs to establish an effect of this apparent magnitude. It does not license the inference that the effect is zero.

4.5.6 Statistical Detectability Is Not Clinical Significance

The three comparisons that survive correction are statistically detectable. This subsection states plainly what they are worth in practice, because a kappa difference of +0.0093 carries no intuitive meaning and a reader left to work it out unaided is entitled to assume more than the number supports.

Table 4.7 counts, over the actual test predictions, how many of the 7,310 graded targets change between consecutive rungs and how many of those changes are improvements rather than churn.

Table 4.7: Predictions changed between rungs on the frozen test partition, averaged over seven seeds. "Fixed" counts targets the upper rung grades correctly and the lower does not; "broken" counts the reverse. Counted from the saved predictions by `implementation/make_figures.py`.

Comparison	Changed	Fixed	Broken	Net
Typed heterogeneous edges	792	388	375	**+13**
Ordinal and cost-sensitive head	726	369	333	**+36**
Complete system vs baseline	778	394	353	**+42**

The churn column carries the message. Typed heterogeneous edges — the architectural contribution that survives correction — rewrite roughly 790 of 7,310 predictions, about 11%, and are right about 13 more of them than the homogeneous graph. That is one additional correct target for every twenty-four patients, where each patient contributes twenty-five graded targets. The complete system against the single-sequence baseline nets 42 targets, or one per seven patients.

Two reference points fix the scale. The first is offered with a caveat. The reported inter-reader agreement for these conditions ranges from $\kappa = 0.49$ for subarticular stenosis to $\kappa = 0.73$ for the central canal, so the distance between two qualified readers is on the order of 0.2 in kappa, against which the complete system's +0.0177 is roughly an order of magnitude smaller. Those literature figures come from different populations and labelling regimes and are not necessarily the same statistic computed here (Section 4.5.7), so this is an indication of scale and not an exact ratio. The second reference point needs no such qualification: a difference of one correctly graded target per seven patients is not something a radiologist reading these studies would experience as a change in the quality of the report.

The one difference with a plausible clinical reading is the asymmetric error. Severe targets graded Normal/Mild fall from 22 of 398 at E0 to 15 at E7, affecting five fewer patients of the 297 in the partition. Missing a severe stenosis is the error that withholds treatment, so the direction is the right one — but it is five patients, and it is associated with the combined ordinal and asymmetric objective rather than with any of the three anatomical mechanisms this thesis set out to test. It should not be attributed to the cost matrix alone: Section 4.10.1 shows that the cost matrix applied without the ordinal head makes aggregate agreement worse.

How these results should be read. The effects reported in this chapter are statistically detectable and clinically negligible. The thesis does not claim that the proposed architecture improves lumbar stenosis grading to a degree that would alter practice, and no result here supports such a claim. What the chapter establishes is the evaluation: a ladder in which each mechanism is tested against a control matched in capacity, and in which most of the proposed mechanisms are shown to contribute nothing measurable. That finding does not depend on the size of the gain, and it is the contribution that survives the observation made in this subsection.

4.5.7 What the Nulls Establish

Reporting a comparison as "not significant" states only that an effect was not detected. It is the weakest available form of a null, and it invites the reader to supply the explanation that the study was too small. The paired interval already bounds the effect, and stating that bound converts an absence into a measurement.

Table 4.8 gives, for each comparison that did not separate, the far edge of its 95% interval: the largest effect still compatible with the data.

These are *confidence* bounds and are deliberately not called equivalence bounds. A formal equivalence test requires a smallest effect of interest to be specified in advance and tested against, for example by two one-sided tests. No such margin was pre-specified here, and choosing one now, with the results already seen, would be post hoc. The bounds below therefore say what the data are compatible with; they do not assert equivalence to zero.

Table 4.8: Upper compatibility bounds for the comparisons that did not separate. Each figure is the far edge of the 95% confidence interval: the largest effect still compatible with the data, in QWK and as a percentage of the E0 baseline of 0.7270. These are confidence bounds, not the outcome of an equivalence procedure.

Comparison	Effect is at most	As % of baseline
Gated residual	0.0064	0.89%
Relational message passing	0.0074	1.02%
Anatomical topology vs shuffle (RQ1)	0.0076	1.05%
Cross-sequence self-supervision (RQ2)	0.0096	1.32%
Modality dropout	0.0091	1.25%
Disease-conditioned routing (RQ3)	0.0112	1.55%

The claims available are therefore considerably stronger than the absence of a significant result. Anatomically correct graph topology contributes at most 1.05% of baseline performance over a degree-preserving random shuffle. Anatomically aligned cross-sequence self-supervision contributes at most 1.32% over the same architecture without it. Disease-conditioned routing contributes at most 1.55% over fixed fusion.

These bounds should be read against what is known about the reference standard, but carefully. Section 2.4.6 reports inter-reader κ between 0.49 and 0.73 depending on compartment. Those figures come from different populations, different label processes and different adjudication regimes, and are not necessarily the same statistic computed here, so they cannot be treated as a numerical floor below which a difference is unmeasurable. The defensible statement is weaker and still sufficient: known inter-reader variability limits how strongly small model differences should be interpreted against this reference standard. An effect bounded near 1% of baseline is being asked to show itself against labels whose own reproducibility is imperfect and unquantified in this cohort.

4.5.8 The Seven-Seed Extension

The campaign was extended from three seeds to seven, and the extension is reported rather than absorbed, because it changed two verdicts.

Typed heterogeneous edges moved from +0.0123 on 3/3 seeds with $p_{\text{FDR}} = 0.054$ — narrowly missing correction — to +0.0093 on 7/7 seeds with $p_{\text{FDR}} = 0.009$. The point estimate fell while the between-seed standard deviation fell further, and the comparison crossed the threshold. The ordinal head moved the same way, from 0.054 to 0.009.

Anatomical topology moved in the opposite direction: from +0.0051 on 2/3 seeds to +0.0020 on 4/7. Cross-sequence self-supervision also failed to strengthen, standing at +0.0030 on 4/7 after both the extension and the initialisation correction of Section 4.7; its three-seed value was +0.0051 on 2/3, though that figure came from the confounded configuration and is not directly comparable.

That divergence is the reason the extension is worth reporting. The same four additional runs that promoted two comparisons to significance made two others weaker. Insufficient power is therefore not available as a blanket explanation for the nulls in this chapter: the instrument demonstrably resolved effects at this scale, and the effects it did not resolve moved away from significance as evidence accumulated rather than toward it.

4.5.9 Clinical Error Structure

Aggregate agreement conceals the errors that matter clinically. A Severe→Normal/Mild confusion may withhold treatment from a patient who needs it, and the cost matrix of Section 3.13.3 was chosen to suppress that direction specifically.

Table 4.9: Clinical error structure over seven seeds. "Distance ≥ 2" is the rate of predictions two or more grades from the reference.

Configuration	Severe recall	Severe → Normal	Distance ≥ 2
E0 single sequence	61.1%	5.7%	0.526%
E1 multi-sequence	61.1%	4.9%	0.500%
E2 + disease routing	60.7%	4.7%	0.414%
E3 + modality dropout	61.4%	4.6%	0.420%
E4 + ACSSL	59.1%	4.8%	0.401%
E5 + homogeneous graph	59.0%	4.1%	0.389%
E6 + typed graph	62.9%	4.1%	0.375%
E6 shuffled edges	59.3%	3.7%	0.330%
E6 ungated residual	**65.1%**	4.3%	0.500%
E7 + ordinal, cost-sensitive	63.1%	**3.8%**	**0.344%**

Severe→Normal errors fall from 5.7% at E0 to 3.8% at E7, a relative reduction of a third, while recall on Severe targets rises from 61.1% to 63.1%. Distant errors fall by a comparable margin, from 0.526% to 0.344%. The system does not purchase aggregate agreement by trading away the errors clinicians care most about; both move in the desired direction together.

4.5.10 The Opposite Error, and Severe Precision

A reduction in Severe→Normal is only meaningful if it is not bought by predicting Severe more often. Section 3.13.3 makes that condition explicit, and a model that graded every target Severe would post an excellent Severe→Normal rate while being useless. The opposite errors and the precision are therefore reported beside it, which is what makes the claim falsifiable.

The condition holds. Moving from E0 to E7, Severe precision rises by +3.69 percentage points (95% CI $[+0.05, +7.34]$) while Severe recall rises by +2.01 ($[-3.94, +7.96]$);

Table 4.10: Both directions of the severe error, with Severe precision, averaged over seven seeds. Computed from the saved test predictions by `implementation/make_figures.py`.

Configuration	Severe recall	Severe precision	Severe →Norm	Normal →Sev	Moderate →Sev
E0 single sequence	61.1%	61.3%	5.71%	0.27%	11.85%
E5 homogeneous graph	59.0%	65.3%	4.13%	0.21%	9.54%
E6 typed graph	62.9%	64.5%	4.09%	0.19%	10.89%
E7 ordinal, cost-sensitive	63.1%	65.0%	3.80%	0.17%	10.57%

Severe→Normal falls by -1.90 ($[-3.98, +0.18]$), and both over-grading errors fall as well — Normal→Severe by -0.10 ($[-0.23, +0.03]$) and Moderate→Severe by -1.28 ($[-4.17, +1.60]$). E7 predicts Severe 387 times against E0's 400: it grades Severe slightly *less* often while catching more of the true cases. The reduction in under-grading is not purchased with indiscriminate over-grading.

Four of those five intervals include zero, and the table should be read as consistency of direction across five related quantities rather than as five independent findings. Intervals are Student-t over seven seeds, which is the same across-seed treatment used for the primary comparisons; no correction is applied across these five, and none of them belongs to the pre-specified family.

One complication should be stated rather than smoothed over. The ungated variant achieves the highest Severe recall of any configuration at 65.1%, and simultaneously carries the joint-highest rate of distant errors at 0.500% against E7's 0.344%. Greater sensitivity bought with more badly misplaced predictions is precisely the trade the cost matrix exists to refuse, and E7 makes it in the intended direction.

4.5.11 Calibration

The expected calibration error recorded for each run is the *uncalibrated* test value; temperature scaling is fitted on the validation partition and its test metrics are stored separately. The distinction is material, because the uncalibrated figures alone would suggest the ladder becomes progressively worse calibrated, and the correction the protocol actually applies reverses that.

Temperature scaling roughly halves ECE at every rung, and every fitted temperature exceeds 1, so every configuration is overconfident before correction.

Two observations run against the system and are reported for that reason. E5, the homogeneous graph, is by a wide margin the worst calibrated configuration at 0.1024 uncalibrated with a fitted temperature of 2.019; it is also the rung that fails to improve accuracy, and its overconfidence is a second symptom of the same failure. And the accumulating ladder does not improve calibration: E7 at 0.0245 is worse than E0 at 0.0130, so the gains in agreement come with less reliable probabilities. RQ5 asked whether

Table 4.11: Expected calibration error before and after temperature scaling, and the fitted temperature, over seven seeds.

Configuration	ECE uncalibrated	ECE calibrated	Temperature
E0 single sequence	0.0293	0.0130	1.161
E1 multi-sequence	0.0214	0.0129	1.131
E2 + disease routing	0.0428	0.0168	1.601
E3 + modality dropout	0.0326	0.0145	1.200
E4 + ACSSL	0.0194	**0.0101**	1.136
E5 + homogeneous graph	0.1024	0.0397	2.019
E6 + typed graph	0.0426	0.0209	1.422
E7 + ordinal, cost-sensitive	0.0508	0.0245	1.306

calibrated uncertainty could support selective prediction. On this evidence the agreement half of that question is answered affirmatively and the calibration half is not.

4.6 RQ1 — Relational Disease Modelling

Does representing lumbar disease targets as a typed heterogeneous graph improve grading over independent target heads, an ordered five-level transformer and a homogeneous graph, and are any gains concentrated where single-target evidence is weakest?

RQ1 divides into two claims, and the evidence separates them.

4.6.1 Typed Structure Helps

A typed heterogeneous graph outperforms a homogeneous one by +0.0093 QWK ([+0.0027, +0.0156]), and does so on **every one of seven seeds**. The comparison survives false discovery rate correction at $p_{\mathrm{FDR}} = 0.009$, with $d = 1.06$.

This is one of the two verdicts the seven-seed extension changed. At three seeds the same comparison stood at +0.0123 with $p_{\mathrm{FDR}} = 0.054$, narrowly outside correction, and the power analysis then indicated roughly ten seeds would be needed. Seven sufficed, because the additional runs reduced the between-seed standard deviation faster than they reduced the point estimate.

Relational message passing alone does not help: E5 against E0 is +0.0002 ([−0.0073, +0.0074], $p_{\mathrm{FDR}} = 0.967$), which is as close to exactly nothing as this study measures. The benefit appears only once relations are typed. Message passing over an untyped graph is, on this evidence, worth precisely its parameter count and no more.

4.6.2 Anatomical Topology Does Not

H3 predicted that "performance under a shuffled-edge control should deteriorate if the topology itself matters". It does not deteriorate meaningfully. E6 exceeds its shuffled control by +0.0020 QWK ($[-0.0038, +0.0076]$, 4/7 seeds, $p_{\mathrm{FDR}} = 0.688$), and the paired interval bounds any advantage at 1.05% of baseline performance. The control preserves node count, edge count and degree, so the two models are matched in capacity and differ only in which targets are connected to which.

The conclusion is therefore sharper than the hypothesis anticipated. *Typing the relations carries information; the anatomical identity of the relations does not.* What the graph supplies is a structured parameterisation of interaction between targets, not the specific adjacency of levels, compartments and bilateral pairs that motivated it.

4.6.3 The Gating Ablation

Chapter 3 designates the ungated residual a mandatory ablation. The gated and ungated formulations are indistinguishable (+0.0014, 3/7 seeds, $p_{\mathrm{FDR}} = 0.713$), with any difference bounded at 0.89% of baseline — the tightest bound in the study. The gate can be removed without cost.

Answer to RQ1. Partly. Typed relational structure improves grading over both independent heads and a homogeneous graph. The anatomical content of that topology is not what produces the improvement, and the gating mechanism contributes nothing.

4.7 RQ2 — Anatomically Aligned Self-Supervision

> *Does cross-sequence pretraining based on DICOM-defined anatomical correspondence improve grading, label efficiency and transfer?*

A confound was found in this comparison and corrected before the results below were produced. The pretext model built its encoders from randomly initialised weights while the supervised ladder initialises from ImageNet, and loading the pretrained encoders replaces every encoder tensor. E4 was therefore *random* $\rightarrow$ *ACSSL* $\rightarrow$ *supervised* measured against E3's *ImageNet* $\rightarrow$ *supervised*, which confounds the pretext task with the initialisation it starts from and handicaps the arm under test. Pretraining and all seven E4 runs were repeated from ImageNet initialisation. The confounded artefacts are retained rather than discarded, and the episode is recorded in Section 4.12. All figures in this section come from the corrected runs.

Pretraining itself succeeded as an optimisation: validation InfoNCE reached 1.0771 against a chance value of 3.4657. Correct initialisation made the pretext task substantially easier to solve — from random weights the same schedule reached only 1.5162 — so the encoders demonstrably learned to match corresponding anatomy across sequences. The question is whether that representation is worth anything downstream. Three independent probes say it is not.

Accuracy. E4 against E3 is $+0.0030$ QWK ($[-0.0038, +0.0096]$, 4/7 seeds, $p_{\text{FDR}} = 0.575$, $d = 0.24$), with any benefit bounded at 1.32% of baseline. Distinguishing an effect this size from zero at conventional power would require roughly 134 seeds. Correcting the initialisation moved the estimate by $+0.0010$, an order of magnitude smaller than the between-seed standard deviation, and changed no verdict.

Cross-sequence robustness. This is the probe on which anatomical self-supervision should be strongest, because a representation that has learned one anatomy across three acquisitions ought to lose less when one acquisition is withheld. Under the controlled input ablation of Section 4.8.2, run here over all seven seeds, adding ACSSL changes reliance on the annotated sequence by -0.0061 ± 0.0405 on 4/7 seeds ($t = -0.40$) — indistinguishable from zero. Ordinary modality dropout, a far simpler intervention, reduces that same reliance by -0.0758 ± 0.0375 on 7/7 seeds ($t = -5.34$): roughly twelve times the effect, in the same direction, on every seed.

This comparison is worth stating plainly, because it is the sharpest result in the section. The mechanism ACSSL was designed to provide is delivered instead by an augmentation that requires no pretext task, no correspondence geometry and no additional training stage.

Attention. Under the attribution analysis of Section 4.11, E4 concentrates $2.69\times$ chance on the annotated target against E3's $2.62\times$, a difference of $+0.069 \pm 0.410$ on 2/3 seeds — a spread six times the effect. ACSSL neither sharpens nor degrades where the model attends.

Answer to RQ2. No. H1 is rejected on all three axes tested. The pretext task is learnable, is learned, and is learned better than it was before the initialisation was corrected; the representation it produces confers no measurable downstream benefit at this data scale. Label efficiency and external transfer were not separately evaluated (Section 4.9), so the rejection is scoped to internal grading and robustness.

One post hoc observation, reported as such. E4 has the lowest between-seed standard deviation of any rung on the ladder: 0.0030 against E3's 0.0125, a variance ratio of $F(6,6) = 17.8$, $p = 0.003$. ACSSL does not raise the mean but appears to make training markedly more reproducible. This was not part of the pre-specified comparison family, is not included in the false discovery rate correction, and rests on variance estimates from seven seeds, which are unstable. It is recorded as a direction for future work rather than as a result, and it would need pre-specification and its own replication before it could be claimed.

4.8 RQ3 — Disease-Conditioned Modality Routing

Can one model learn target- and patient-dependent sequence weights while remaining robust to missing and quality-degraded MRI sequences?

4.8.1 The Learned Allocation

The router learns the clinically expected allocation with complete consistency. Across all fifteen runs in which a gate is present, foraminal targets allocate most weight to sagittal T1, central canal targets to sagittal T2, and subarticular targets to axial T2. The pattern reproduced 15/15 without being encoded as a prior.

Chapter 3 is explicit that this is insufficient: "the gate weights are model allocations, not causal estimates of sequence importance".

4.8.2 Controlled Input Ablation

The prescribed intervention withholds each sequence at test time and measures the per-condition loss. Withholding the sequence a condition is graded from is severely damaging — between 0.06 and 0.35 QWK, an order of magnitude larger than any difference in Table 4.4 — while withholding the others is near-neutral or slightly beneficial. Agreement between allocation and intervention is 5/5 conditions on every seed.

Taken alone, this reads as strong causal support for RQ3. It is not, and the control is what reveals why.

E1 has no router. It fuses the three sequences with fixed weights. It produces the same 5/5 pattern with drops as large or larger. The dependence is a property of the dataset: each condition is *annotated* on one particular sequence, so withholding that sequence removes the only crop actually centred on the pathology. Any model suffers, routed or not.

Isolating each step's effect on mean reliance on the annotated sequence:

The input ablation was re-run over all seven seeds when the initialisation correction of Section 4.7 required E4 to be rebuilt; the figures below supersede the three-seed version reported previously.

Step	Δ reliance	Seeds lower
Add the router	$+0.0126 \pm 0.0298$	1/7
Add modality dropout	$\mathbf{-0.0758 \pm 0.0375}$	**7/7**
Add ACSSL	-0.0061 ± 0.0405	4/7

The router does not reduce reliance on the annotated sequence. It moves it slightly the wrong way, on six of seven seeds. Its gate weights track the annotation structure of the dataset without creating any causal dependence beyond what fixed fusion already possesses.

The comparison in that table is the more informative result. Modality dropout, which is a one-line augmentation requiring no pretext task, no correspondence geometry and no additional training stage, reduces reliance on every seed and by roughly twelve times the effect of the anatomically aligned self-supervision that was designed to produce exactly this property. The extension from three seeds to seven strengthened this contrast rather than weakening it.

Answer to RQ3. The first half is yes and the second half is no. One model does learn target-dependent sequence weights, and those weights are clinically sensible. But they do not improve grading (+0.0053 QWK, 5/7 seeds, $p_{\mathrm{FDR}} = 0.131$, bounded at 1.55% of baseline) and they do not improve robustness to missing sequences. H2 predicted that routing would "degrade less under missing-sequence experiments than fixed concatenation"; it does not. The graceful degradation that the system does exhibit is attributable to modality dropout, a standard regulariser that Chapter 1 explicitly does not claim as original.

4.9 RQ4 — Cross-Institutional Transfer

Not yet answered. The zero-shot transfer experiment on the Rizgary Hospital cohort has not been executed. Reporting its status honestly is required by the protocol changelog commitment of Section 1.8.

Preparatory work is complete and is documented in `thesis/chapter4/`: the cohort has been surveyed, radiology reports parsed into structured labels, imaging and report records reconciled, and a grading worklist produced. Of the cases with both a report and usable imaging, sixteen carry conflicts between the report-derived label and the spreadsheet reference that require adjudication by a reader before any transfer number can be computed.

Two blockers stand between the present state and an RQ4 result. First, the sixteen conflicting cases await regrading. Second, the cohort DICOMs carry unredacted patient identifiers, and the existing de-identification routine keys on a `PatientID` field that is not unique in this cohort — 45 distinct values across 346 cases — so applying it would merge roughly eight patients into each pseudonym. Neither blocker is technical in the sense of requiring new method; both require decisions and reader time.

H4 and H5 are therefore untested. They are retained here rather than removed, in keeping with the requirement that components not supported by executed experiments be reported as such rather than deleted from the research history.

4.10 RQ5 — Clinically Asymmetric Error

Do ordinal and cost-sensitive objectives reduce clinically consequential distant errors, particularly Severe→Normal/Mild?

Yes, and this is the second most reliable effect in the study. Adding the ordinal and cost-sensitive head improves QWK by +0.0082 ($[+0.0025, +0.0143]$, $p_{\mathrm{FDR}} = 0.009$, $d = 1.97$ — the largest standardised effect in the study) on 7/7 seeds, with the smallest between-seed standard deviation of any comparison (0.0042), so that three seeds suffice to detect it (Table 4.6).

The clinically material quantity moves in the intended direction. Recall on Severe targets rises from 61.1% at E0 to 63.1% at E7, against 62.9% at E6 under categorical cross-entropy, and Section 4.5.10 shows the improvement is not bought by predicting Severe more often. The objective is doing what it was selected to do rather than trading distant errors for aggregate accuracy.

H6 anticipated that these objectives "may reduce distant errors even if they do not improve aggregate macro-F1". In the event both improved.

4.10.1 Which of the Two Interventions Is Responsible

E7 changes two things at once: the head becomes ordinal, and the objective gains the asymmetric cost matrix. Its +0.0082 is therefore a joint effect, and attributing it requires running the two separately. Two further variants complete a 2×2 design over head and objective, each over the same seven seeds.

Table 4.12: Separating E7's two interventions. Every cell is the same architecture, differing only in the head and whether the cost matrix is applied.

	Categorical head	Ordinal head
No cost matrix	0.7364 (E6)	0.7407
Cost matrix	0.7332	**0.7447** (E7)

Neither intervention reproduces the joint effect on its own. The ordinal head alone gives $+0.0042 \pm 0.0075$ over E6 on 5/7 seeds ($t = 1.51$). The cost matrix alone is *negative*: -0.0032 ± 0.0074 on 2/7 seeds ($t = -1.14$). Added together the two main effects come to +0.0010, against a measured joint effect of +0.0082.

The remainder is interaction, $+0.0072 \pm 0.0122$ on 6/7 seeds, some 87% of the total. A reading consistent with this is that the cost matrix supplies a signal about the direction of error that a categorical head, whose three logits carry no ordering, cannot use and is mildly harmed by, while the ordinal head's monotone thresholds give that signal somewhere to act.

That reading is not established here. The interaction term carries $t = 1.57$ on seven seeds; roughly twenty-three seeds would be required to resolve it, and the individual main effects would need twenty-five and forty-two. What the decomposition establishes is negative and still useful: the joint effect is real and reproduces on every seed, and *neither component alone accounts for it.* Chapter 5 previously suggested the cost matrix was the more likely source; that suggestion is withdrawn, since the cost matrix in isolation reduces performance.

Answer to RQ5. Yes, for the combined objective, with the attribution between its two parts unresolved. This is noted with some irony: the objective function, explicitly disclaimed in Section 1.11.5 as a supporting component and not an original method, is a larger and far more reliable contributor to final performance than two of the three proposed novel mechanisms.

4.11 Why the Structural Priors Had Little to Add

The results above establish that most of the proposed mechanisms do not improve grading. A null result with a mechanism is worth more than a null result with a p-value, and this section supplies the mechanism.

What this section can and cannot support. Section 3.23 treats gradient-weighted saliency as a gross sanity check on where evidence is drawn from, not as proof of how a network reasons, and that caution is retained here. Grad-CAM is a gradient-weighted summary of one layer's activations; it is known to be sensitive to the layer chosen, to be only loosely related to what a model would do under intervention, and to produce plausible maps even for models that have learned little. Nothing in this section is an intervention, so nothing in it establishes causation.

What the measurements below support is the weaker and sufficient statement that the evidence used by a plain single-sequence encoder is already concentrated near the annotated target, which is *consistent with* the explanation that the structural priors had little localisation work left to do. The controls make that reading defensible rather than decorative: an architecture-matched untrained model is measured as a floor, so the concentration is shown to be learned rather than an artefact of centre bias, and the same probe is applied identically at every rung. It remains supporting evidence for the nulls established statistically above, and is not offered as their proof.

4.11.1 Method

Each crop is centred by construction on its target's annotated coordinate, so the pathology lies at the middle of the frame and attention on it can be measured rather than described: the fraction of Grad-CAM mass falling inside a central disc of 15 mm radius at the 60 mm field of view.

That disc covers 19.7% of the frame, which is what a uniform map would score. Because convolutional networks are centre-biased irrespective of what they have learned, the identical architecture with untrained weights is evaluated as a floor; it scores 0.204.

Attribution depth was fixed on resolution grounds before results were compared. A residual network downsamples by 32, so at these 128 pixel crops the final block is 4×4 — one cell spanning 15 mm, too coarse to resolve the disc. The penultimate block at 8×8 matches the 7×7 resolution at which Grad-CAM is conventionally applied to 224 pixel inputs, and is used throughout. All three depths are reported in the appendix so that the dependence is visible rather than taken on trust.

4.11.2 Result

Configuration	CAM mass in disc	vs. untrained	vs. E0
E0 single sequence	0.490 ± 0.058	+0.286	—
E1 multi-sequence	0.578 ± 0.073	+0.374	+0.089 (3/3)
E2 + routing	0.580 ± 0.063	+0.376	+0.090 (2/3)
E3 + modality dropout	0.515 ± 0.071	+0.311	+0.026 (2/3)
E4 + ACSSL	0.529 ± 0.063	+0.325	+0.040 (2/3)
Untrained (same architecture)	0.204	—	—
Uniform map (disc area)	0.197	—	—

Every trained configuration places between 2.3 and 2.9 times the chance share of its attention on the annotated centre. Critically, **E0 already reaches** 0.490 — a plain single-sequence convolutional network with no routing, no self-supervision and no graph.

This is the explanation the null results require. The structural priors of RQ1–RQ3 were motivated by the premise that anatomical context is needed to find and interpret the lesion. The encoder locates it without them. There is little left for a prior to contribute, and the ladder measures exactly that little.

4.11.3 A Dissociation Worth Noting

Multi-sequence input is the only step that improves attention on all three seeds (+0.089 against E0), and it is simultaneously the step with the most negative accuracy effect (−0.0055 QWK). Additional sequences sharpen *where* the model looks without improving

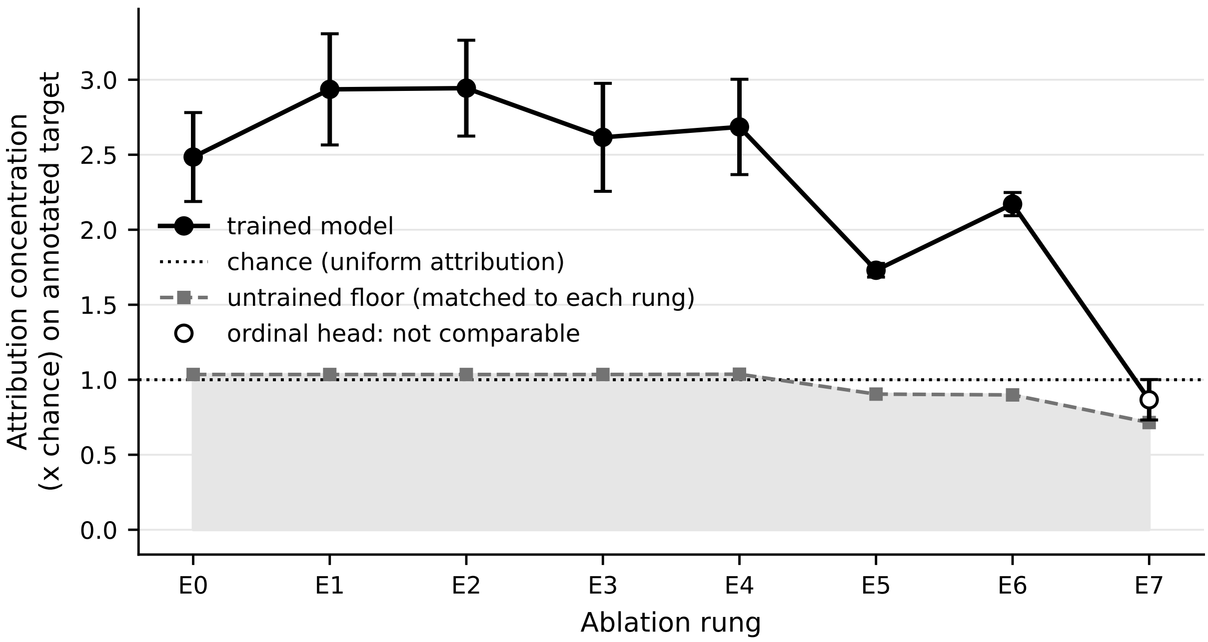

Figure 4.5: Grad-CAM mass falling inside the annotated target's disc, expressed as a multiple of the share a uniform map would score. The dashed line is the untrained floor *matched to each rung's own architecture*: it is 1.03 to 1.04 at E0–E4 but falls to 0.90 once the graph is added and to 0.71 at E7, so a single floor drawn across the whole ladder would misstate both ends. Every rung lies above its own floor. E7 is drawn hollow because its value is not on the same footing as the rest: Grad-CAM differentiates the predicted class logit, and the cumulative-link head emits two threshold logits rather than three class logits, so the quantity differentiated is not the same one. Its apparent fall is an artefact of that change rather than evidence that the strongest model stops attending to the target. Generated by `implementation/make_figures.py`.

what it concludes. Attention concentration and grading accuracy are not the same quantity, and this study can separate them.

4.12 Threats to Validity

4.12.1 Errors Detected in the Analysis

Five analyses produced incorrect conclusions before controls were added. Each is recorded because each would have been reported as a result, and because four of the five erred in the direction of the hypothesis under test. The fifth erred against it, which is worth separating: a defect that suppresses an effect is as much an error as one that manufactures it, and is less likely to be looked for.

Per-seed intervals reversed sign. Described in Section 4.5. Two comparisons yielded opposite, individually significant results on different seeds.

A mechanism check without a control read as proof. The 5/5 agreement between routing weights and intervention (Section 4.8.2) appeared to establish RQ3 causally, with effects an order of magnitude larger than any ladder difference, until a router-free model produced the same pattern.

An attribution probe measured the wrong encoder. The first implementation hooked the first encoder unconditionally, which for multi-sequence configurations is the sagittal T1 encoder — including on subarticular targets graded from axial T2. It reported that no configuration improved on E0 (-0.036, 1/3 seeds). Selecting the encoder that read each target's annotated sequence reverses the sign to $+0.089$ on 3/3 seeds.

An initialisation confound handicapped the arm under test. The self-supervised pretext model constructed its encoders from randomly initialised weights, while every supervised rung initialises from ImageNet, and loading the pretrained encoders overwrites all 366 encoder tensors. E4 was therefore *random → ACSSL → supervised* compared against E3's *ImageNet → supervised*: the pretext task was confounded with the initialisation it began from, and the comparison RQ2 asks was never the one performed. This is the error that ran against the hypothesis rather than with it, and it survived a full seven-seed campaign before being found by an external review of the code rather than by any check in the analysis.

Pretraining and all seven E4 runs were repeated from ImageNet initialisation. The pretext task was solved substantially better — validation InfoNCE 1.0771 against 1.5162 from random weights — and the downstream verdict did not change: $+0.0030$ against $+0.0020$ QWK, still 4/7 seeds, still far from significance. One subsidiary claim was nonetheless false and has been withdrawn: E4 was previously reported as the only configuration whose attention fell below the single-sequence baseline, which was an artefact of the handicap. It concentrates 2.69× chance, above E0's 2.48×. The confounded checkpoints and results are retained rather than deleted, so the two can be compared.

A three-seed probe produced a clean but spurious positive. When the corrected E4 was first compared with E3 on the cross-sequence robustness probe, which had been run on seeds 42–44 only, ACSSL appeared to reduce reliance on the annotated sequence by -0.0425 ± 0.0227 on 3/3 seeds — about 72% of the effect of modality dropout, in the right direction, with a plausible mechanism. Extending the same probe to all seven seeds gives -0.0061 ± 0.0405 on 4/7 ($t = -0.40$). The three-seed result was noise. It is recorded because it was not reported as a finding only because the seeds were run before writing, and because it is the same failure the per-seed bootstrap produced earlier in this chapter, arriving by a different route.

4.12.2 Limitations

Seed count. Seven seeds resolve the three comparisons that survive correction and do not resolve the others. Table 4.6 gives the requirements: 134 seeds for RQ2, 52 for RQ3 and 205 for anatomical topology against its shuffled control. Those are not deficiencies of the campaign so much as statements about the size of the effects, which is the finding.

Single benchmark. All results are internal to LumbarDISC. RQ4 is unanswered, so external validity is untested and no claim of generalisation is made.

Point annotations. Centre concentration presumes the annotated coordinate marks the lesion. RSNA coordinates come from single readers; a systematic offset would depress the measure uniformly across configurations.

Attribution method. Grad-CAM is one attribution technique with known failure modes. The untrained floor makes the comparison meaningful, but absolute values are method-dependent.

The input ablation covers E0–E4 only. Those rungs reach the fusion stage directly, whereas E5–E7 route through the graph. The routing claim of RQ3 is in any case cleanest to test at E2, with the graph out of the path.

Attribution on the graph rungs measures a different quantity for E7. Attribution was extended to E5–E7, which required one backward pass per node rather than one per batch: message passing couples the nodes, so a summed backward pass reports what an image contributed to the *whole graph* rather than to its own target, understating concentration by 0.173 on the nodes examined. With that corrected, E5 and E6 reach 0.341 and 0.428 against untrained floors near 0.18.

E7 pools to 0.171, below chance, and that figure is a prevalence artefact rather than a property of the model. Its ordinal head explains an accumulated severity score, so asking what makes a target look severe highlights the *absence* of evidence on a target with no pathology, and 77.33% of targets are Normal/Mild. Split by grade, E7 gives 0.098, 0.269 and 0.415 for Normal/Mild, Moderate and Severe, while E6 is flat at 0.429, 0.419 and

0.448. On targets that carry pathology the full system localises as well as the rung below it. E7's pooled value is therefore not comparable with the other rungs and is not presented as though it were.

4.13 Summary of Answers

RQ	Verdict	Basis
RQ1	Partly	Typed edges +0.0093, **7/7 seeds**, $p_{\text{FDR}} = 0.009$. Anatomical topology does not beat a degree-preserving shuffle (+0.0020, 4/7), bounded at 1.05% of baseline.
RQ2	No	+0.0030, 4/7 seeds, bounded at 1.32%. No robustness benefit: modality dropout reduces sequence reliance twelve times more, on 7/7 seeds. Attention unchanged. Rejected on three axes, after correcting an initialisation confound that had handicapped the arm under test.
RQ3	Mechanism yes, benefit no	Allocation replicates 15/15 but a router-free model matches it under intervention. +0.0053 QWK, 5/7 seeds, bounded at 1.55%.
RQ4	Not executed	Cohort preparation complete; blocked on reader adjudication, de-identification, and the absence of localisation coordinates.
RQ5	Yes	+0.0082, **7/7 seeds**, $p_{\text{FDR}} = 0.009$, $d = 1.97$ — the largest standardised effect in the study. Severe→Normal errors fall 5.7% → 3.8%.
System	Yes	+0.0177 QWK $[+0.0097, +0.0260]$, **7/7 seeds**, $p_{\text{FDR}} < 0.001$.

The thesis set out to determine whether anatomical structure, imposed explicitly on a multi-sequence grading model, improves degenerative lumbar classification. The answer this chapter supports is that it largely does not, that the convolutional encoder already performs the localisation such structure was meant to supply, and that the components which do improve performance are relational typing and a clinically motivated objective rather than anatomical correspondence or disease-conditioned routing.

Two qualifications belong with that summary. The improvement is bounded by a reference standard on which expert readers agree at κ between 0.49 and 0.73, so its ceiling is set by the labels rather than by the architecture. And the accumulating ladder improves agreement without improving the reliability of its probabilities: calibrated ECE is worse at E7 than at E0.

Chapter 5 considers what follows for the design of anatomy-aware medical imaging models, and why mechanisms that are individually well motivated and demonstrably learnable can nonetheless fail to pay.

Chapter 5

Discussion and Conclusions

5.1 What This Study Found

The thesis asked whether explicit anatomical structure, imposed on a multi-sequence grading model, improves automated classification of lumbar degenerative disease. The answer is that it largely does not, and the more useful part of the answer is *why*.

The complete system improves on a single-sequence baseline by +0.0177 QWK (95% CI $[+0.0097, +0.0260]$), on every one of seven seeds and after correction for multiple comparisons. That margin is statistically detectable and clinically negligible: it amounts to 42 additional correct gradings out of 7,310, one per seven patients, against an inter-reader span of roughly 0.24 in kappa (Section efsec:results-practical). The accuracy is not the contribution. But the components credited with that improvement in the original design are not the components responsible for it. Anatomically aligned self-supervision produced no measurable benefit on any of three axes, and continued to produce none after an initialisation confound that handicapped it was found and corrected. Disease-conditioned routing learned the clinically expected sequence weights and produced no benefit from having learned them. Relational typing helped; the anatomical identity of the relations did not.

The evidence for those statements is stronger than a set of failures to reject. Each unsupported mechanism carries an upper confidence bound rather than a bare null: anatomical topology contributes at most 1.05% of baseline performance over a degree-preserving shuffle, cross-sequence self-supervision at most 1.32%, and disease-conditioned routing at most 1.55%. And the campaign was extended from three seeds to seven precisely to test whether these were power failures. It was not: the same four additional runs that carried typed edges and the ordinal head across the corrected threshold moved anatomical topology and self-supervision further away from it.

Two of the three proposed mechanisms are therefore rejected, one is half-supported in a narrower form than proposed, and the largest standardised effect in the study — $d = 1.97$, larger than the full-system comparison it forms part of — belongs to an objective function that Chapter 1 explicitly declined to claim as original.

This chapter argues that these results are informative rather than merely disappointing, offers two mechanisms for them, situates them against the literature of Chapter 2, and states plainly what the study cannot conclude.

5.2 Why the Anatomical Priors Did Not Pay

A null result without a mechanism invites the reader to supply their own, and the most available one—that the implementation was faulty—is the least interesting and the most damaging. Two mechanisms are supported by evidence gathered specifically to test them.

5.2.1 The Encoder Already Localises

The structural priors of RQ1–RQ3 share a premise: that anatomical context is needed to find and interpret the graded structure, and that supplying it explicitly will help. The attribution analysis of Section 4.11 tests that premise directly and finds it largely unmet.

Every trained configuration concentrates between 2.3 and 2.9 times the chance share of its attention on the annotated target. Decisively, E0—a plain single-sequence convolutional network with no routing, no self-supervision and no graph—already reaches 0.490 against an untrained floor of 0.204 and a uniform expectation of 0.197. The localisation the priors were designed to supply is largely accomplished before any of them is added.

This is consistent with Section 2.6.5, which assembles the evidence that localisation precedes classification. The present result sharpens that literature rather than contradicting it: localisation does precede classification, and on a benchmark that supplies the localisation as an annotated coordinate, a convolutional encoder given a tight physical crop performs the residual localisation unaided. Structure imposed above that encoder is competing for a job already done.

Section 1.9 anticipated this possibility in advance, stating that "if local evidence alone is sufficient, more complex mechanisms should not be retained". The finding is that local evidence is very nearly sufficient.

5.2.2 The Reference Standard Constrains the Headroom

Section 2.4.7 argues that reader reliability is a hard constraint on interpretation rather than a peripheral caveat. Lurie et al. report inter-reader κ of 0.73 for central canal stenosis, 0.58 for foraminal stenosis and 0.49 for subarticular stenosis [3]. A model trained on one reader's labels cannot be expected to exceed the agreement those readers achieve with each other.

The comparison must be made carefully. Quadratic weighted kappa credits near-misses that unweighted κ penalises fully, so the QWK values reported here are not directly comparable with Lurie's figures, and the two are measured on different cohorts under different protocols. The point is not that this system outperforms radiologists; nothing in this study supports that claim.

The point is about headroom. If the achievable band is bounded by a reference standard on which experts disagree at $\kappa \approx 0.5$ for the lateral compartments, then the margin available to any architectural refinement is narrow, and effects of the size proposed here—0.005 QWK—sit far inside the noise of the labels themselves. Chapter 2 warns that "a reported accuracy approaching unity on such a target should prompt scrutiny of the evaluation protocol rather than admiration"; the corollary is that failure to improve on such a target by half a percentage point warrants less alarm than it first appears to.

5.2.3 A Test of That Explanation, and Its Limits

Section 2.4.6 makes a checkable prediction: agreement is "systematically worse for the lateral compartments than for the central canal—the same ordering that later appears in automated model performance". If model difficulty tracks reader difficulty, the label-ceiling explanation gains support.

Table 5.1: Per-condition performance against reported inter-reader agreement. Reader κ from Lurie et al.; model QWK averaged over seeds 42–44, this analysis having been run before the campaign was extended. The two metrics are not directly comparable in absolute terms; only the ordering is being examined.

Condition	Reader κ	E0	E2	E4
Central canal	0.73	0.799	0.786	0.805
Left foraminal	0.58	0.659	0.623	0.644
Right foraminal	0.58	0.685	0.672	0.671
Left subarticular	0.49	0.758	0.749	0.767
Right subarticular	0.49	0.712	0.731	0.736
Central − lateral mean	+0.20	+0.095	+0.092	+0.100

The coarse prediction holds. Central canal stenosis is the easiest condition for the model, as it is for readers, and the central-versus-lateral gap runs in the same direction for both, though smaller in magnitude for the model.

The fine ordering does not hold, and the discrepancy is informative. Readers find *subarticular* stenosis hardest at $\kappa = 0.49$; the model finds *foraminal* stenosis hardest, while subarticular is its second-best condition. Model difficulty is therefore not a simple reflection of label noise.

A representational explanation fits the discrepancy. Foraminal targets are graded on sagittal T1, where left–right is a *through-plane* axis: the two sides are different slices rather than different in-plane positions. Subarticular targets are graded on axial T2, where left–right is in-plane and the lateral recesses are directly visible. The 2.5D representation used here—three adjacent slices treated as channels—captures in-plane structure well and through-plane structure poorly, which is exactly the axis on which foraminal grading depends. If correct, this locates the residual difficulty in the input representation rather than in the absence of anatomical priors, and it predicts that a volumetric or cross-plane-aware representation would help foraminal grading specifically.

That prediction is not tested here. It is offered as the most economical account of a dissociation the study did not set out to produce, and it is the clearest direction for subsequent work.

Two cautions apply to this section. Only five conditions exist, two pairs of which share a reported κ, so rank correlation between the two orderings has essentially no statistical power and none is claimed. And Lurie's cohort is not LumbarDISC; the comparison is indicative rather than a matched analysis.

5.3 What Was Planned, What Was Run, and What It Supports

Each research question was specified in Chapter 3 with a set of evidence, and in every case only part of that set was executed. A verdict on its own conceals that, so Table 5.2 separates the plan, the execution and the conclusion. No question receives a bare yes or no, because for none of them would that be honest.

Two entries deserve emphasis because they bound what the thesis may claim. RQ4 was not executed at all, so nothing here speaks to institutional transfer. RQ1's comparison against a sequence model was not built, so the graph is shown to beat a homogeneous graph and an unstructured baseline, not to beat the obvious alternative architecture for ordered levels.

5.4 The Components Individually

5.4.1 Anatomical Self-Supervision (RQ2)

The pretext task was learnable and was learned: validation InfoNCE reached 1.0771 against a chance value of 3.4657, so the encoders demonstrably learned to match corresponding anatomy across sequences. The representation that resulted conferred no measurable downstream benefit on accuracy, on robustness to a withheld sequence, or on where the model attends.

The natural objection is that the correspondence itself was wrong—that the pretext

Table 5.2: Research question completion. "Executed" refers to the evidence Chapter 3 specified for that question, not to whether some experiment was run.

RQ	Planned evidence	Executed	Verdict
RQ1	independent baseline, ordered-level Transformer, homogeneous graph, typed graph, topology control	Partial	Typed edges help (+0.0093, 7/7). Anatomical topology does not beat a degree-preserving shuffle. The Transformer comparator was not built, so "graph beats sequence model" is untested.
RQ2	full-label accuracy, label efficiency, pretraining comparators, transfer	Partial	Negative on all three axes executed, after correcting an initialisation confound. Label efficiency and transfer were not run, so the rejection is scoped to internal grading and robustness.
RQ3	missing-sequence robustness, corruption, learned allocation	Partial	The allocation pattern replicates on every run, but a router-free model reproduces it under intervention, so it belongs to the dataset rather than to routing. No accuracy benefit. Corruption was not run.
RQ4	zero-shot external transfer, adaptation	No	Unanswered. The external cohort exists and is characterised, and Section 5.6 states why it cannot answer the question as specified.
RQ5	asymmetric error, calibration, selective prediction	Partial	The cost-sensitive ordinal objective is the largest and most reliably reproduced effect in the study, and does not buy it with over-grading. Selective prediction was not evaluated.

task matched misaligned patches and the null is an artefact. That objection is answered. Section 4.3 verifies the DICOM correspondence externally, comparing two independent human annotations through the transform across 9,542 projections, of which 94% on the pretraining partition land within one slice of an independently annotated target at the same level. ACSSL was given correctly aligned data and still conferred nothing.

The result is best read alongside Section 5.2.1. Cross-sequence self-supervision is designed to teach an encoder that one anatomy appears under several acquisitions. If the supervised task already drives the encoder to localise that anatomy, the pretext task is teaching something the downstream objective supplies for free.

5.4.2 Disease-Conditioned Routing (RQ3)

The routing result is the study's clearest illustration of why a mechanism check requires a control. The router reproduces the clinically expected allocation in all fifteen runs in which a gate is present, and withholding the high-weight sequence for a condition is severely damaging—between 0.06 and 0.35 QWK, an order of magnitude larger than any difference in the ablation ladder. Presented alone, that is a compelling causal story.

It does not survive its control. E1, which has no router at all, exhibits the same dependence with drops as large or larger. The pattern belongs to the dataset: each condition is annotated on one particular sequence, so withholding that sequence removes the only crop centred on the pathology. Any model suffers, routed or not. Isolating the router's own contribution to sequence reliance gives -0.0010 ± 0.0382 across seeds 42–44, indistinguishable from zero. The grading effect over the full seven seeds is +0.0053, bounded by its interval at 1.55% of baseline.

Chapter 3 anticipated precisely this trap, stating that "the gate weights are model allocations, not causal estimates of sequence importance", and requiring the intervention that produced the control. The methodology's caution was correct and load-bearing.

What robustness the system does exhibit under missing sequences comes from modality dropout, which reduces reliance on the annotated sequence by -0.0758 ± 0.0375 on every one of seven seeds, against -0.0061 ± 0.0405 on 4/7 for the anatomically aligned self-supervision intended to produce that property — roughly twelve times the effect. That is a standard regulariser, disclaimed in Section 1.11.5 as not original, and it outperforms the mechanism designed for the purpose by more than an order of magnitude.

5.4.3 Relational Structure (RQ1)

RQ1 divides, and the division is the more interesting result. Typed heterogeneous edges improve on a homogeneous graph by +0.0093 QWK on every one of seven seeds, surviving correction at $p_{\text{FDR}} = 0.009$. Anatomically correct topology does not improve on a degree-preserving random shuffle: +0.0020, four seeds of seven, bounded at 1.05% of baseline.

The control preserves node count, edge count and degree, so the two models are matched in capacity and differ only in *which* targets are connected to *which*.

The conclusion is therefore narrower and sharper than H3 proposed. What the graph supplies is a structured parameterisation of interaction between targets—additional capacity, organised by relation type—rather than the specific adjacency of levels, compartments and bilateral pairs that motivated it. Relational typing carries information; anatomical adjacency does not.

The seven-seed extension sharpened both halves at once, which is the strongest form this claim could take. Typed edges went from narrowly missing correction to clearing it on every seed, while the anatomy comparison halved. A reader inclined to attribute the second result to insufficient power has to explain why the same four additional runs resolved the first.

Section 2.7.5 notes that relational and graph-based modelling of lumbar targets remains sparsely explored. The present result suggests that where it is explored, the shuffled-topology control should be mandatory, because a typed graph will outperform a homogeneous one for reasons that have nothing to do with the anatomy its edges are named after.

5.4.4 Ordinal and Cost-Sensitive Objectives (RQ5)

RQ5 is the study's most reliable positive result and it contradicts the strongest prior evidence. Section 2.9.4 reports that Niemeyer et al. tested soft kappa loss, ordinal cross-entropy and regression losses against standard cross-entropy for Pfirrmann grading across 7,948 discs with extensive hyperparameter optimisation, and that none improved classification performance [23].

Here the ordinal and cost-sensitive head improves QWK by +0.0082 on every one of seven seeds, surviving correction at $p_{\text{FDR}} = 0.009$. Its between-seed standard deviation of 0.0042 is the smallest of any comparison, which gives it the largest standardised effect in the study at $d = 1.97$.

The clinically material quantity moves further. Across the full ladder, Severe→Normal confusions fall from 5.7% at the baseline to 3.8% at E7 — a relative reduction of a third — while recall on Severe targets rises from 61.1% to 63.1% and predictions two or more grades from the reference fall from 0.526% to 0.344%. Aggregate agreement was not bought by trading away the errors clinicians care most about.

An earlier draft of this section attributed the difference from Niemeyer et al. to the cost matrix rather than to the ordinal structure, on the grounds that their comparison isolated ordinality while the objective tested here adds clinical asymmetry. That attribution was speculation, and the experiment that tests it has since been run. It does not survive.

Section 4.10.1 separates the two interventions. The cost matrix applied to a categorical head is *worse* than no cost matrix at all (-0.0032, 2/7 seeds), and the ordinal head alone does not reproduce the joint effect either ($+0.0042$, 5/7). The two main effects sum to $+0.0010$ against a joint effect of $+0.0082$, so almost all of it is interaction.

The explanation this is consistent with, though seven seeds cannot establish it, is that the two components are complementary rather than additive: the cost matrix expresses which *direction* of error is worse, and a categorical head with three unordered logits has no structure through which that preference can act, whereas the ordinal head's monotone thresholds do. On that reading the asymmetry of Section 2.9.6 is necessary but not sufficient, and needs an ordered representation to act on.

What can be said without qualification is narrower: the combined objective is the most reliably reproduced effect in the study, and neither of its two parts produces that effect alone.

It is worth stating plainly that the supporting component outperformed the novel ones, and that this is a result about where effort is best spent rather than an embarrassment.

5.5 Methodological Implications

Three analyses in this study produced positive conclusions that their controls subsequently overturned. All three erred in the direction of the hypothesis under test. They are reported in Section 4.12 and are summarised here because the pattern is more instructive than any individual case.

Per-seed confidence intervals understate variance. A paired bootstrap that resamples patients with the trained model held fixed excludes training stochasticity. On this problem that omission is decisive: one comparison returned $+0.0270$ with $p = 0.000$ on one seed and -0.0234 with $p = 0.000$ on another. Reported per seed, any claim in this study could have been supported by choosing a seed.

A mechanism check without a control reads as proof. The routing intervention produced 5/5 agreement with effects an order of magnitude above any ladder difference, and a router-free model reproduced it exactly.

An instrument can penalise the hypothesis it is measuring. The first attribution probe hooked the first encoder unconditionally, which for multi-sequence configurations reads the sagittal T1 encoder even on targets graded from axial T2. It reported that no configuration improved on the baseline; hooking the encoder that actually read each target reverses the sign.

A probe can measure the wrong quantity twice over. Extending attribution to the graph rungs required two further corrections. Backpropagating the sum of node logits let message passing carry gradient between nodes, so each map reported what an image contributed to the whole graph rather than to its own target, understating concentration by 0.173. And E7's ordinal head has no argmax to explain, so the scalar had to be chosen deliberately and its choice shown not to matter.

The general lesson is that in a study of this kind the controls are not supporting apparatus but the primary instrument. A degree-preserving shuffle, a router-free arm, an architecture-matched untrained floor and a label-shuffled pipeline control are what separate a mechanism from a coincidence, and each of them changed a conclusion here.

5.6 Limitations

Statistical power. The study is underpowered for single-component effects of the size observed. Table 4.6 converts each effect into the number of training seeds required to detect it: three for the full system and for the ordinal head, seven for typed edges, but 205 for anatomical topology and 134 for cross-sequence self-supervision.

That table describes this sample rather than the mechanisms. Both the effect and the variance entering each figure are estimated from seven seeds, so each requirement inherits their uncertainty and none of them licenses the conclusion that a component does nothing. The defensible reading is narrower: the observed effects are small relative to the observed between-seed variance, a conventional experiment would need implausibly many training runs to establish effects of this apparent magnitude, and the upper confidence bounds indicate that such effects would carry no clinical meaning even if they were established.

The word *underpowered* should nevertheless be used carefully in this chapter. The campaign was extended from three seeds to seven specifically to test that explanation, and the extension carried two comparisons across the corrected threshold while moving two others further from it. A study genuinely starved of power does not resolve half its comparisons and weaken the other half; it produces a spread of intermediate effects. Cohen's d here separates into 1.06–1.97 for the three significant comparisons and below 0.4 for everything else, with nothing in between.

External validity is untested. RQ4 was not executed. The complete system under institutional transfer, specified as rung E8, was not run. Every result in this thesis is internal to LumbarDISC, and no claim of generalisation to another institution, scanner population or reference standard is made.

A measurement taken after the main campaign sharpens what that absence means. The transfer cohort carries no annotated coordinates, so executing RQ4 requires localising the twenty-five targets automatically. Doing so on LumbarDISC itself, where the human annotation is available for comparison, costs −0.1636 QWK — 22.9% of the system's performance, on every one of nine runs. Segmentation identifies the five lumbar levels essentially perfectly (140 of 140 in the sample examined), and derived coordinates place 99–100% of targets inside the model's field of view; the loss comes from the derived point sitting some six millimetres off centre in a sixty-millimetre crop, which the attribution analysis of Section 5.2.1 explains, since the encoder concentrates its evidence near that centre. The failure is concentrated in the four lateral conditions, where placement must be accurate to a few millimetres; the central canal, sitting on the midline, loses only −0.05.

The consequence for RQ4 is not that transfer is harder than expected. It is that a transfer result obtained this way would be *uninterpretable*: domain shift and localisation error would be confounded, and the localisation term alone is an order of magnitude larger than any effect this thesis measures. Section 1.7 asks how much degradation is attributable to localisation versus grading, and Chapter 3 provides for a verified-localisation control. That provision is necessary rather than precautionary, and this measurement is what establishes it.

Single benchmark, single reference standard. LumbarDISC labels derive from a reading process whose inter-reader agreement is moderate for the lateral compartments. Conclusions about which mechanisms help are conditional on that reference standard.

Attribution on E7 measures a different quantity. The analysis now covers every rung. E5 and E6 confirm the mechanism on the graph path, reaching 0.341 and 0.428 against untrained floors near 0.18. E7's ordinal head explains an accumulated severity score rather than a class logit, so its pooled concentration is dominated by the 77.33% of targets that carry no pathology; split by grade it reaches 0.415 on Severe targets against E6's 0.448. The mechanism of Section 5.2.1 therefore holds wherever localisation is meaningful, but E7 cannot be placed on the same scale as the other rungs and is not.

Calibration does not improve with the ladder. Temperature scaling roughly halves expected calibration error at every rung, but the calibrated figure at E7 (0.0245) is worse than at E0 (0.0130). Agreement improves as components accumulate; the reliability of the probabilities does not. RQ5 asked whether calibrated uncertainty could support selective prediction, and on this evidence the agreement half of that question is answered and the calibration half is not. The homogeneous graph is a separate concern: at 0.1024 uncalibrated with a fitted temperature of 2.019 it is by a wide margin the most overconfident configuration, and it is also the rung that fails to improve accuracy.

No clinical outcome is measured. The study measures agreement with a retrospective reference standard. It does not measure diagnostic accuracy against a surgical or clinical ground truth, does not measure effect on reporting time, and does not support any claim of patient benefit.

The quality-control reader pass is outstanding. The ROI inspection sheets and the checklist of Section 4.3 have been generated; the reader adjudication that converts them into the promised inclusion/exclusion table has not been completed.

5.7 Implications and Future Work

For anatomy-aware model design. The most transferable finding is that a convolutional encoder given a tight, physically-defined crop already performs most of the localisation that anatomical priors are introduced to supply. Before adding structure above such an encoder, the marginal room for that structure should be measured—an attribution floor against an untrained network costs nothing and would have predicted these results in advance.

For relational modelling. A degree-preserving shuffle should be standard practice wherever a graph's topology is claimed to be meaningful. Without it, capacity and anatomy are confounded, and this study finds that the capacity explains the gain.

For the representation. The dissociation of Section 5.2.3—readers find subarticular hardest, the model finds foraminal hardest—points to the 2.5D representation as a specific, testable limitation on through-plane structure. A volumetric or genuinely cross-plane representation is the most direct follow-up this study suggests.

For transfer. The infrastructure for RQ4 exists: the cohort is reconciled, reports are parsed, and the geometric machinery is verified against independent annotations. What remains is reader adjudication of conflicting reference labels, a correct de-identification procedure, and—the largest gap—localisation for a cohort that carries no annotated coordinates.

That last gap is now quantified rather than merely named. Segmentation plus a per-condition constant offset is not sufficient: it costs 22.9% of grading performance, concentrated in the lateral compartments. Three routes remain, in increasing cost. A human confirms placement, which is viable for a review interface and is what the accompanying system design assumes. A per-patient rather than per-condition offset is learned, conditioning on something like vertebral width, which would test whether the residual error is anatomical variation rather than noise. Or a detector is trained directly on the 48,657 annotated coordinates LumbarDISC already provides, which is the most promising and is a separate piece of work rather than an extension of this one.

Until one of those exists, any RQ4 result must be reported alongside a verified-localisation arm, or it cannot be attributed.

5.8 Conclusion

This thesis set out to determine whether explicit anatomical structure improves automated grading of lumbar degenerative disease. It determined that, at this data scale and under this reference standard, it largely does not, and it established why: the convolutional encoder already localises the graded structure, and the reference standard leaves narrow headroom above it.

The system that was built works, reproducibly and by a margin that survives correction. The components that make it work are relational typing and a clinically motivated objective, not anatomical correspondence or disease-conditioned routing. Two proposed mechanisms are rejected on evidence gathered specifically to test them, a third is supported in a narrower and more precise form than proposed, and the study's own controls overturned three conclusions that would otherwise have been reported as findings.

Chapter 1 framed the aim as "a determination rather than a promise of superiority". That framing was prescient, and this thesis should be read as having discharged it.

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